

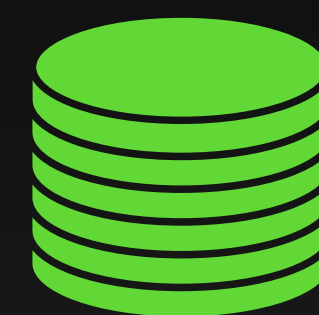
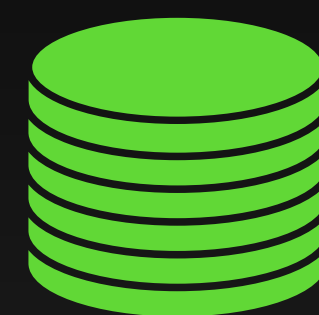
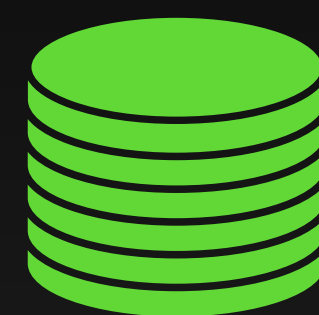
DAG-Based Consensus

Introduction and practical Aspects

Alberto Sonnino

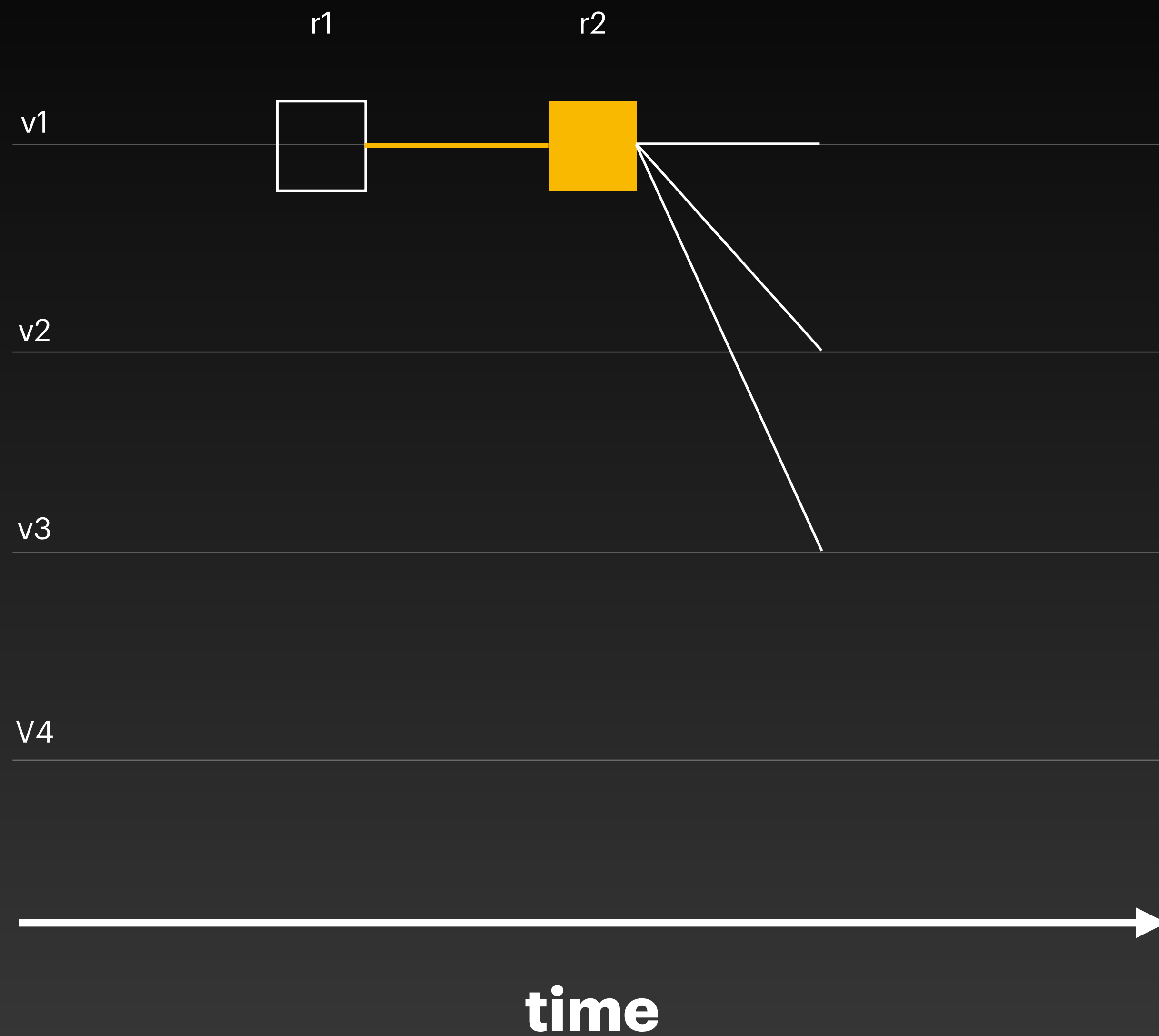


> 2/3

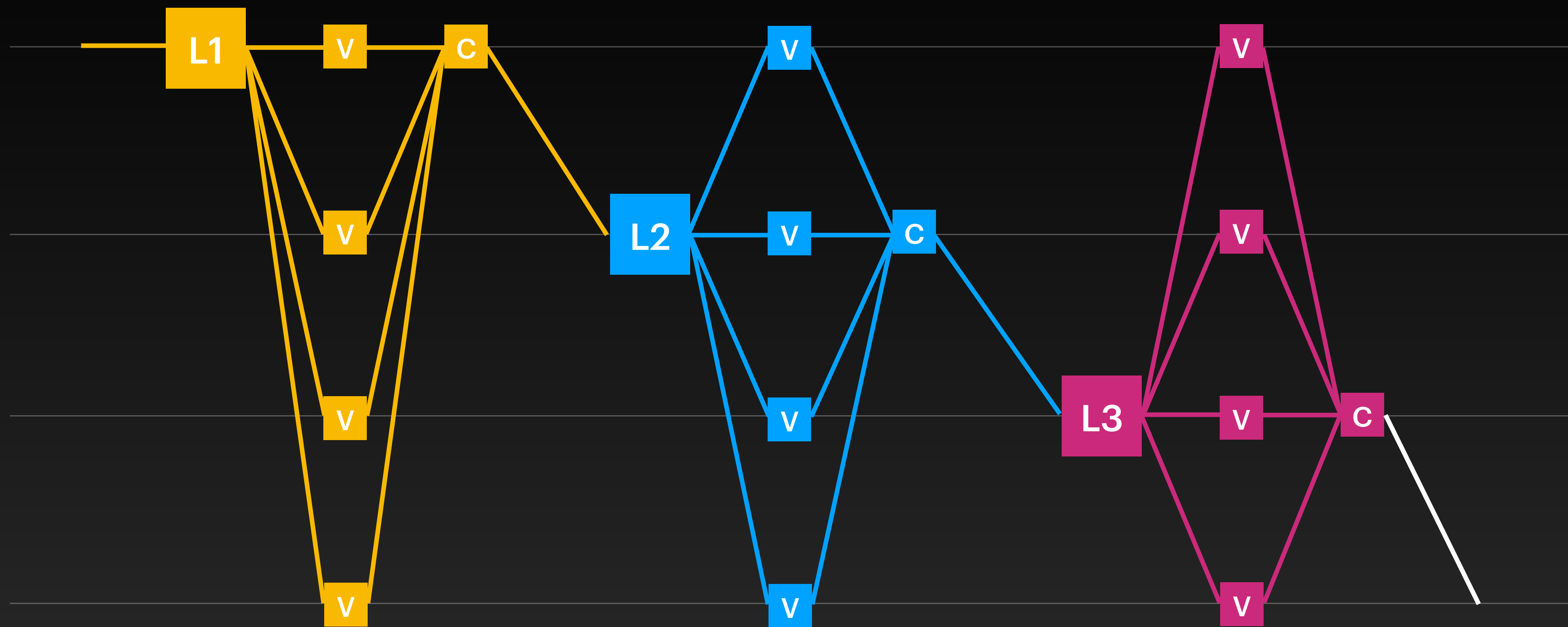


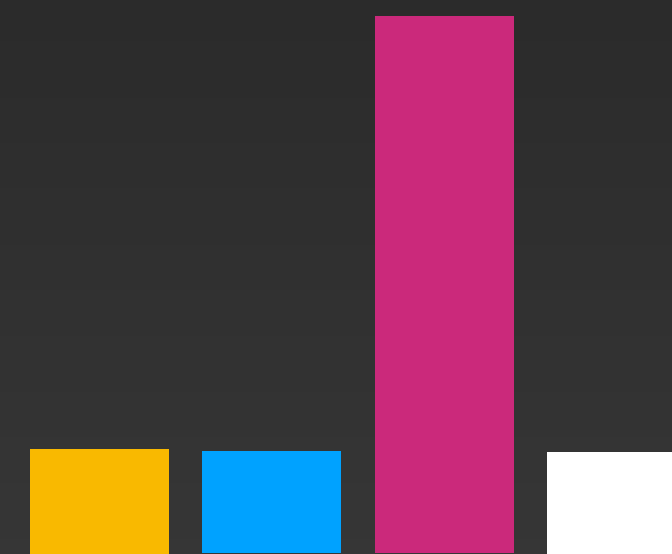
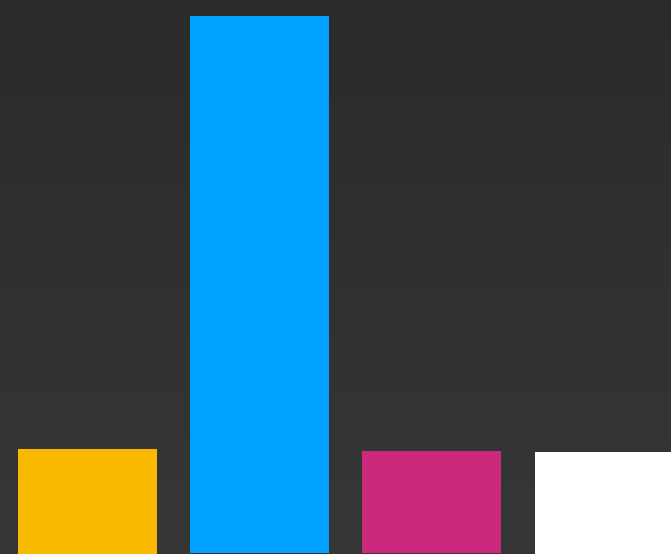
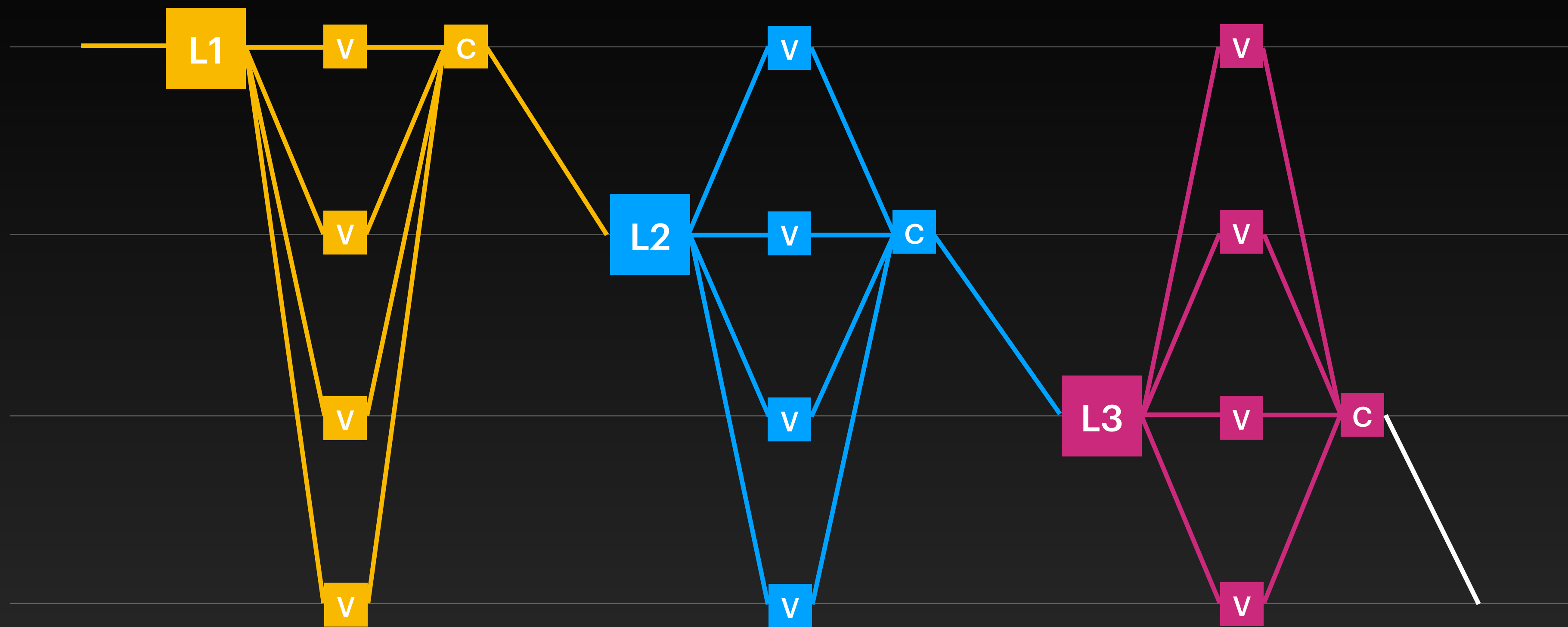
Partial Synchrony

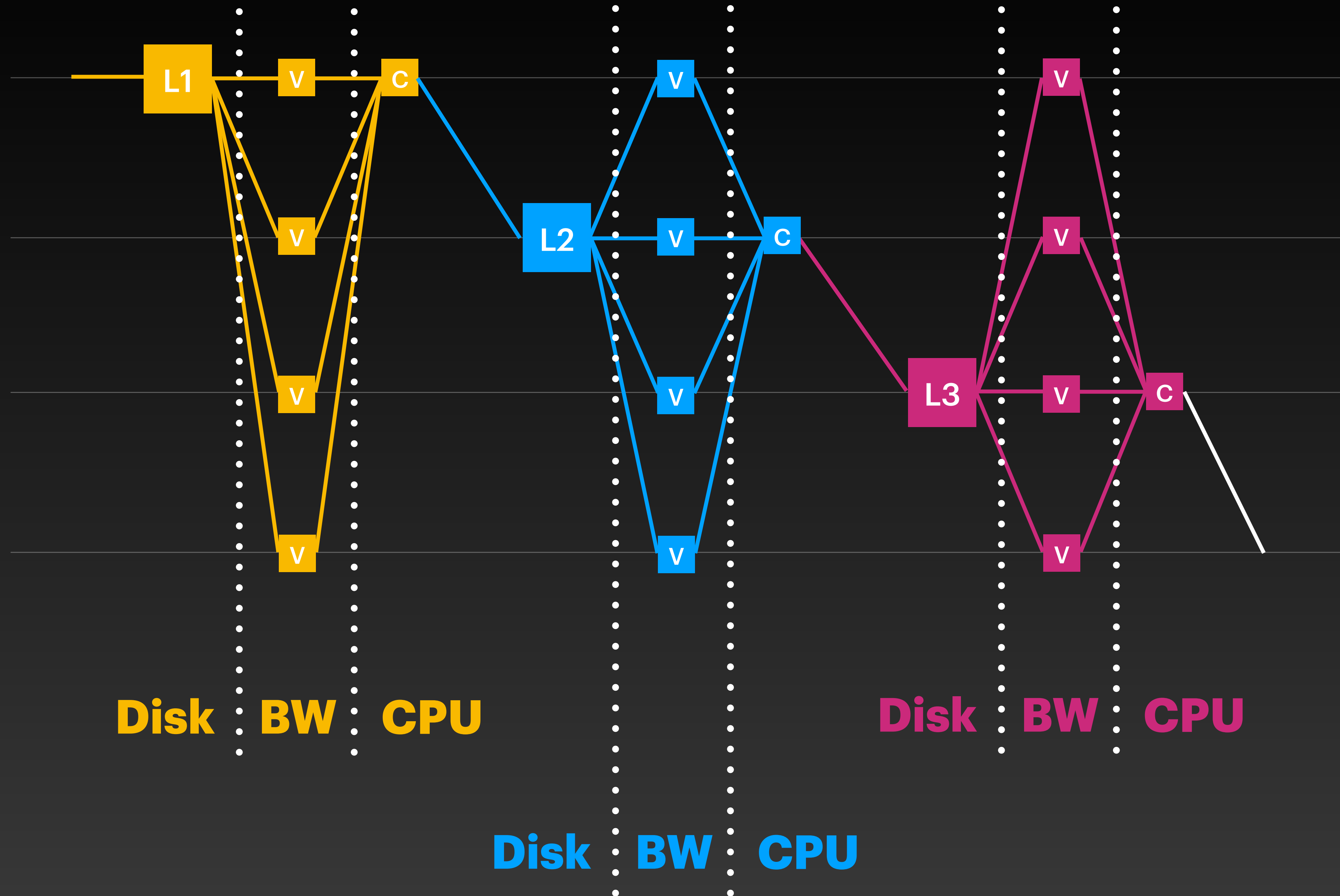


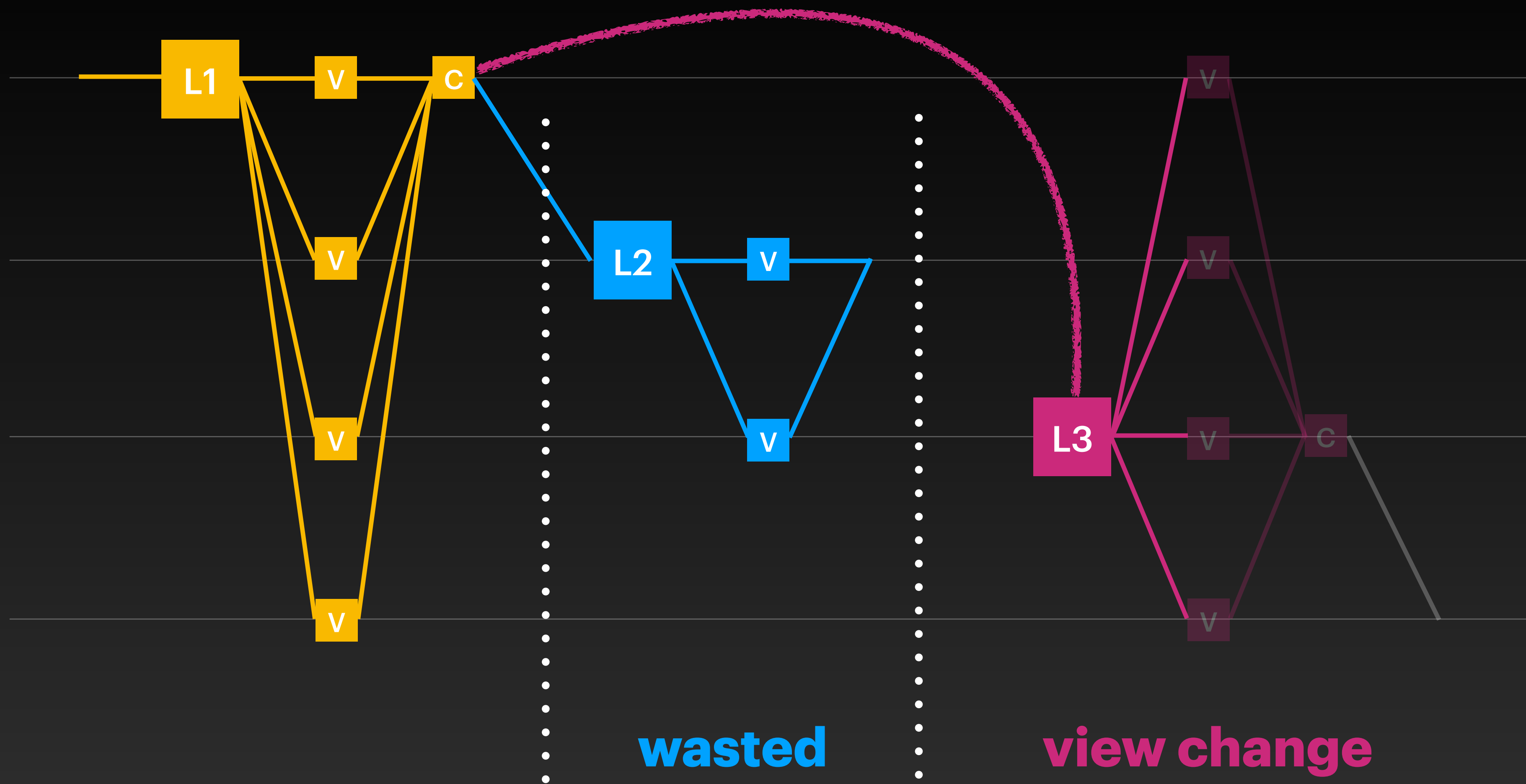


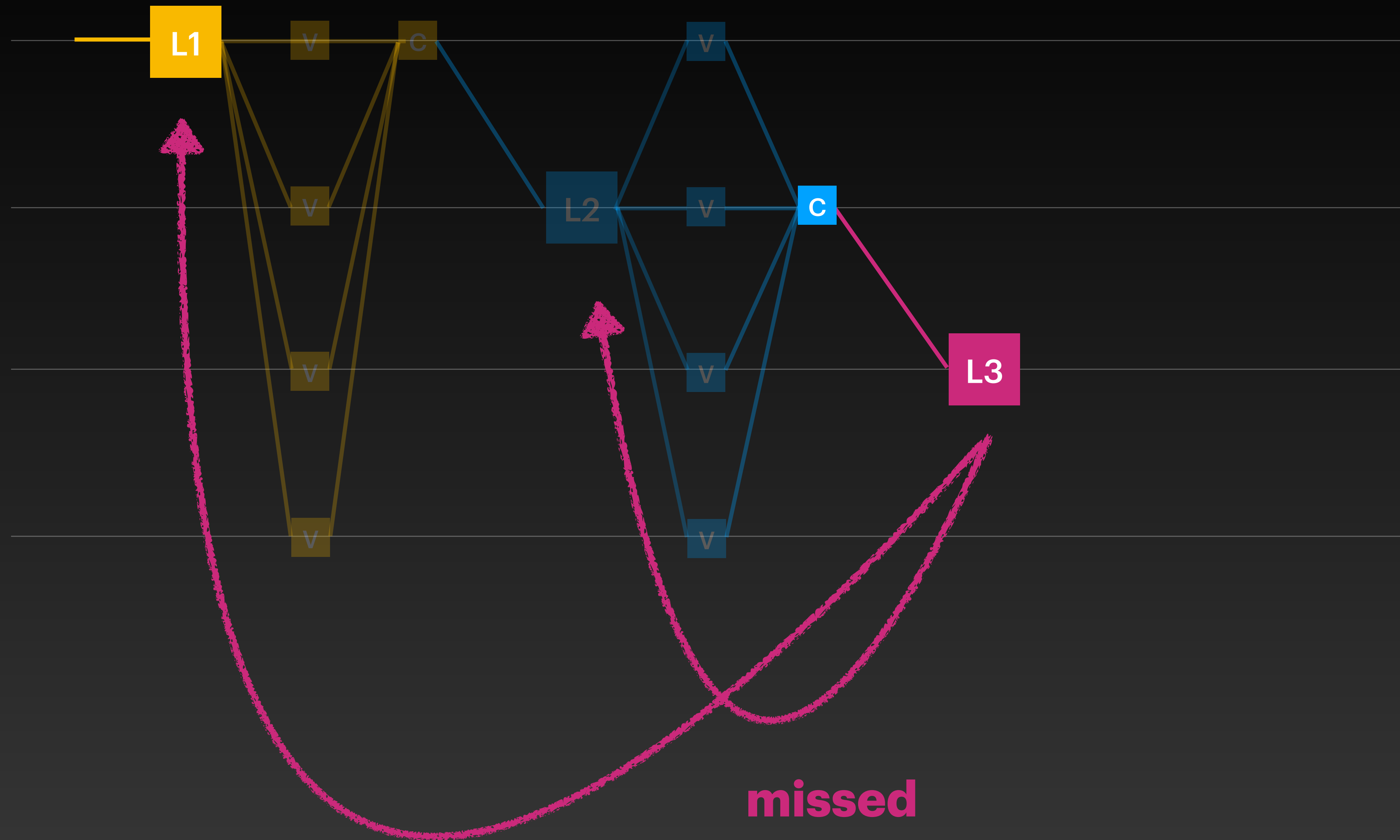
- Round number
- Author
- Payload (txs)
- Parent
- Signature

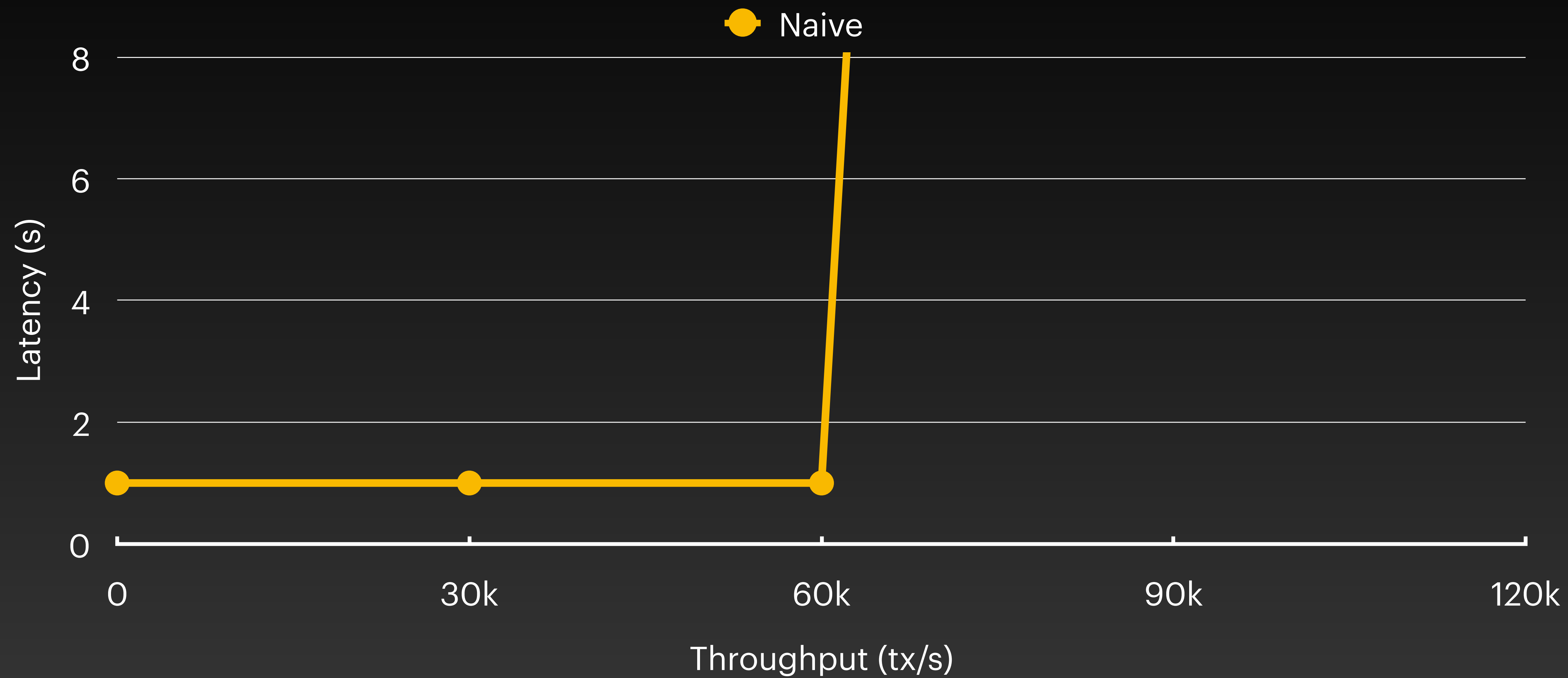


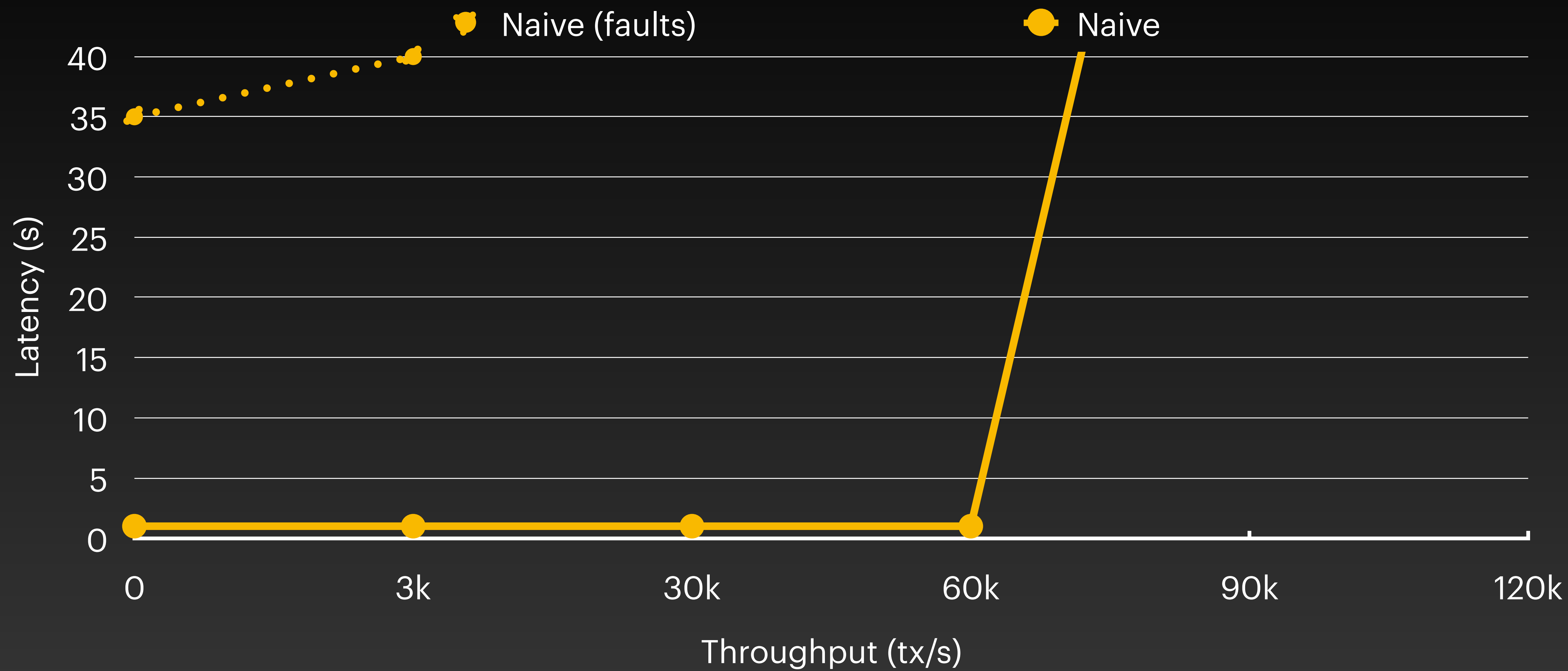


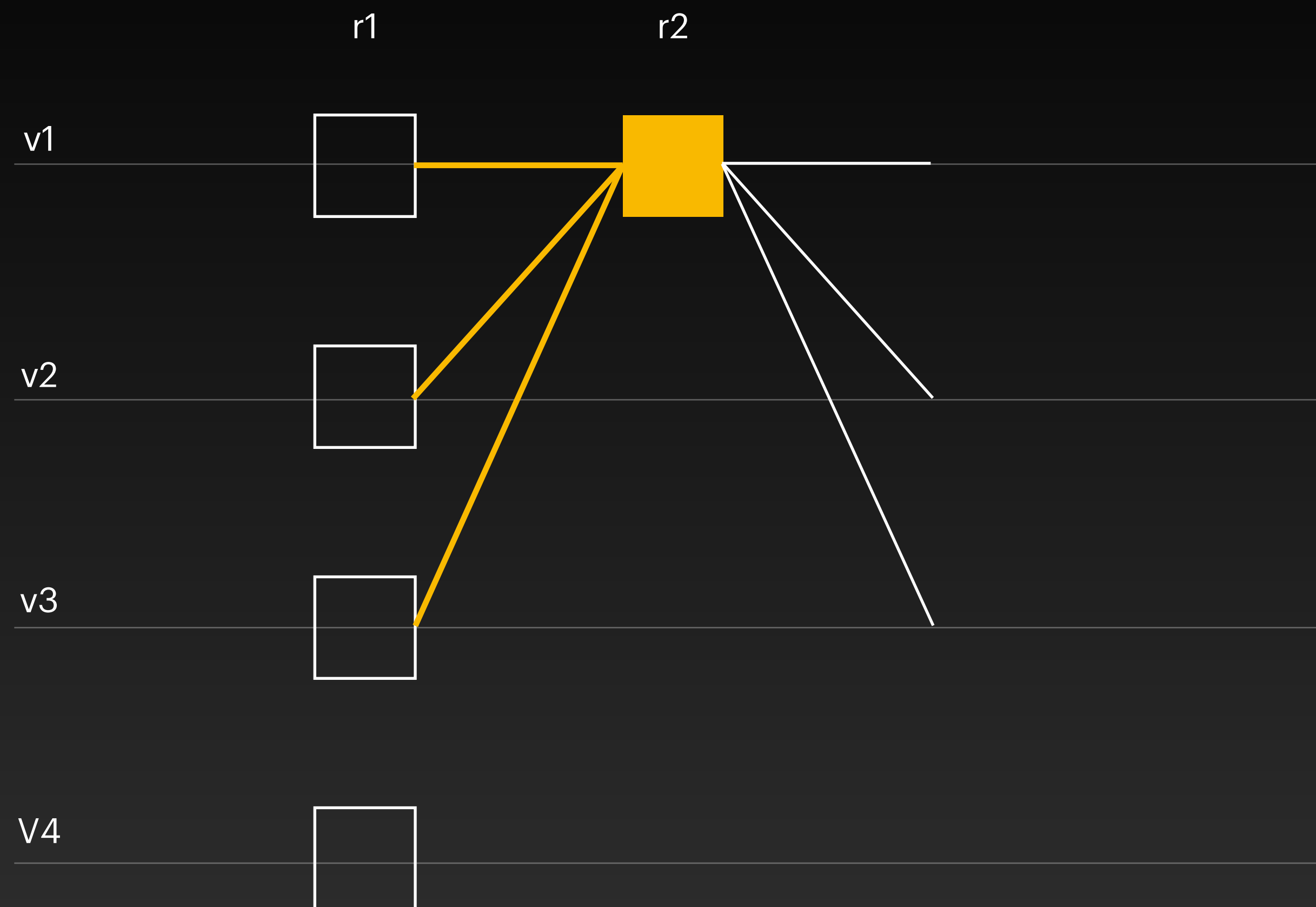






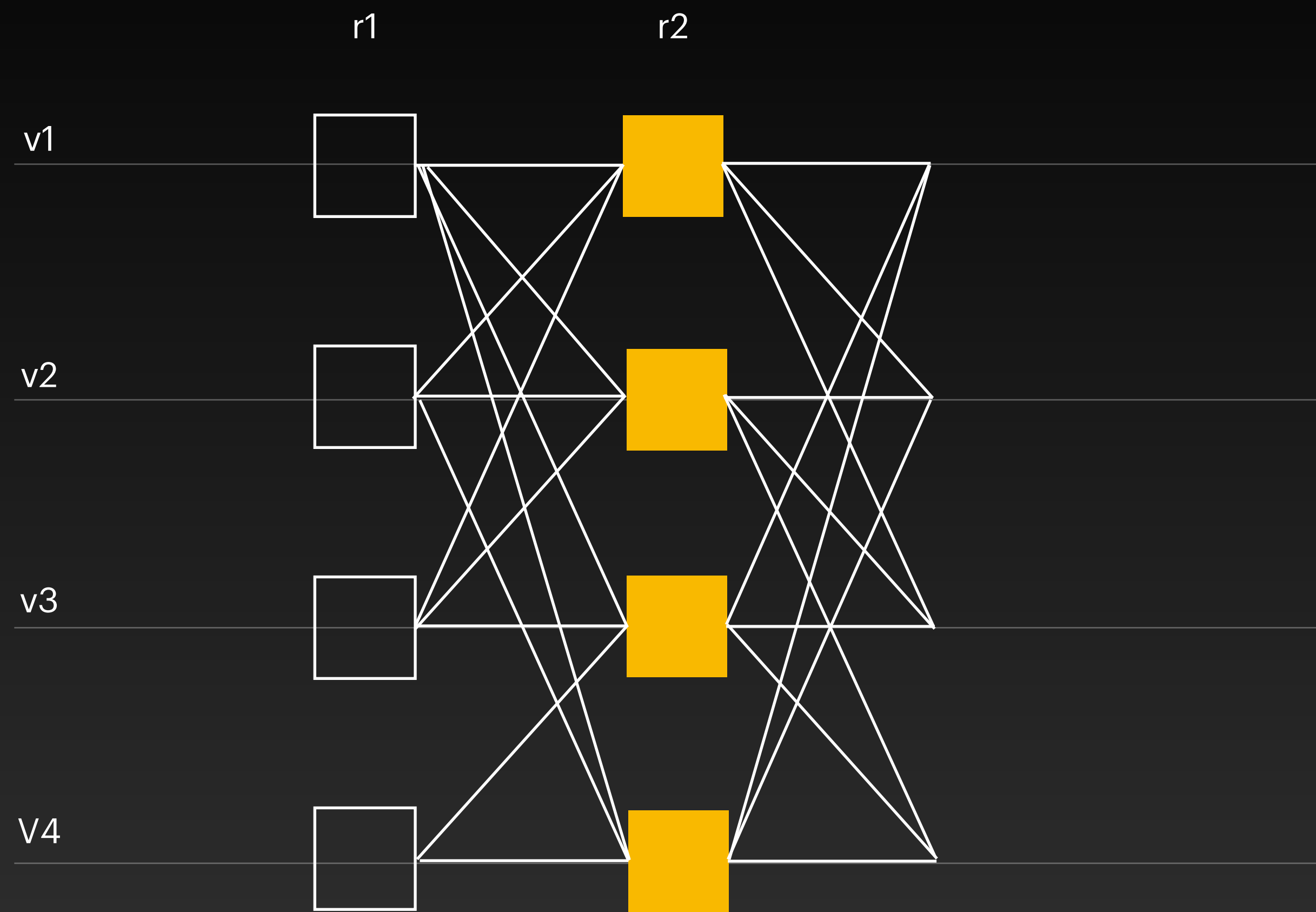






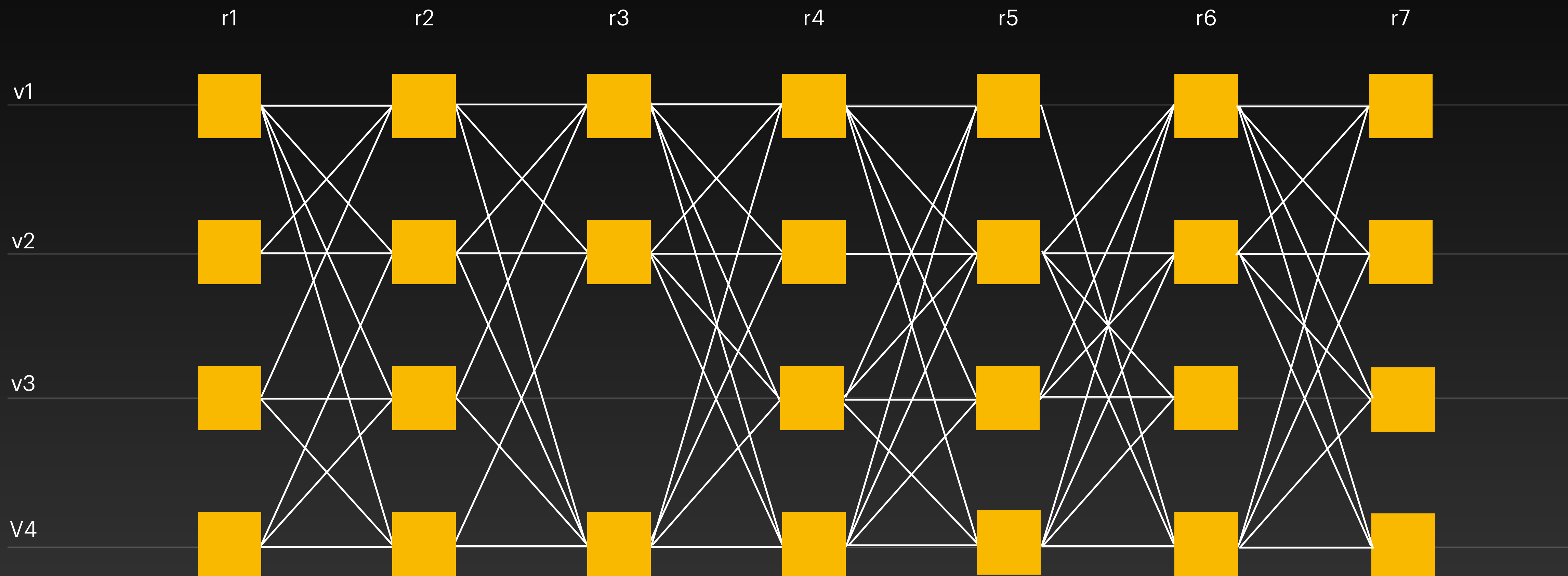
- Round number
- Author
- Payload (txs)
- **Parents**
- Signature

Back Pressure



- Round number
- Author
- Payload (txs)
- Parents
- Signature

Multi Proposer

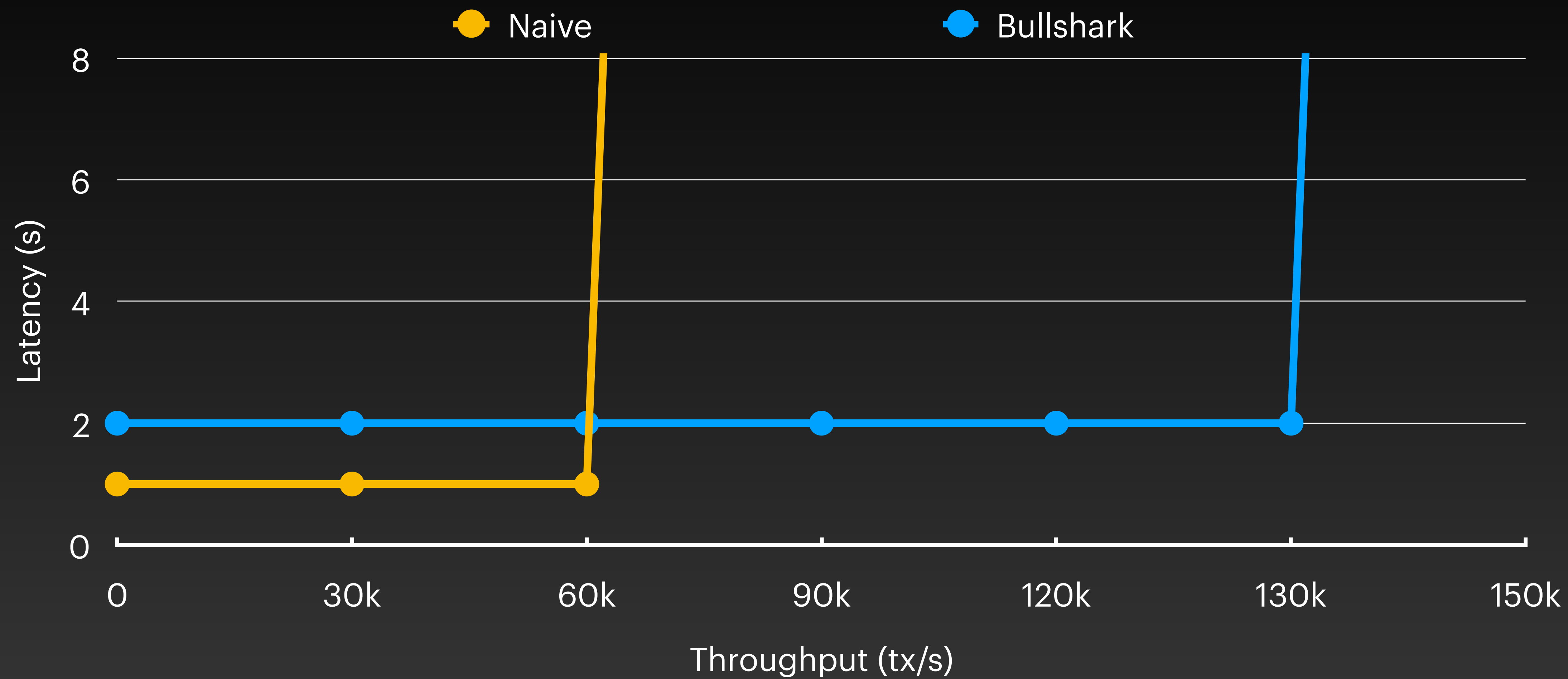


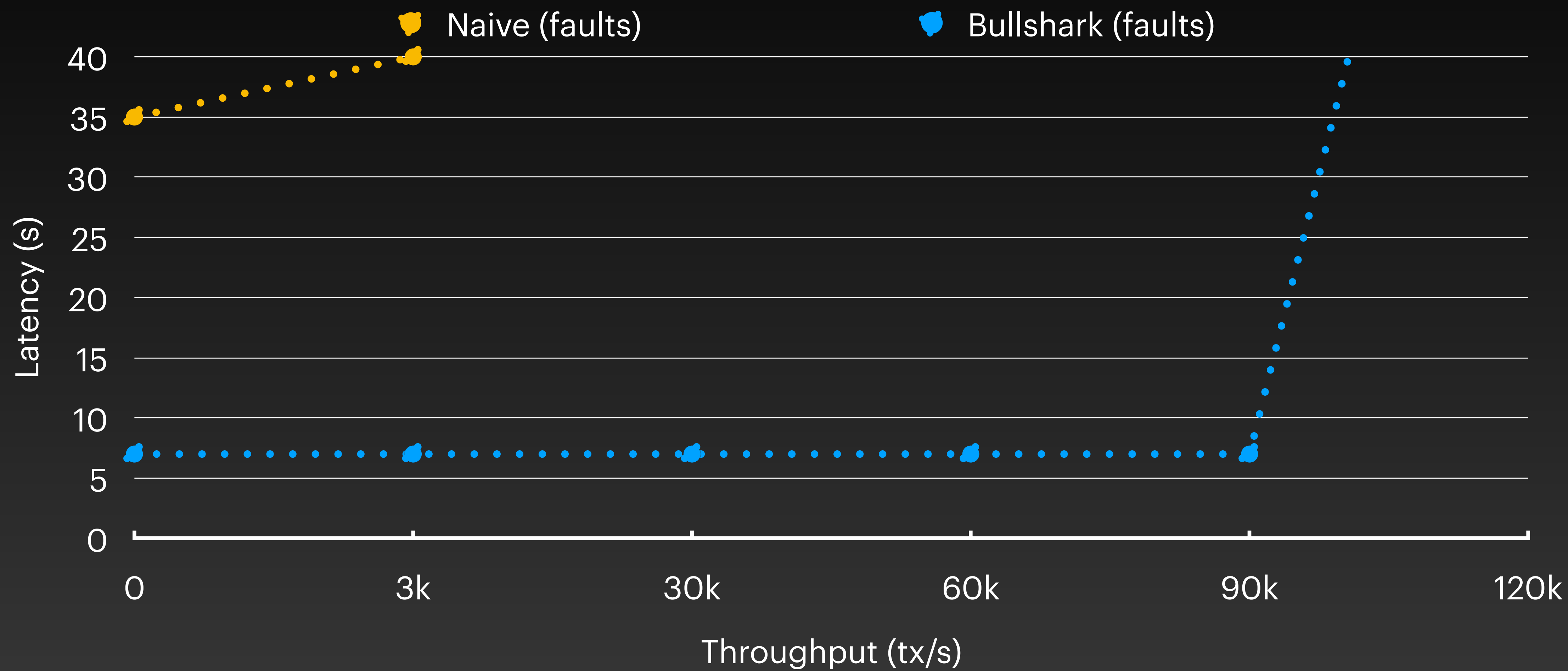
Back Pressure

Robustness

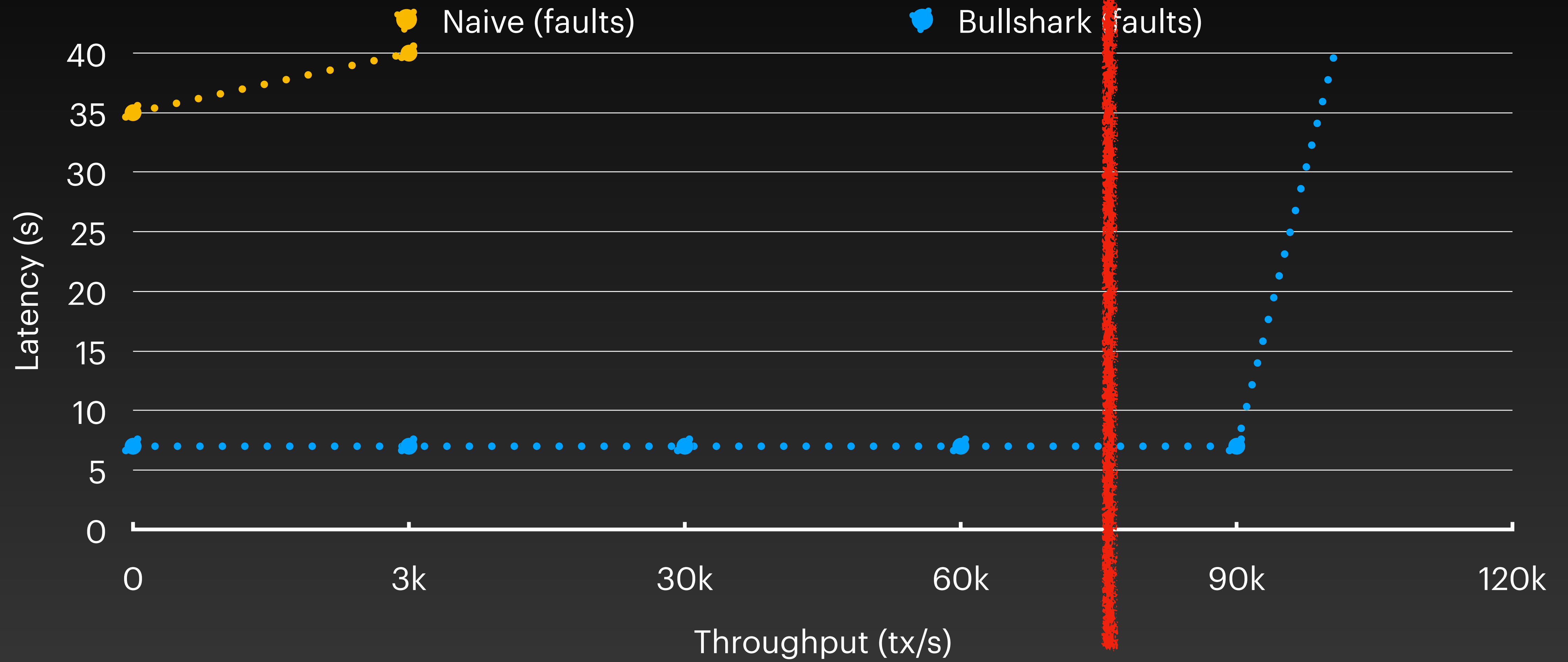
High Resource Utilisation

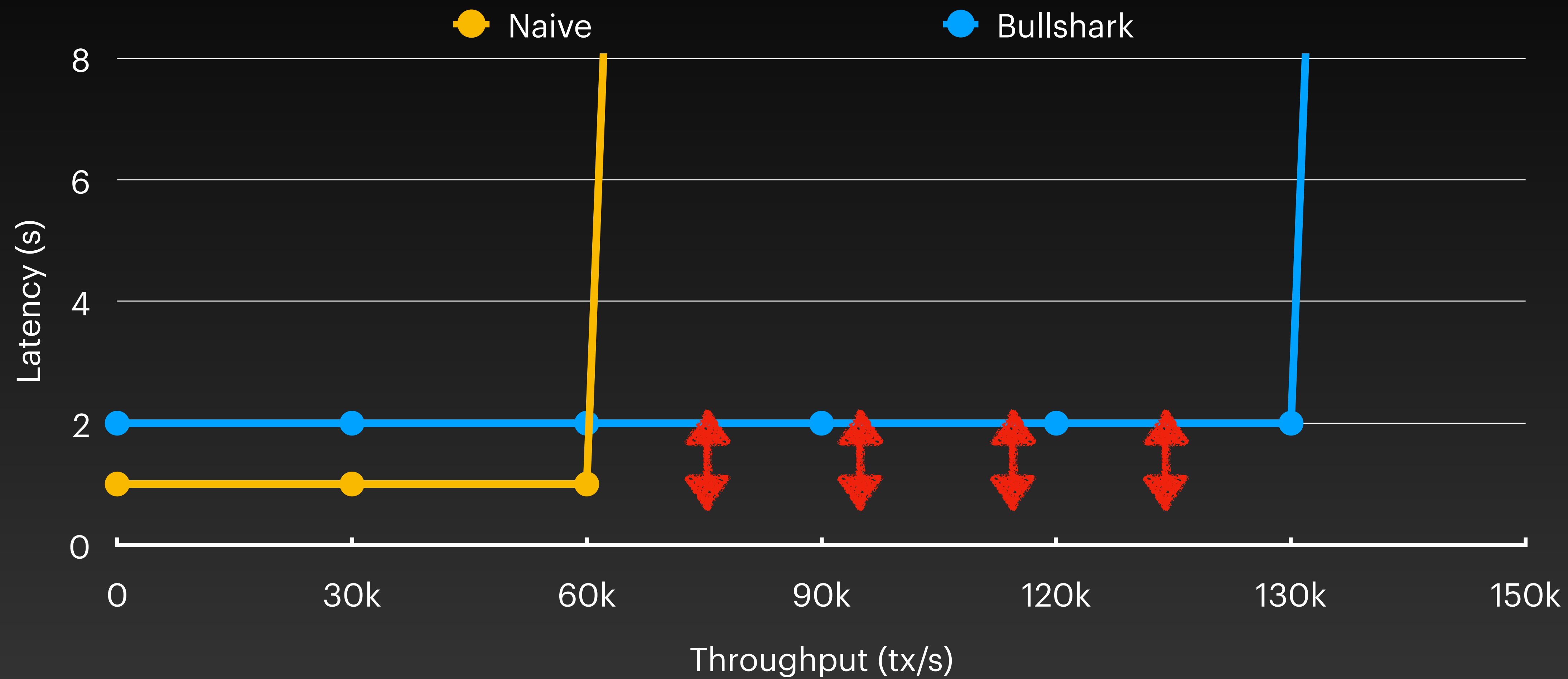
Throughput

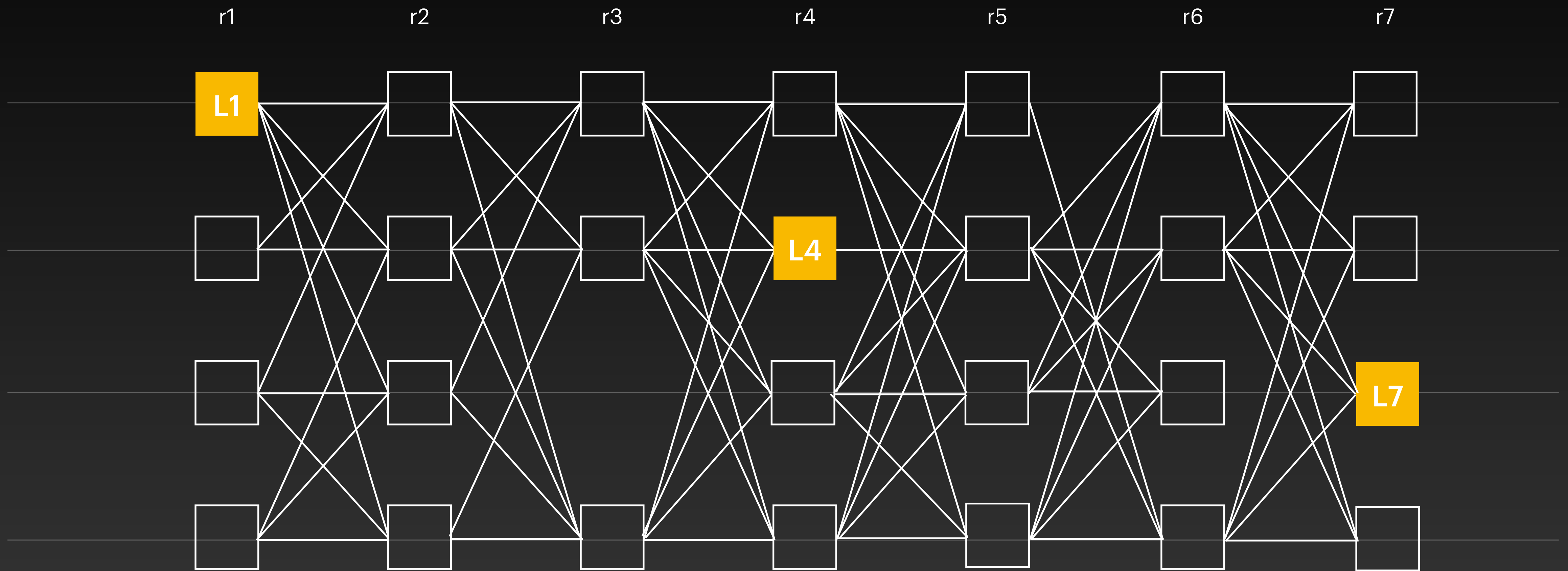




Visa + Master Card







Back Pressure

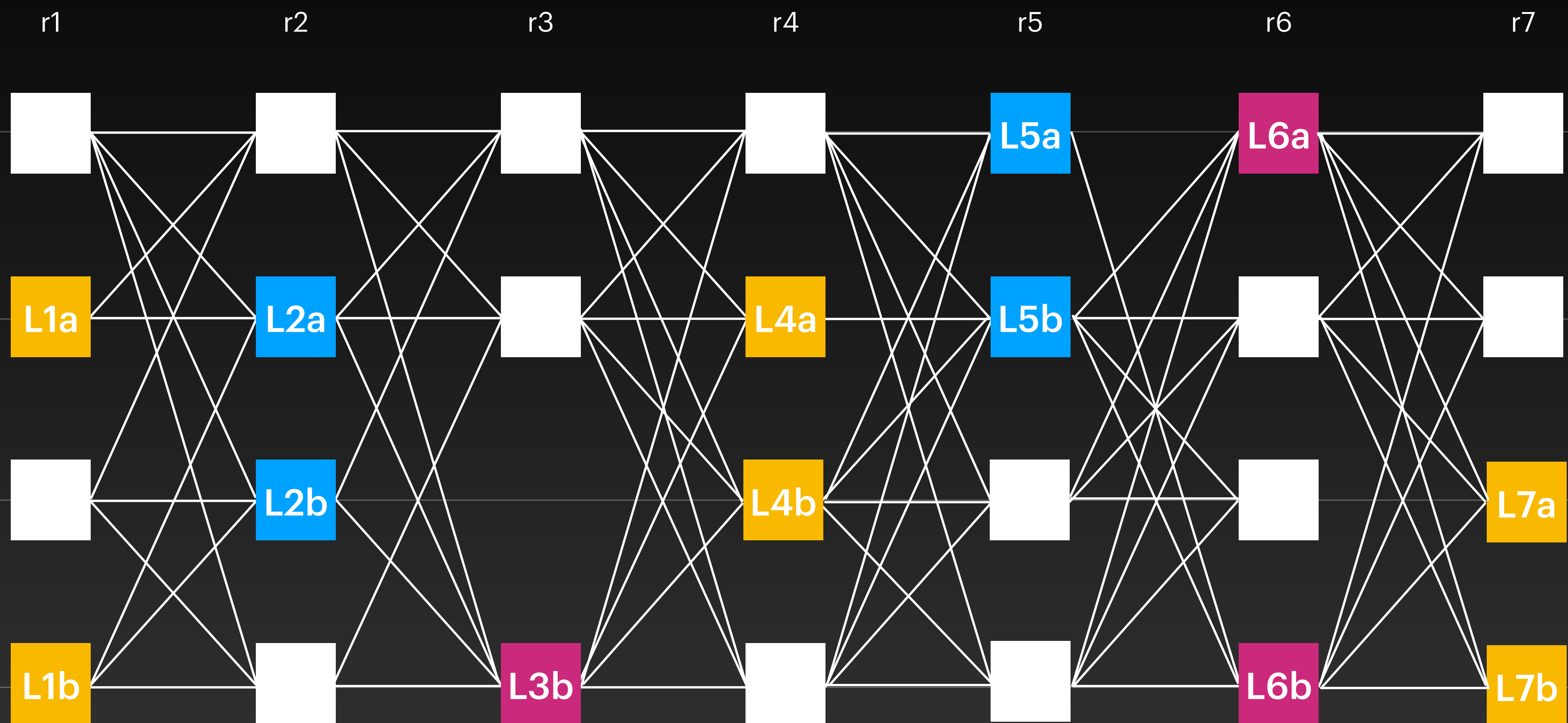
Robustness

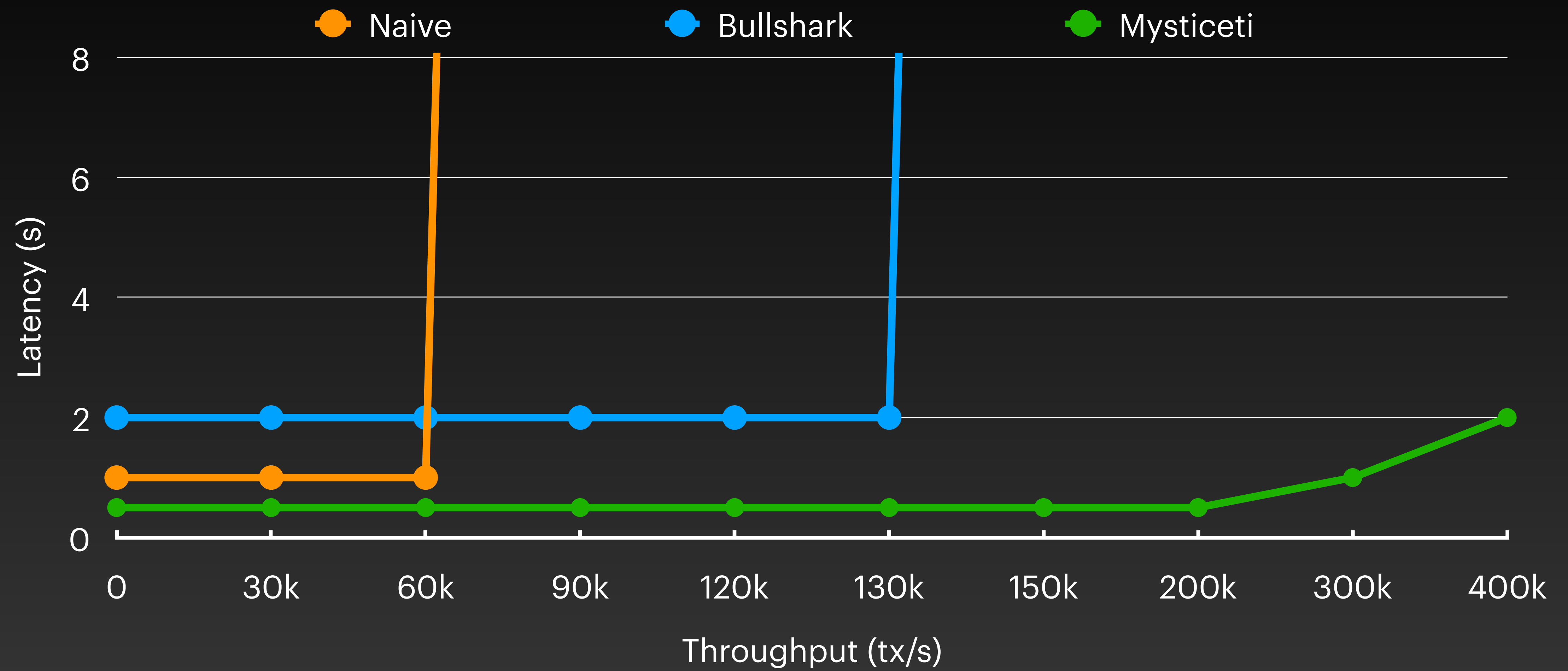
High Resource Utilisation

Throughput

Frequent Commits

Latency



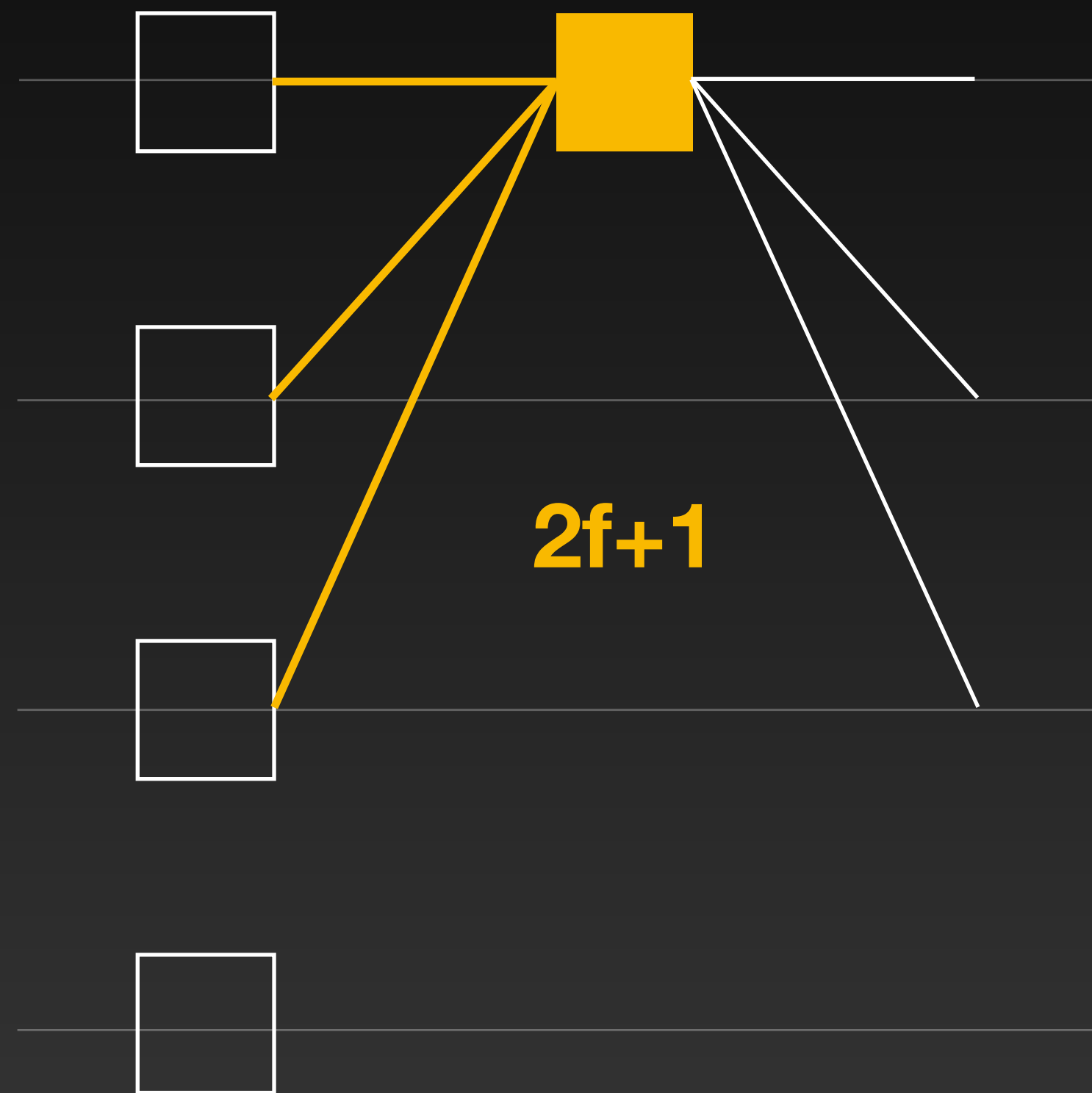


Protocol	Committee	Load/TPS	P50	P95
Bullshark	137	5k	2.89 s	4.60 s
Mysticeti	137	5k	397 ms	690 ms

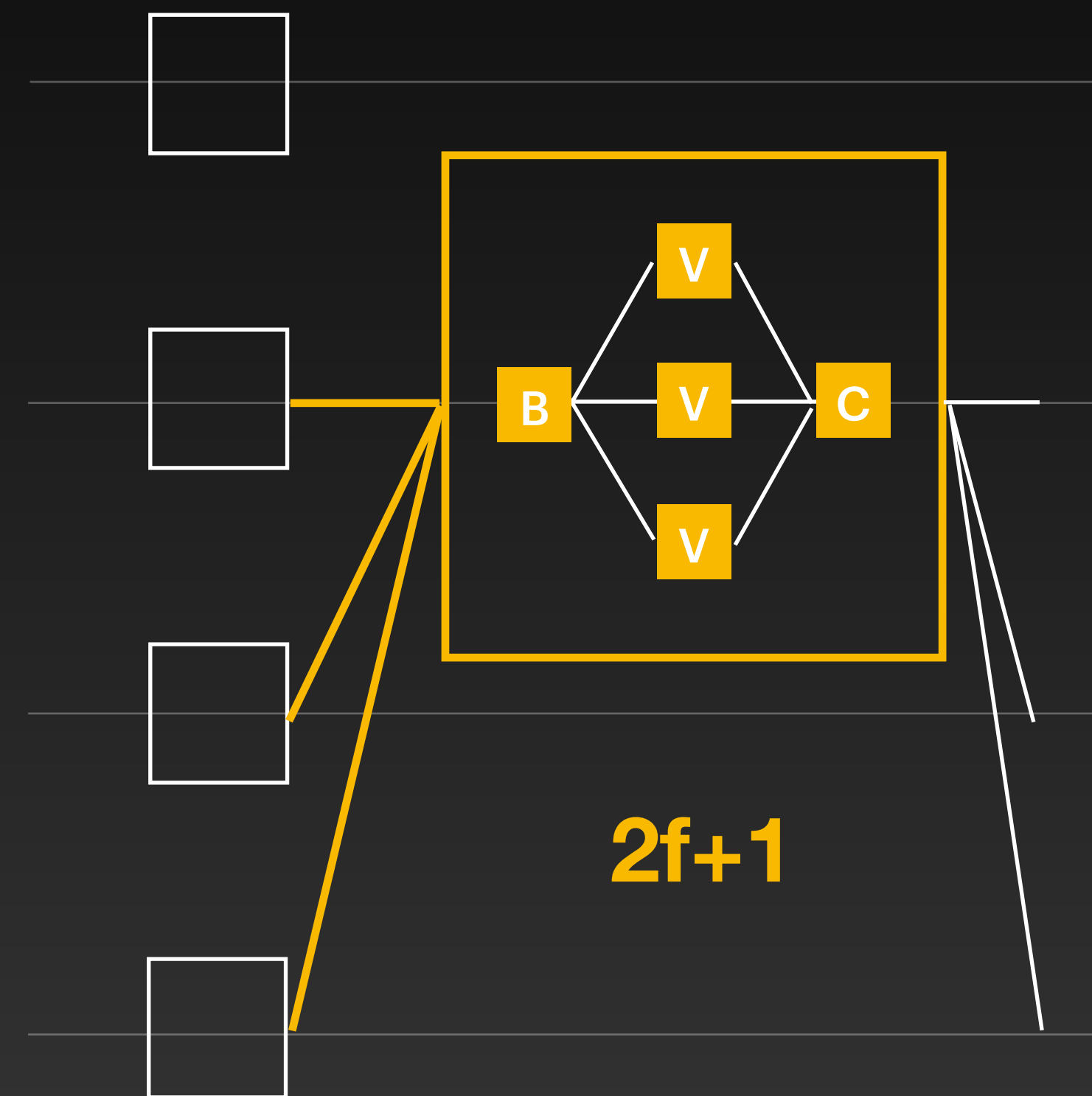
We ran it for 24h: all metrics are fine 🍷

Structured DAGs

Uncertified

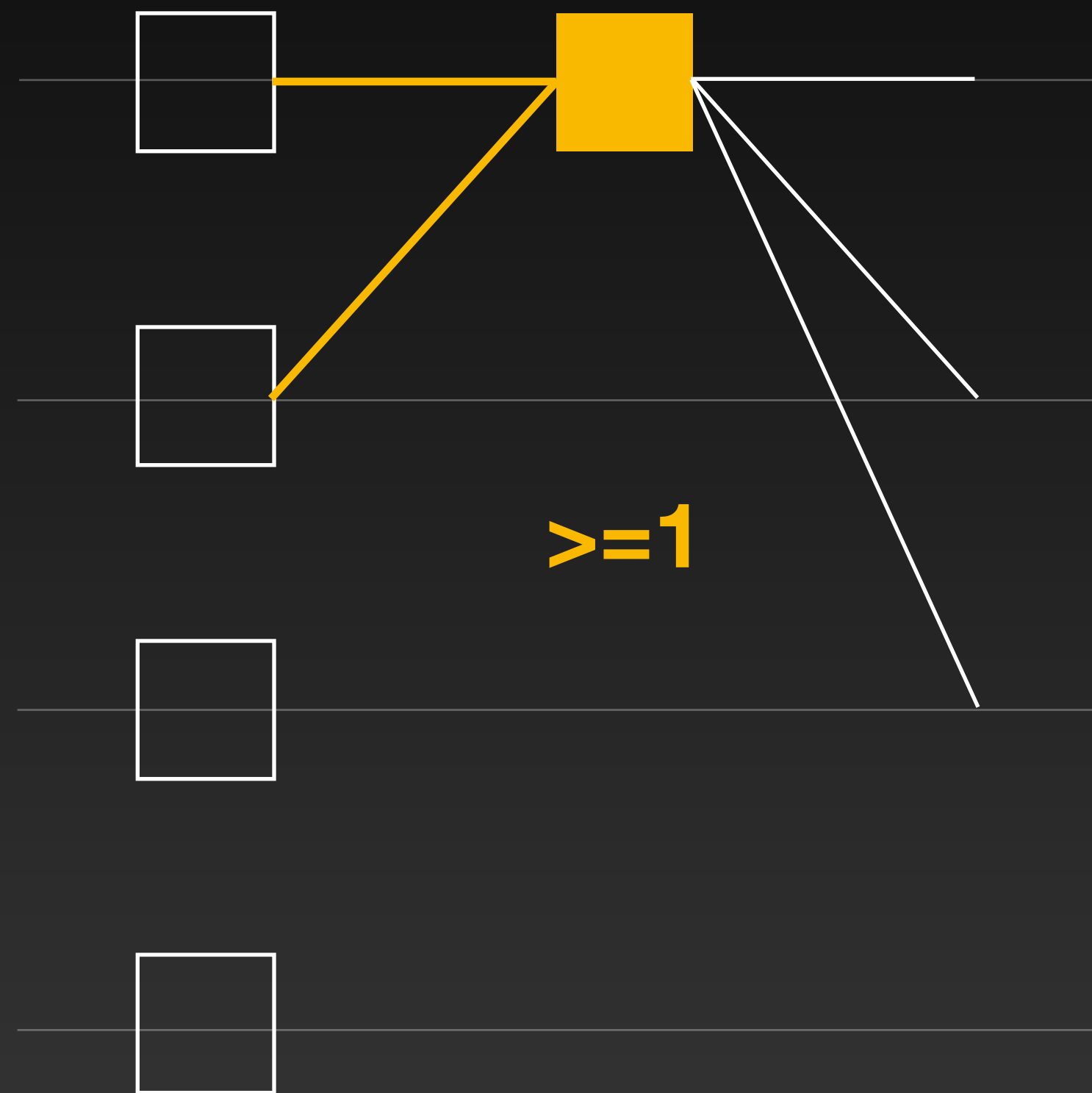


Certified

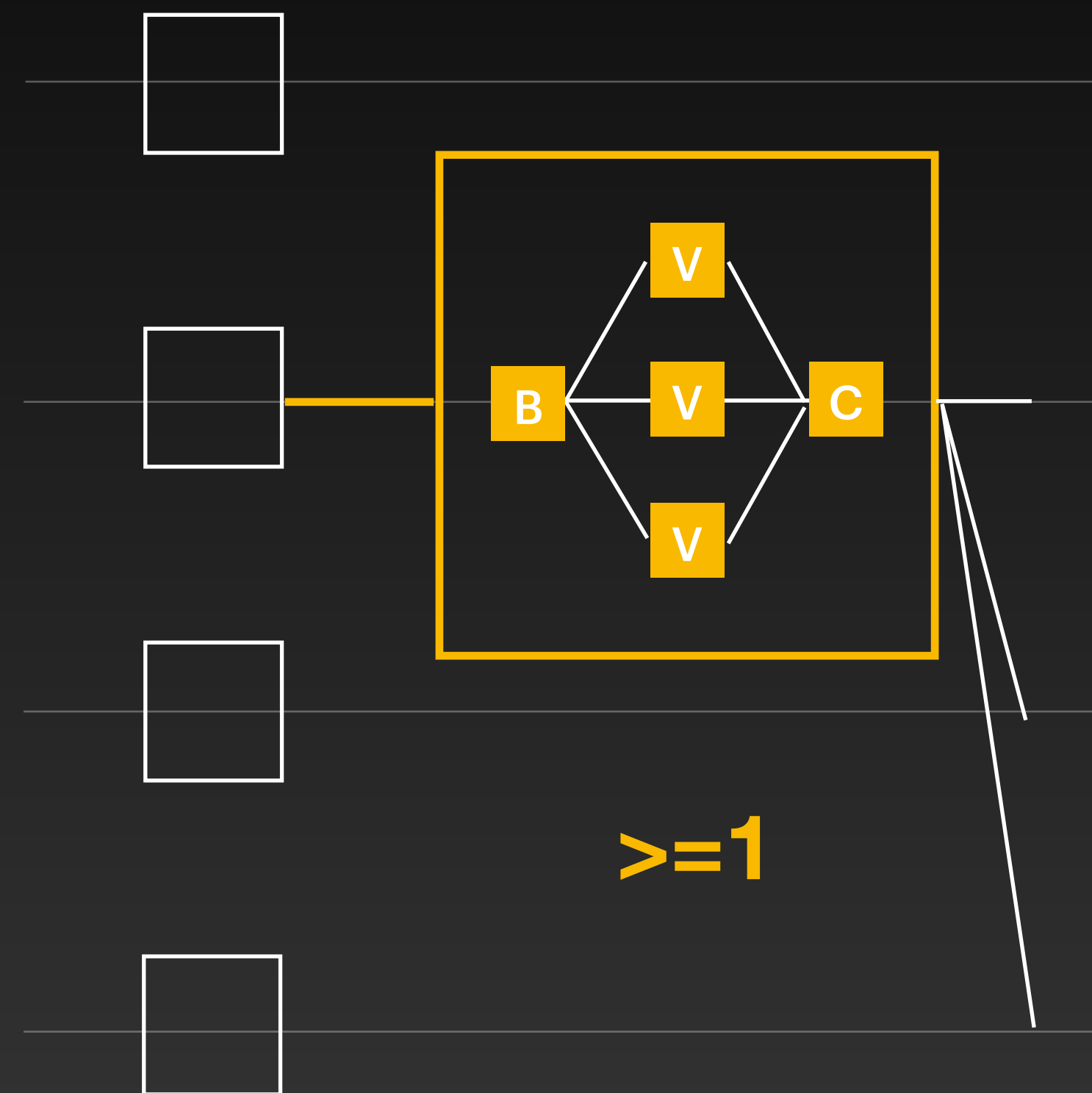


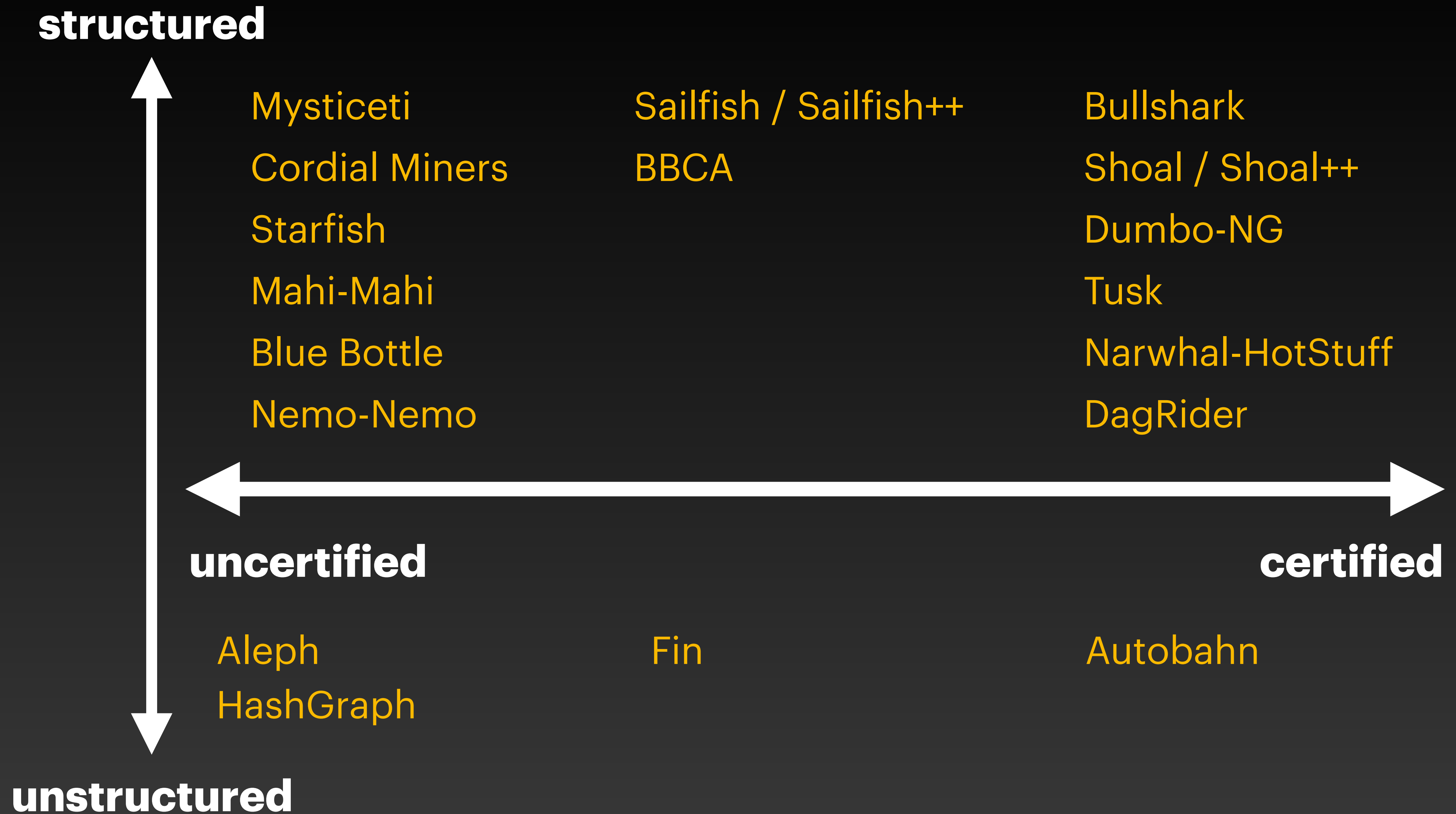
Unstructured DAGs

Uncertified



Certified





Certified

High Robustness • Easier Block Synchronisation

Uncertified

Low CPU • Graceful Crash Faults

Unstructured

Network Flexibility

Cordial Miners

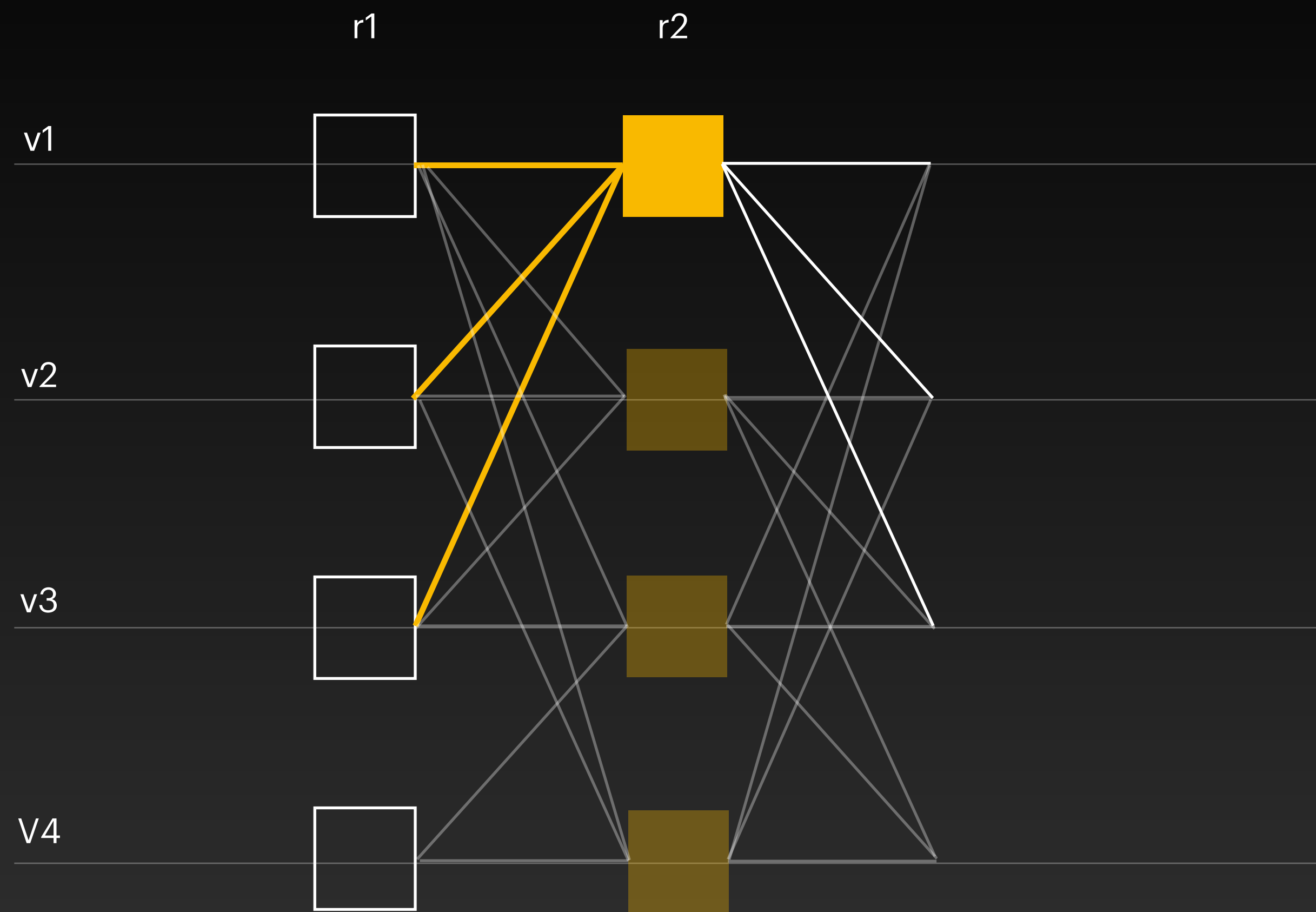
Bullshark

Shoal++

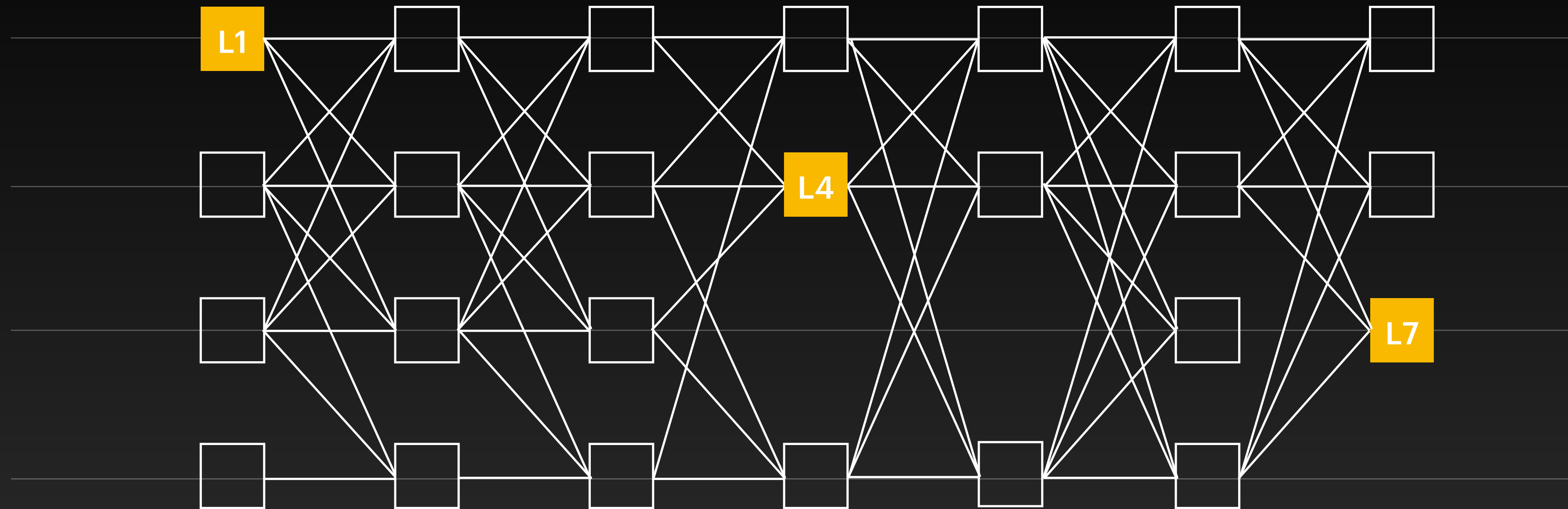
Mysticeti

...

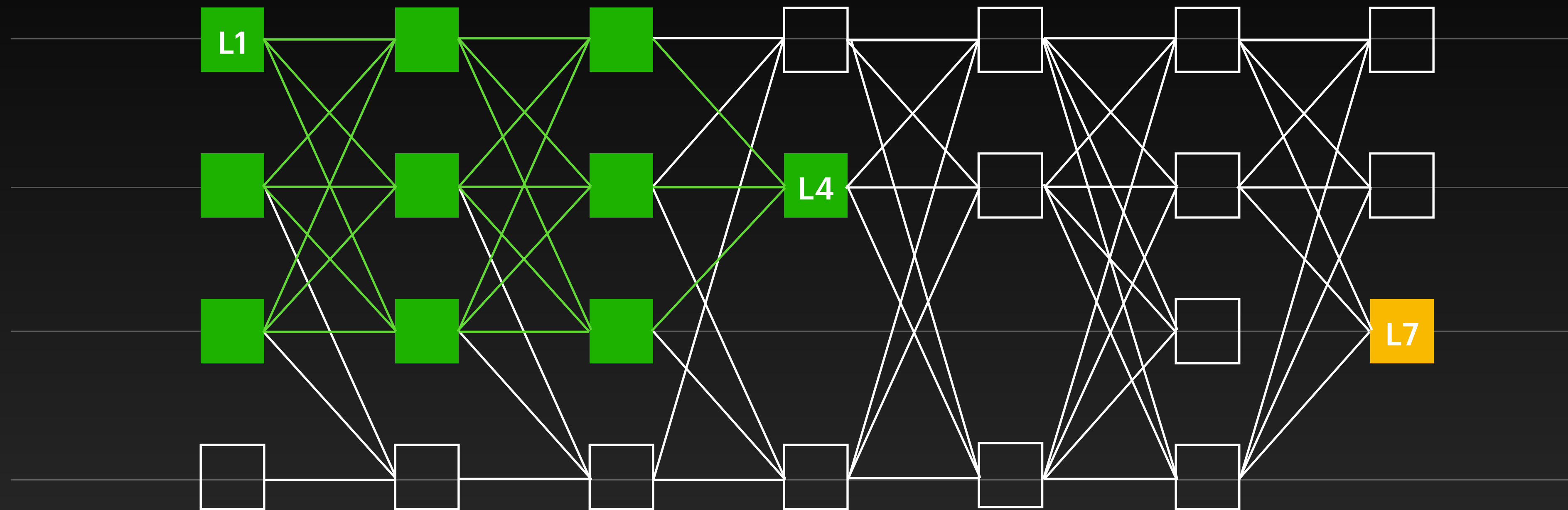
Structured • Uncertified



- Round number
- Author
- Payload (txs)
- **$2f+1$ Parents**
- Signature



- We focus on ordering leaders: L1 L4 L7



- We focus on ordering leaders: L1 L4 L7
- Linearising the sub-DAG is simple

wave 1

wave 2

wave 3

r1

r2

r3

r4

r5

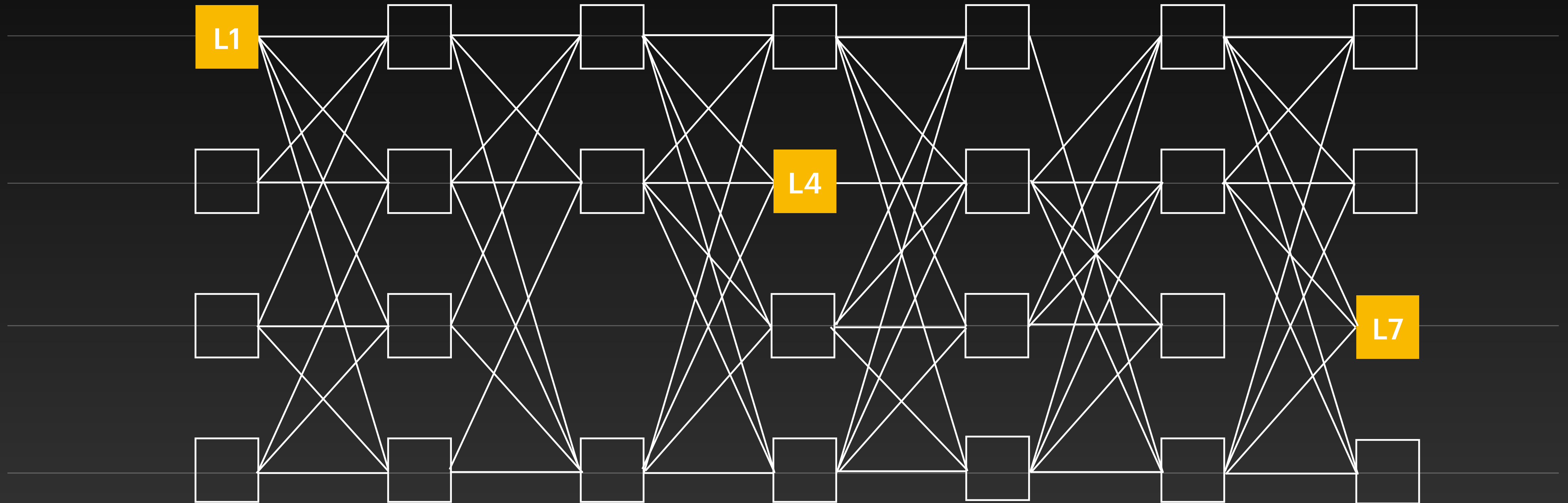
r6

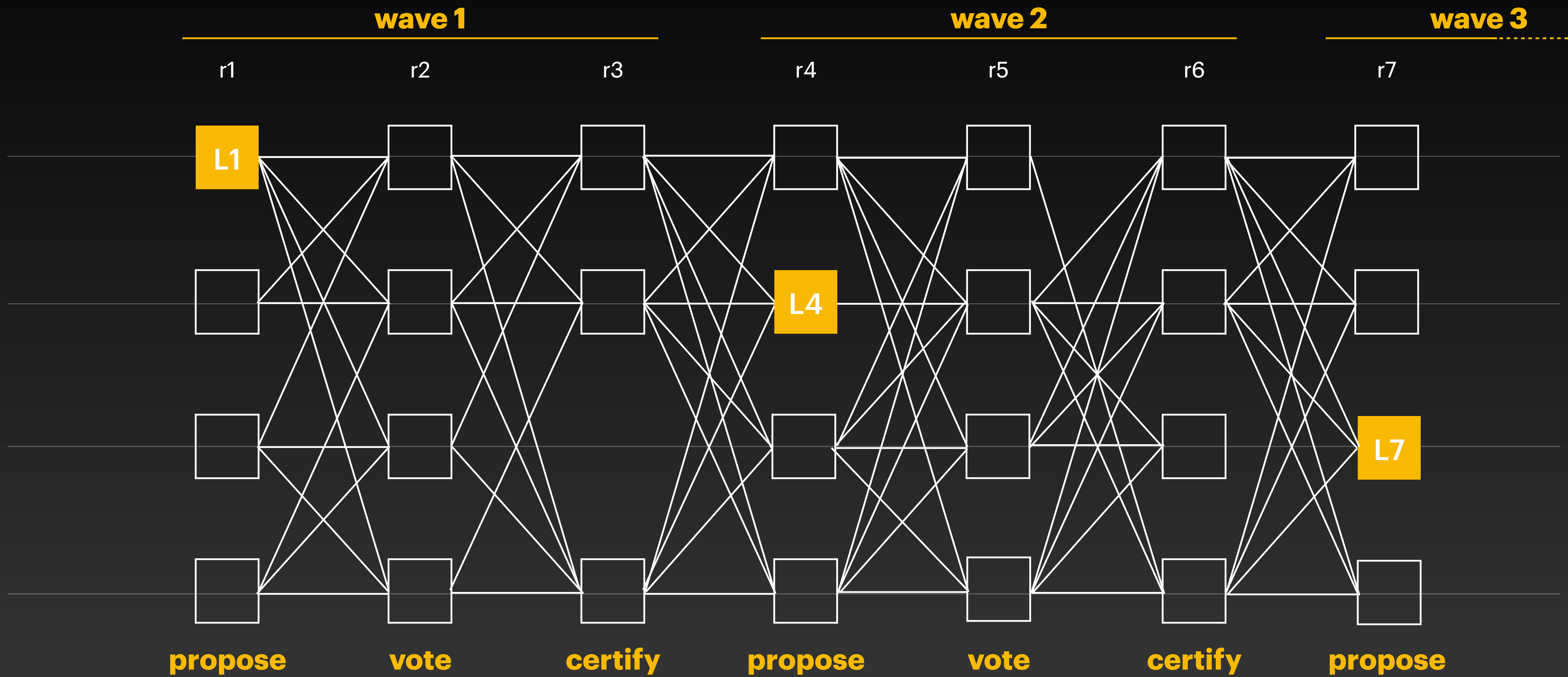
r7

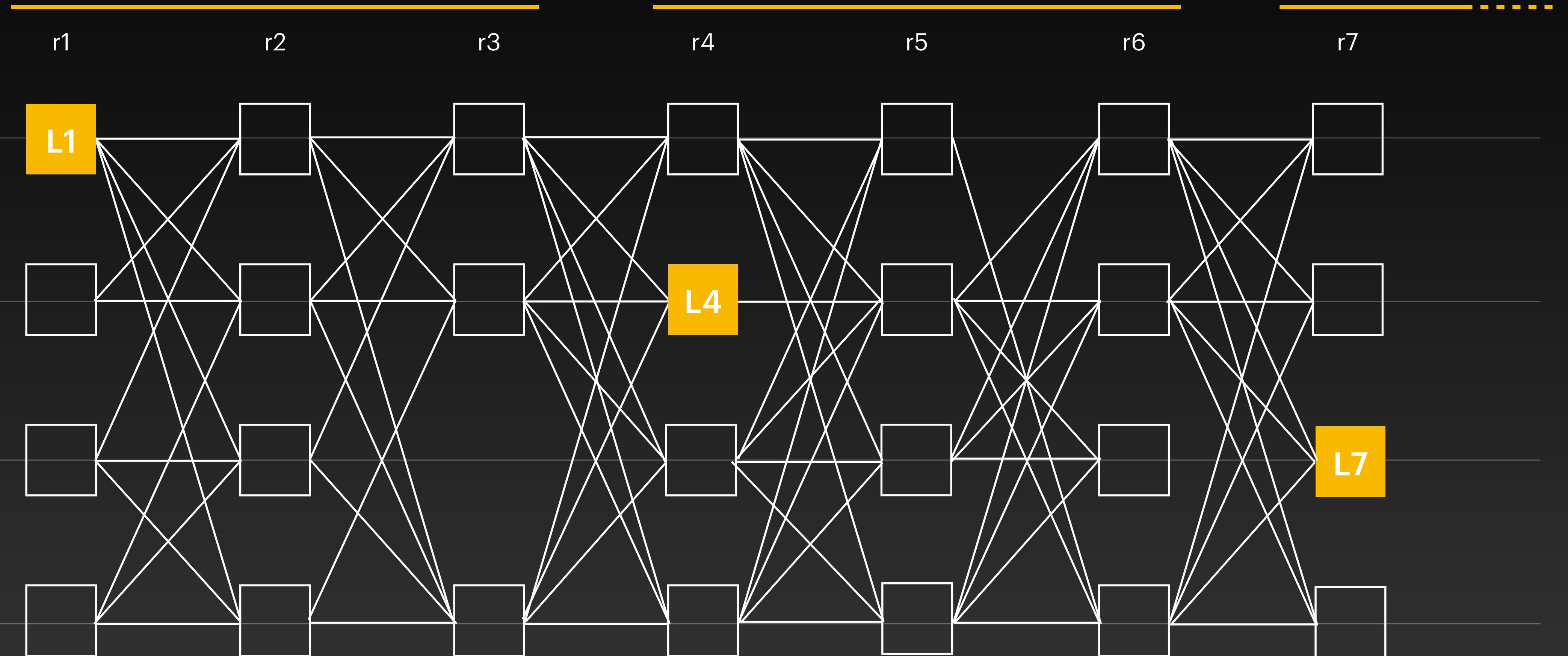
L1

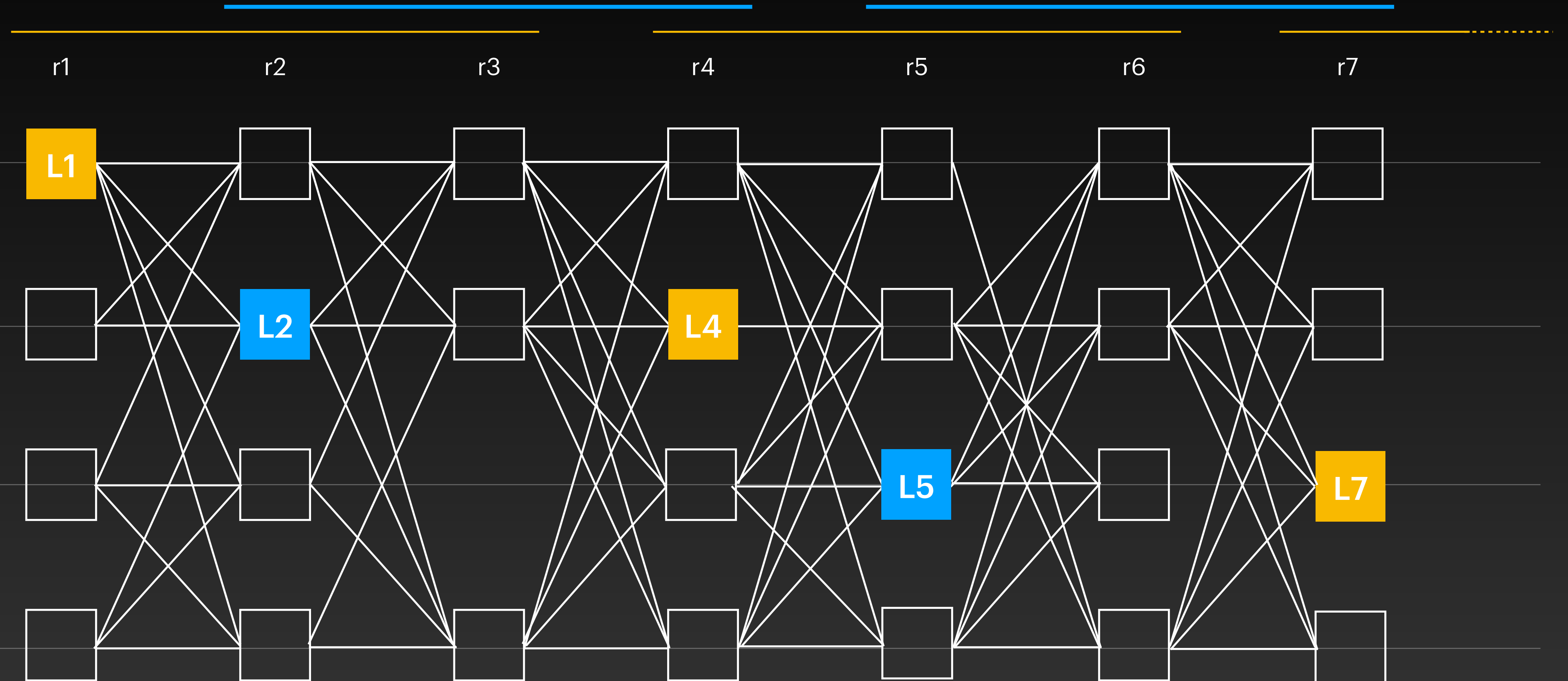
L4

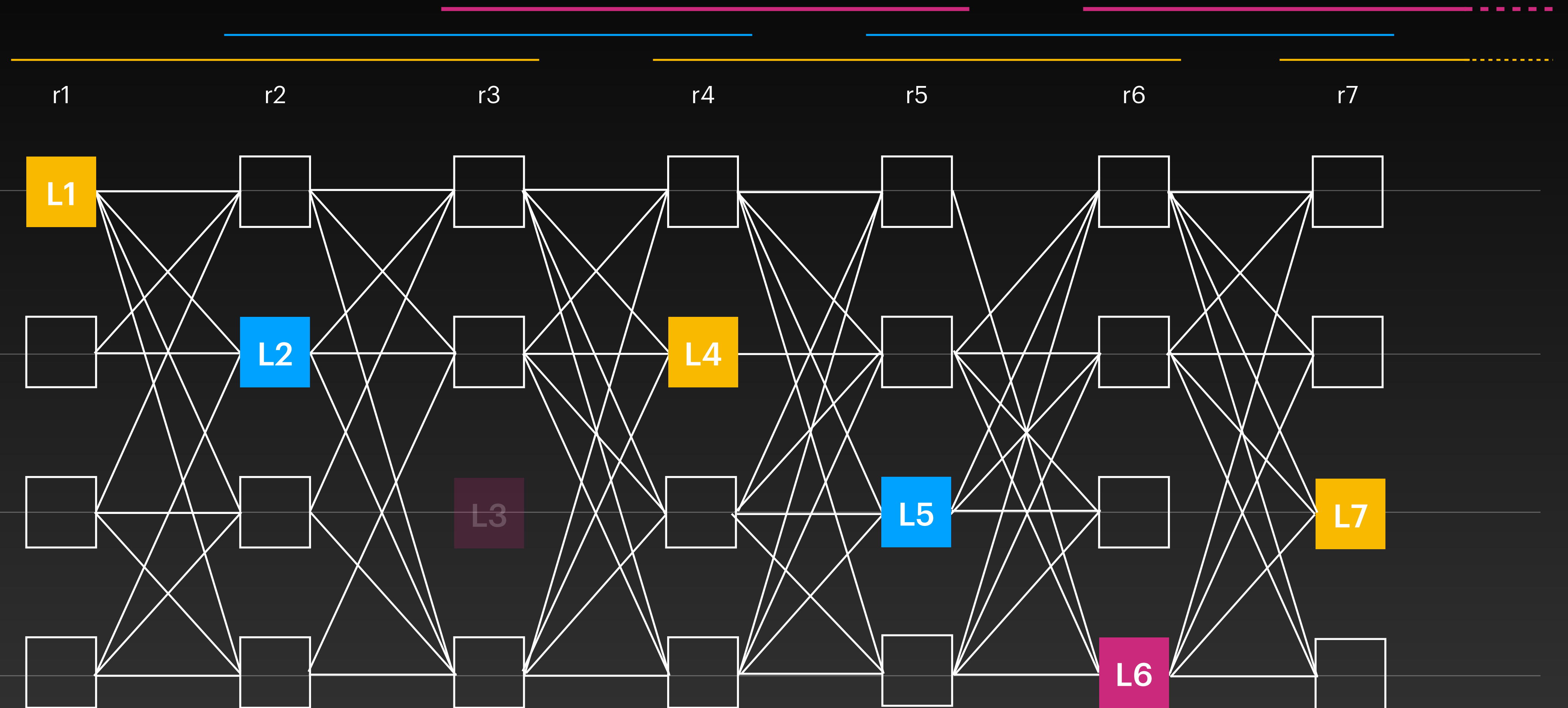
L7

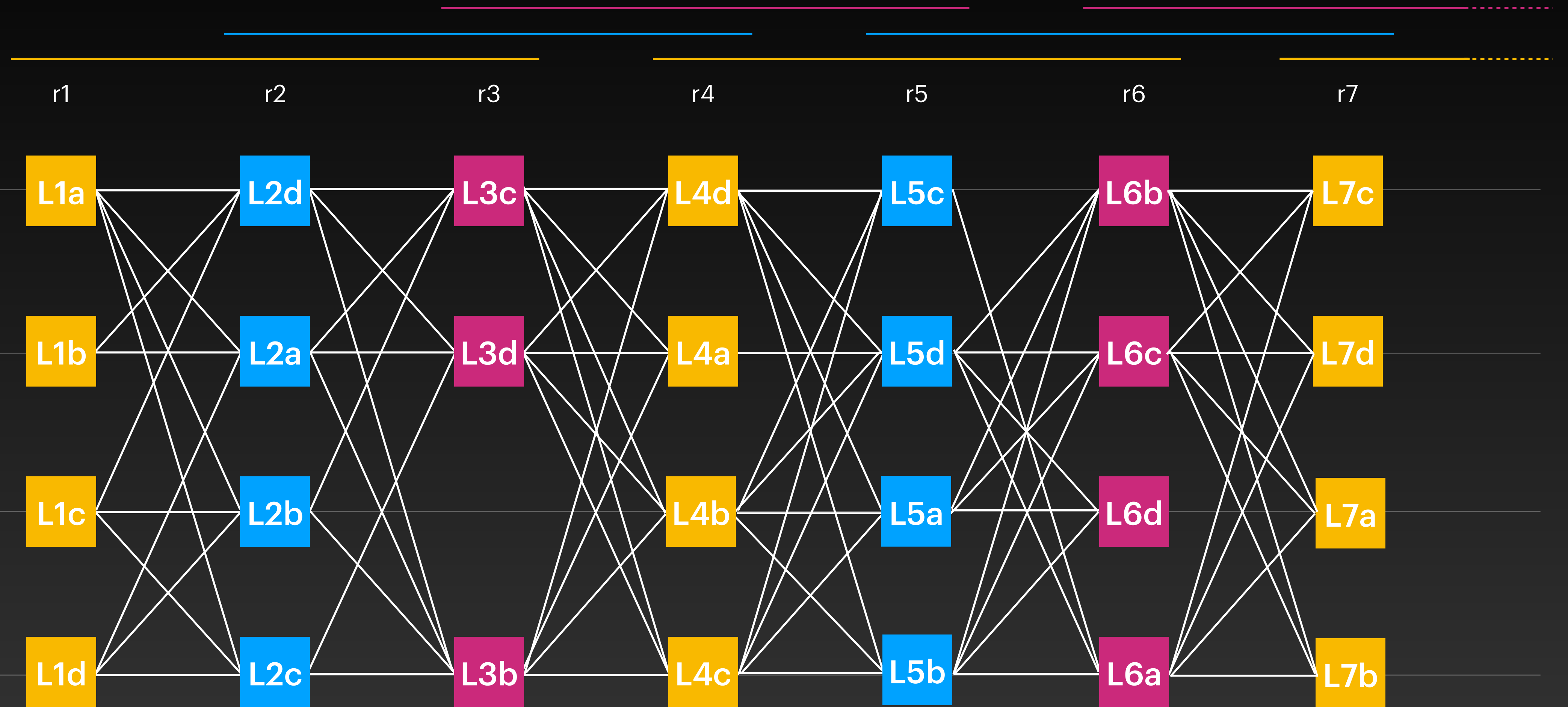


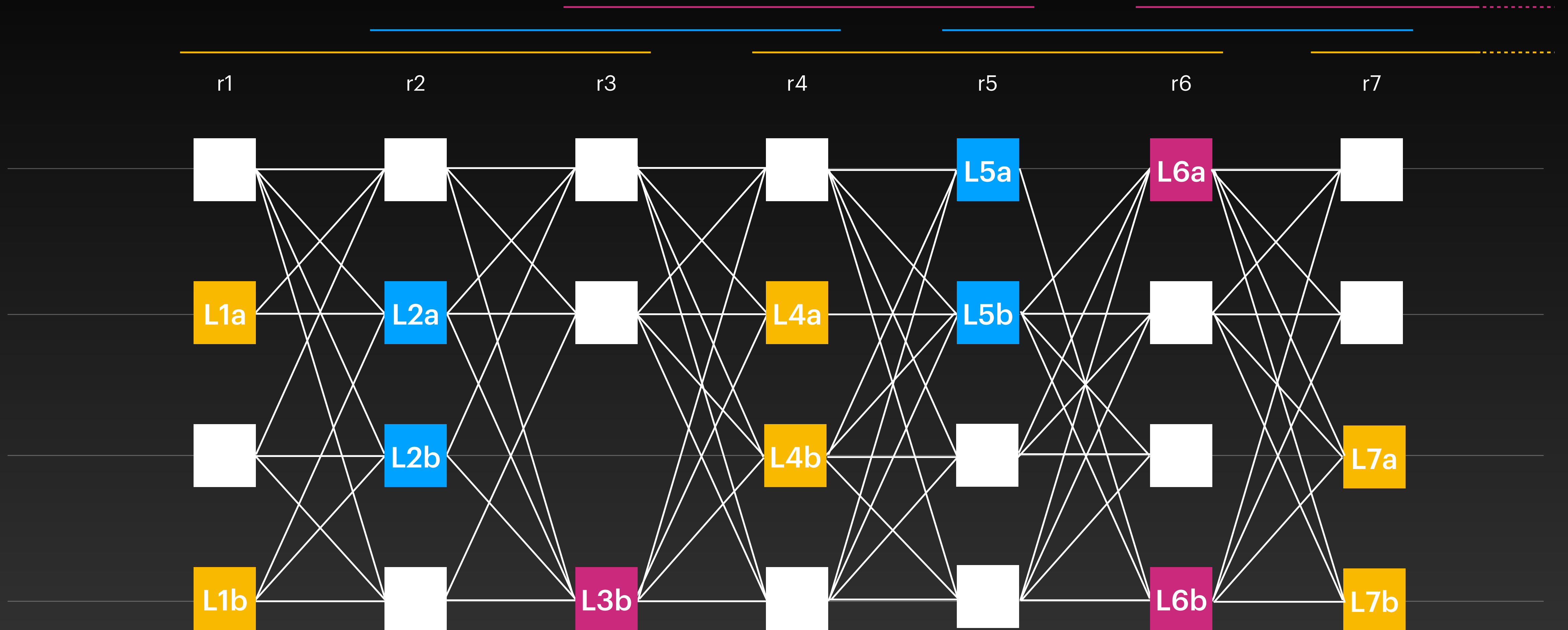


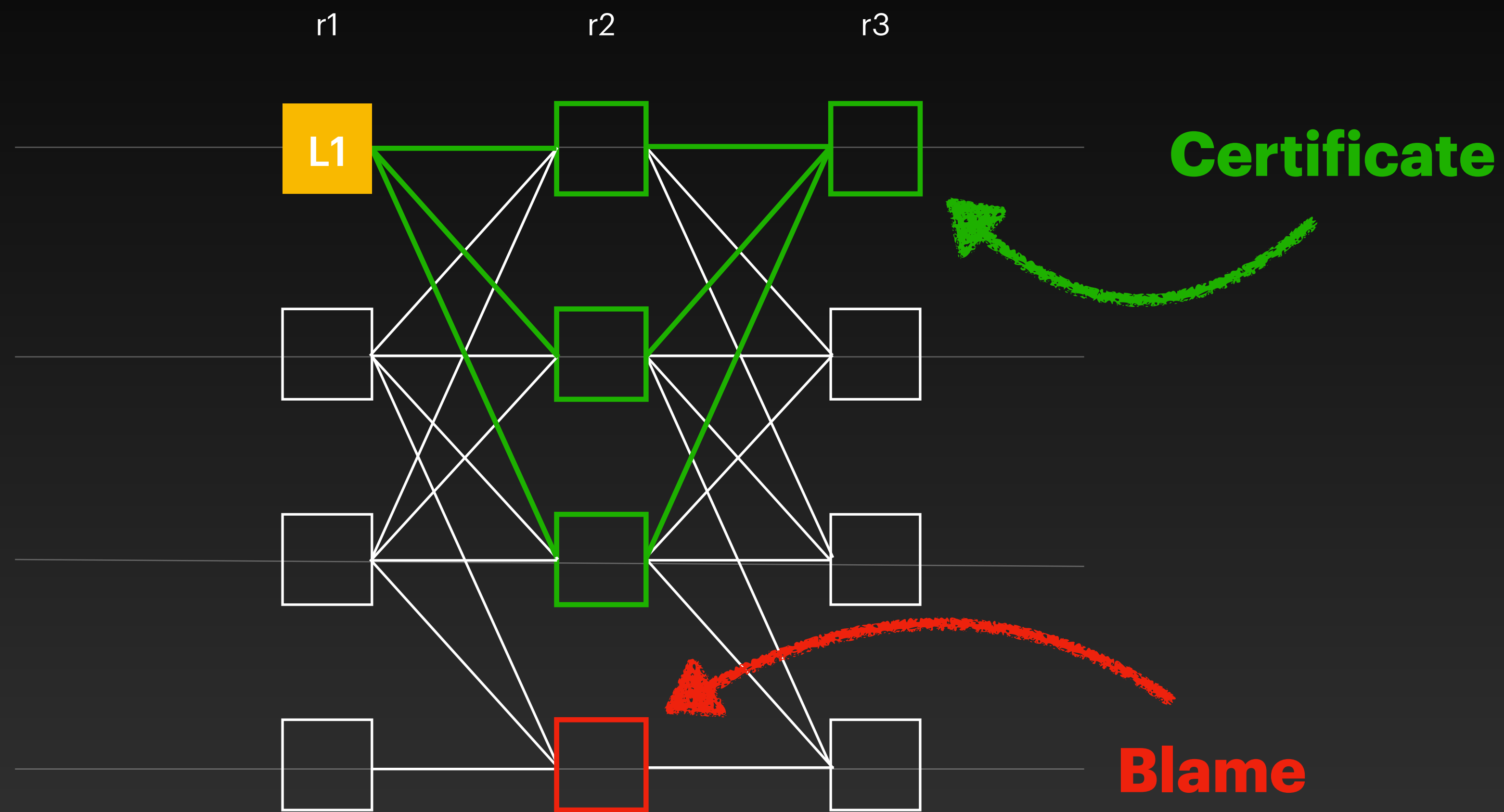












Direct Decision Rule

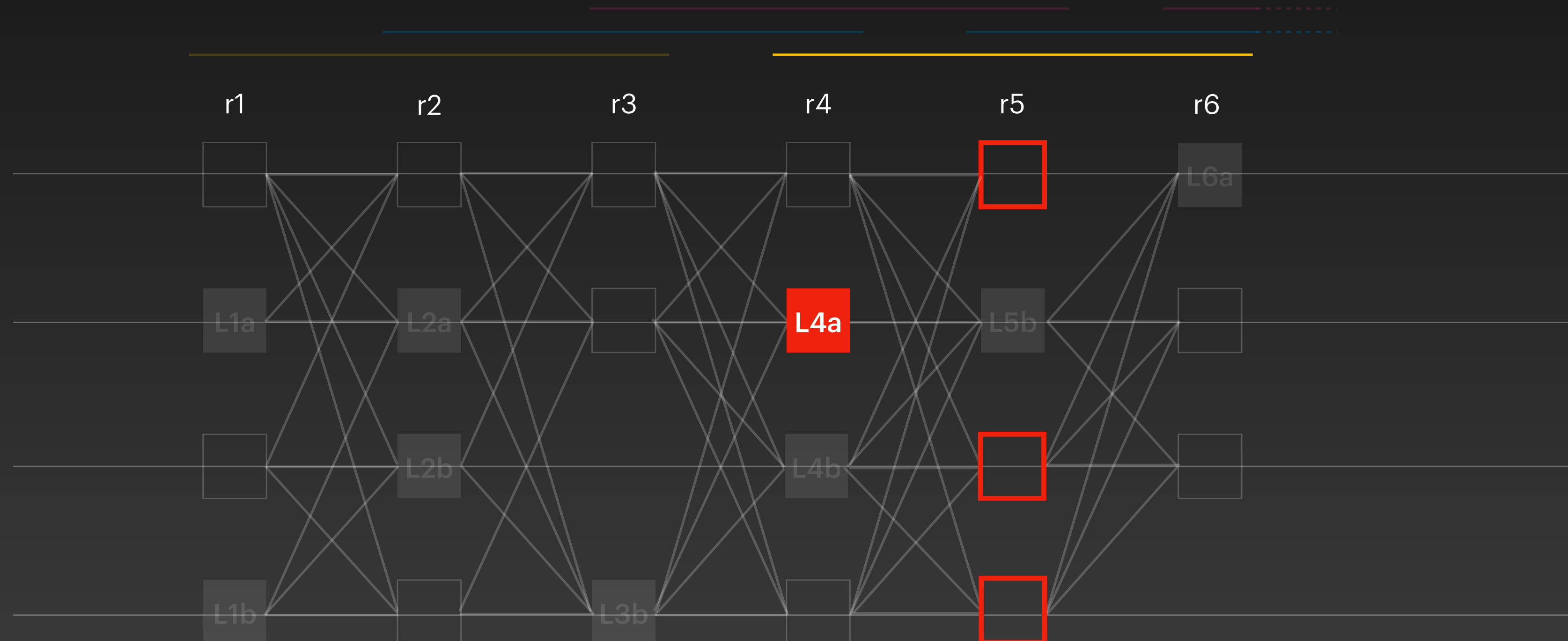
On each leader starting from highest round:

- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise

Direct Decision Rule

On each leader starting from highest round:

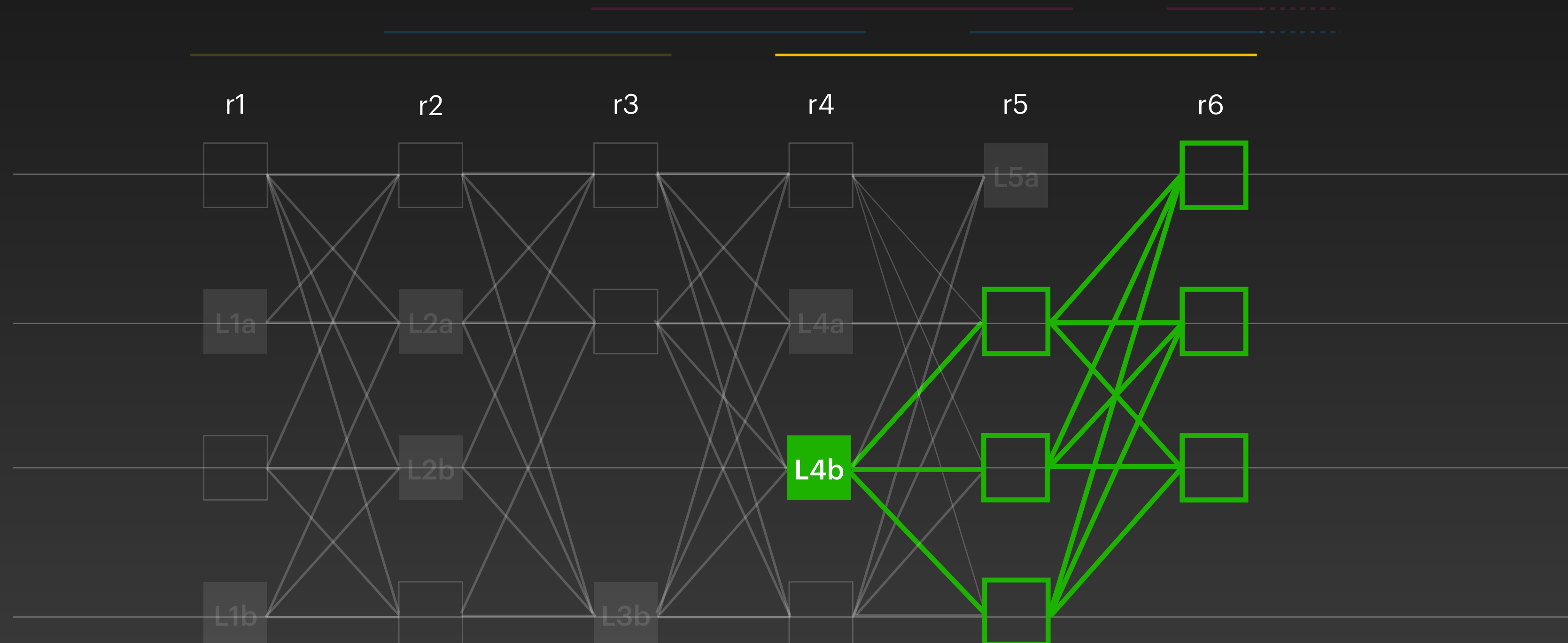
- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise



Direct Decision Rule

On each leader starting from highest round:

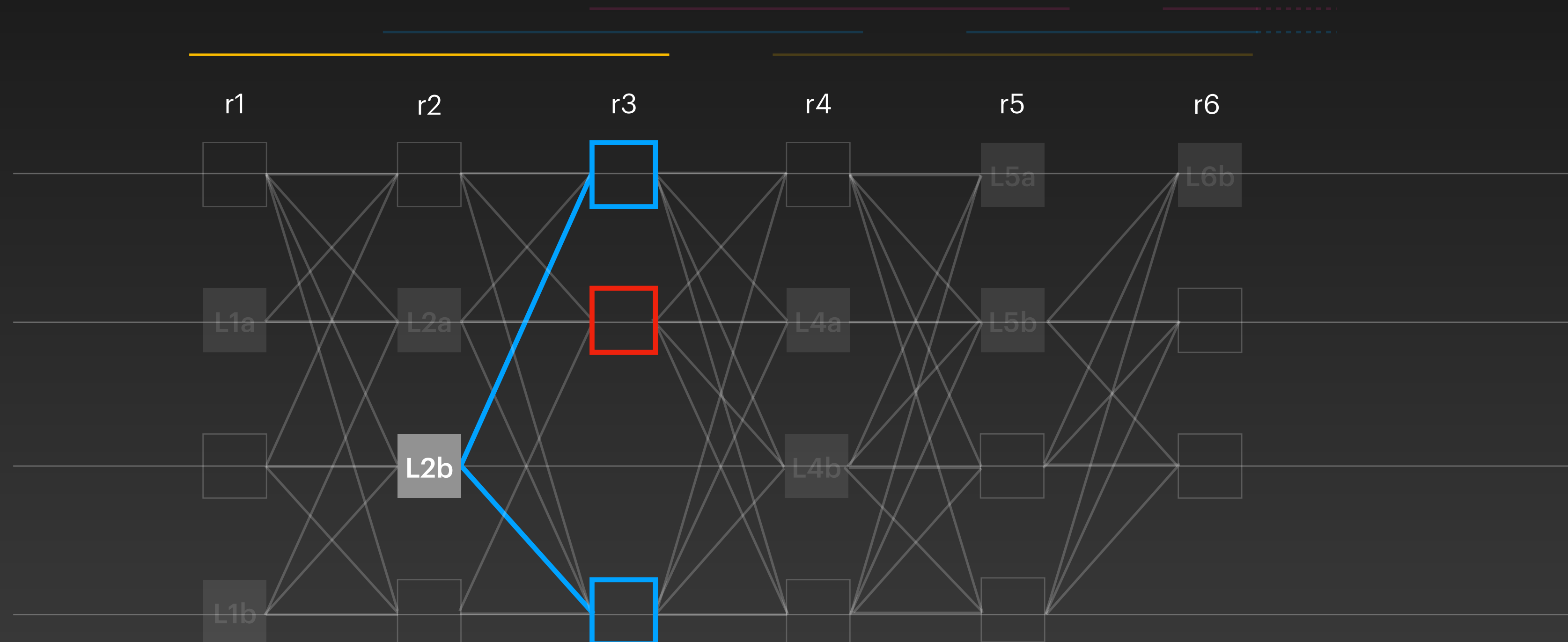
- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise



Direct Decision Rule

On each leader starting from highest round:

- **Skip** if $2f+1$ blames
- **Commit** if $2f+1$ certificates
- **Undecided** otherwise

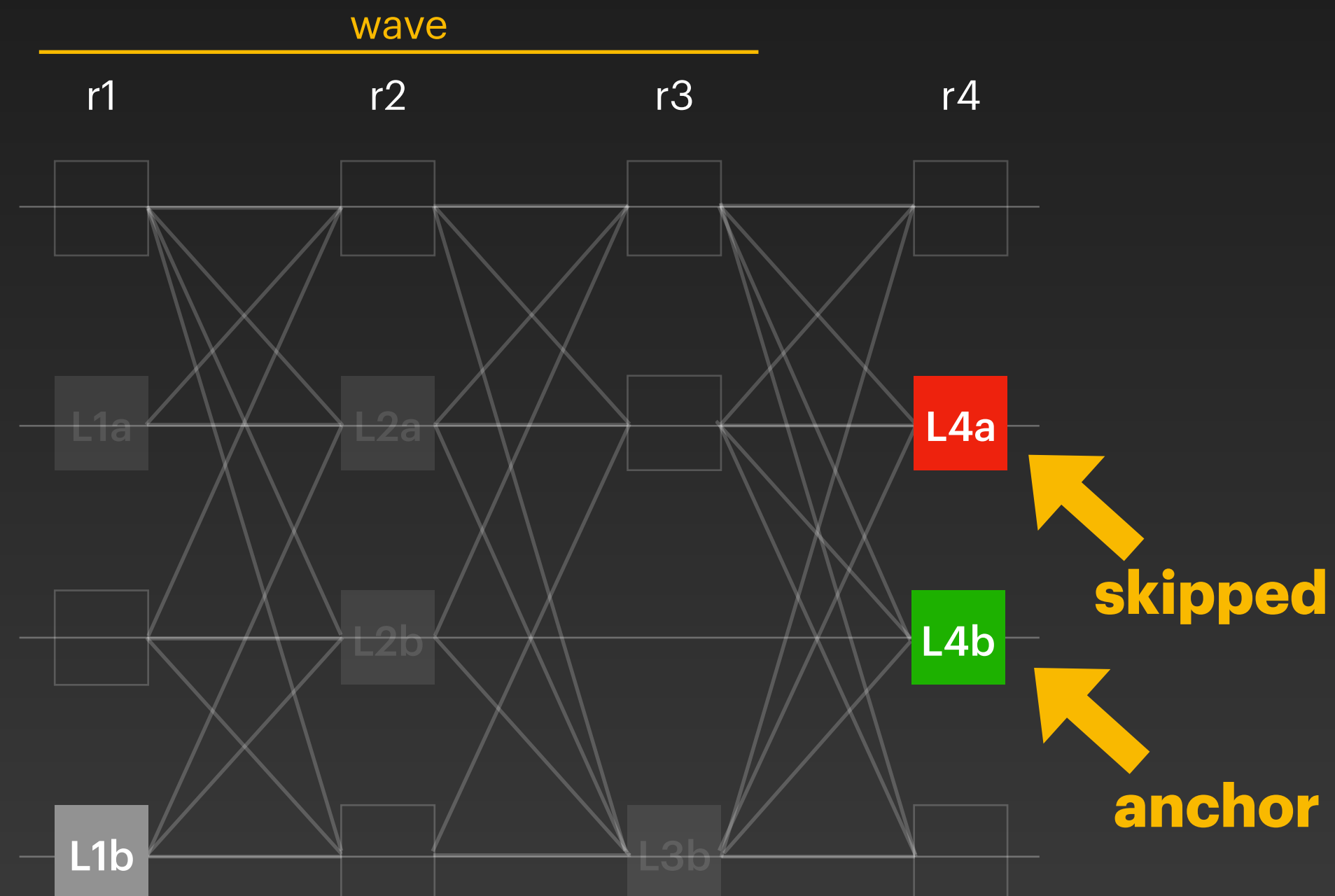


Indirect Decision Rule

Indirect Decision Rule

1. Find Anchor

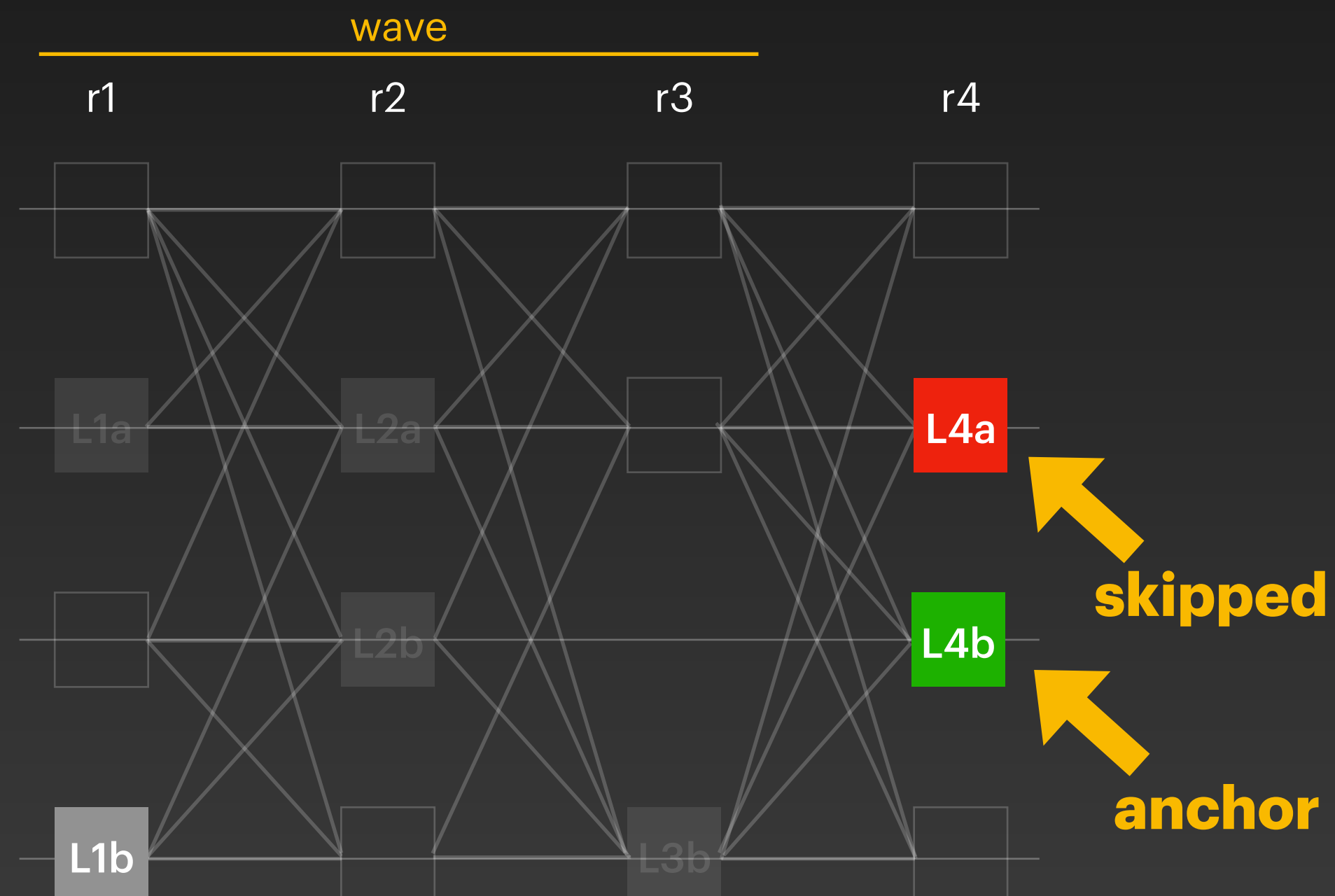
- First block with round $> r+2$ that is **Commit** or **Undecided**



Indirect Decision Rule

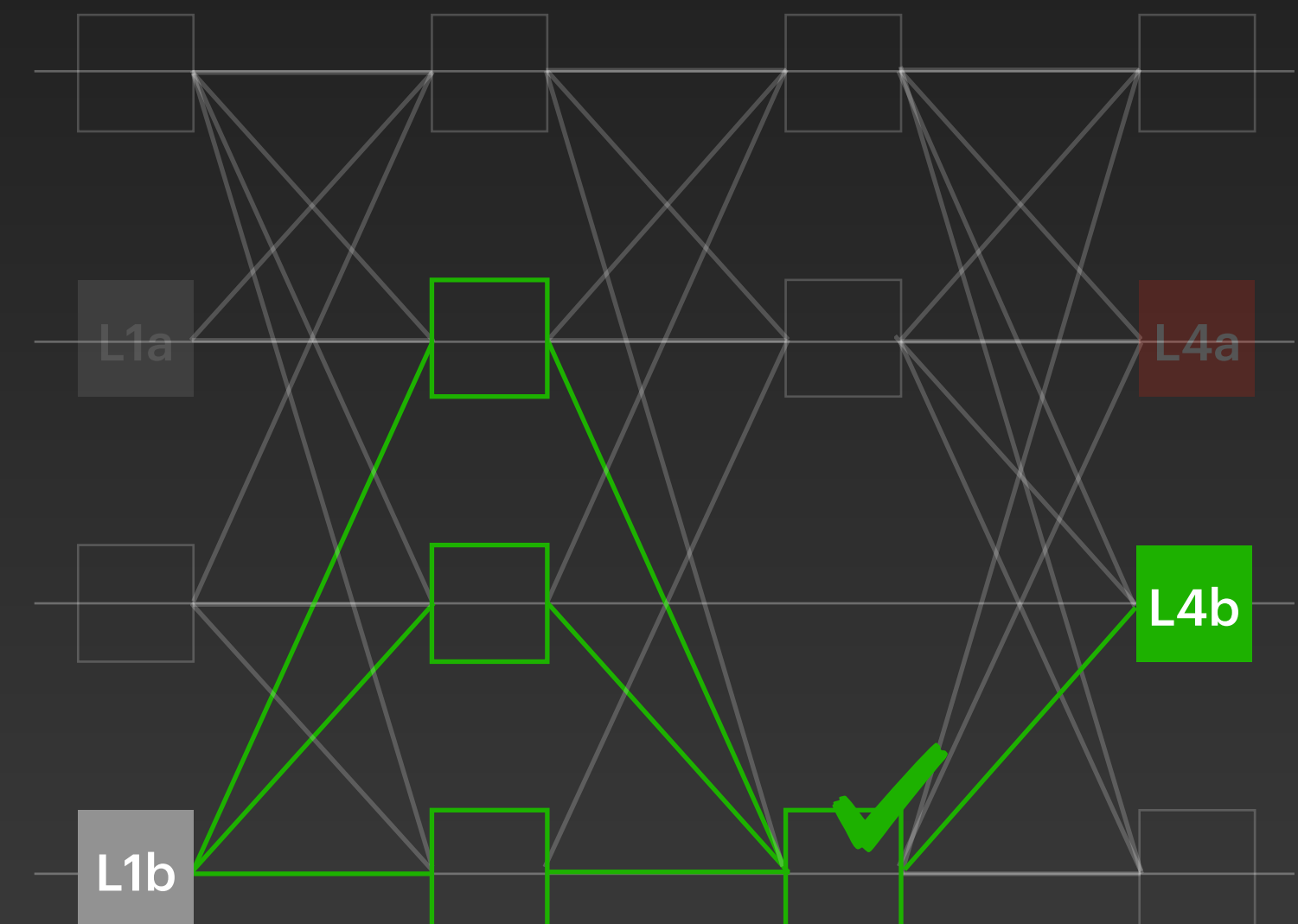
1. Find Anchor

- First block with round $> r+2$ that is **Commit** or **Undecided**

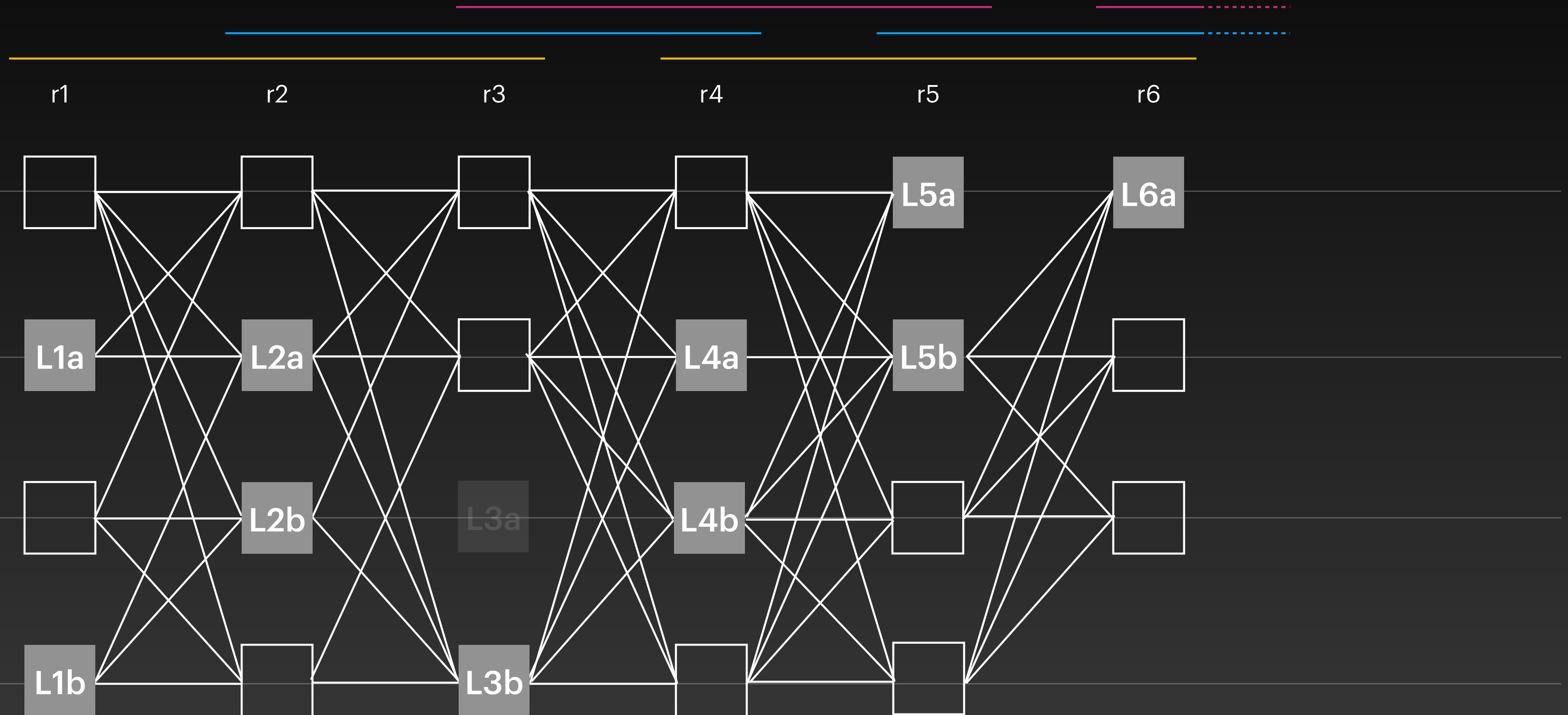


2. Certified link

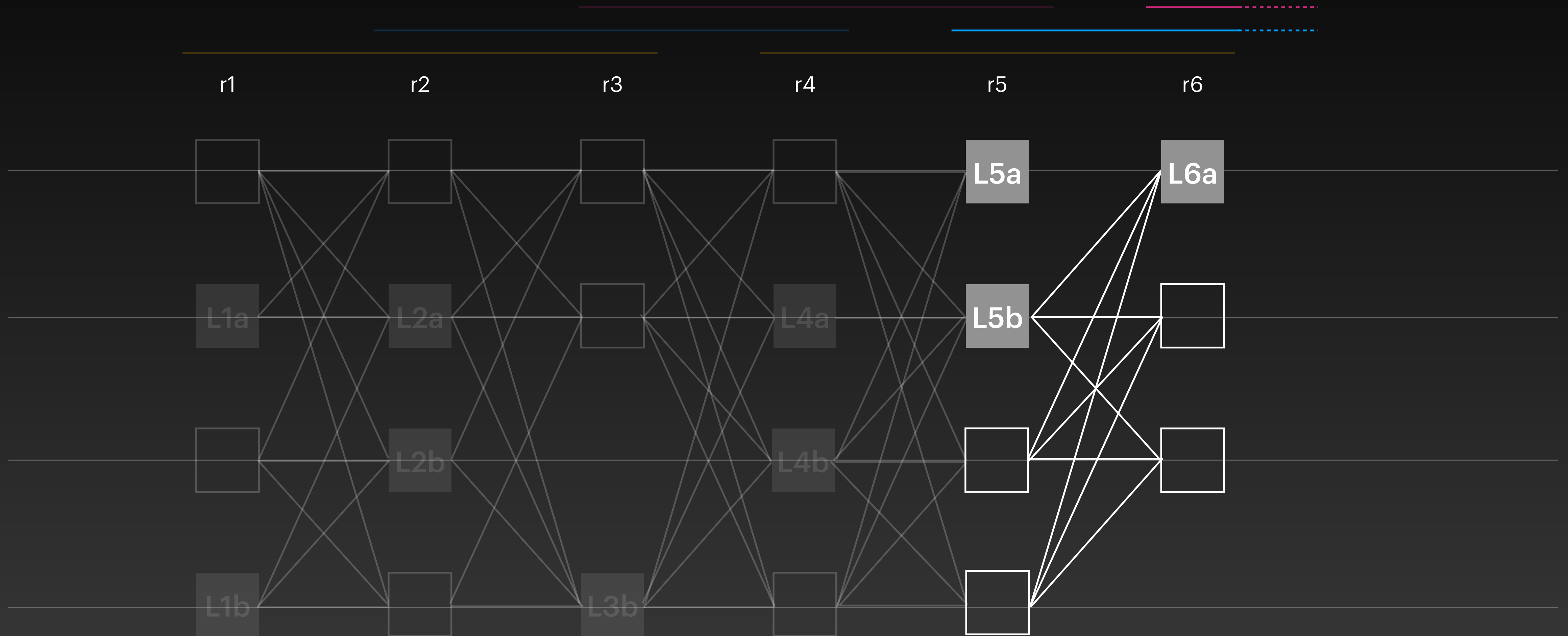
- Commit** if $B \leftrightarrow \text{certified link} \leftrightarrow A$ otherwise **Skip**



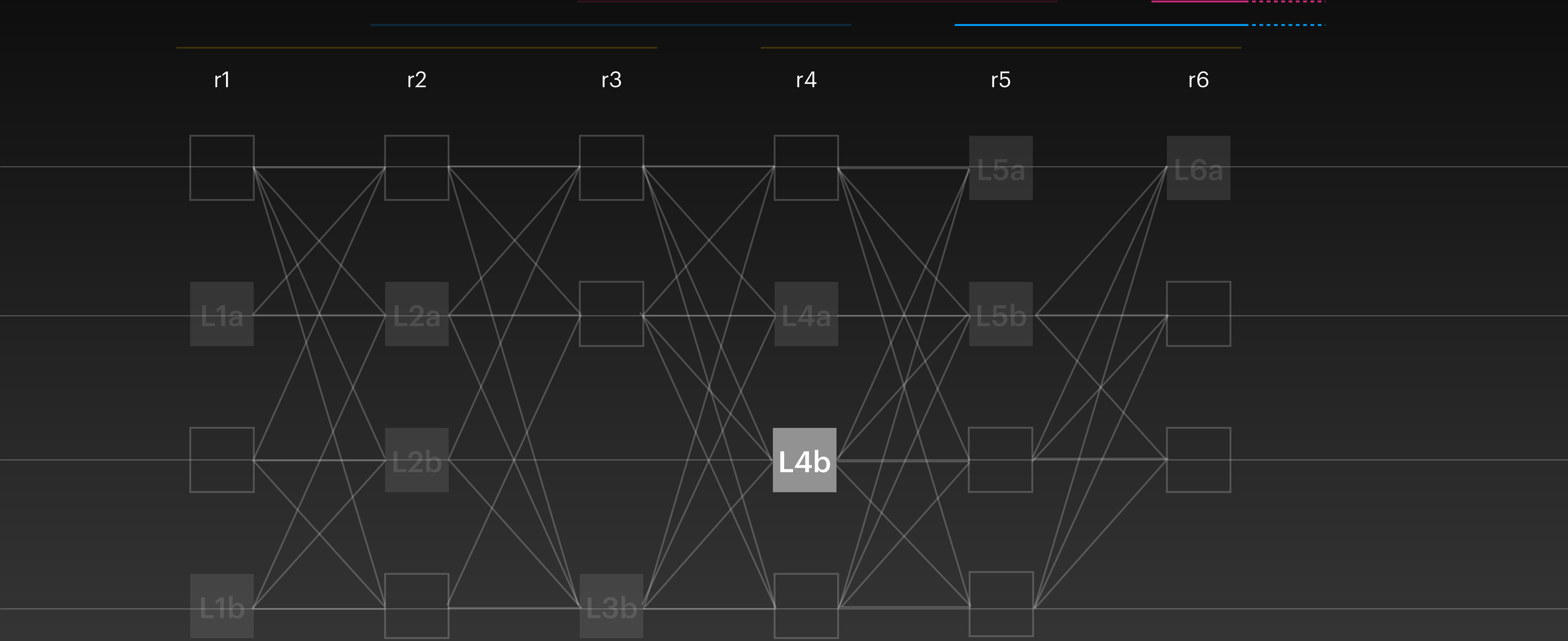
All Start at Undecided



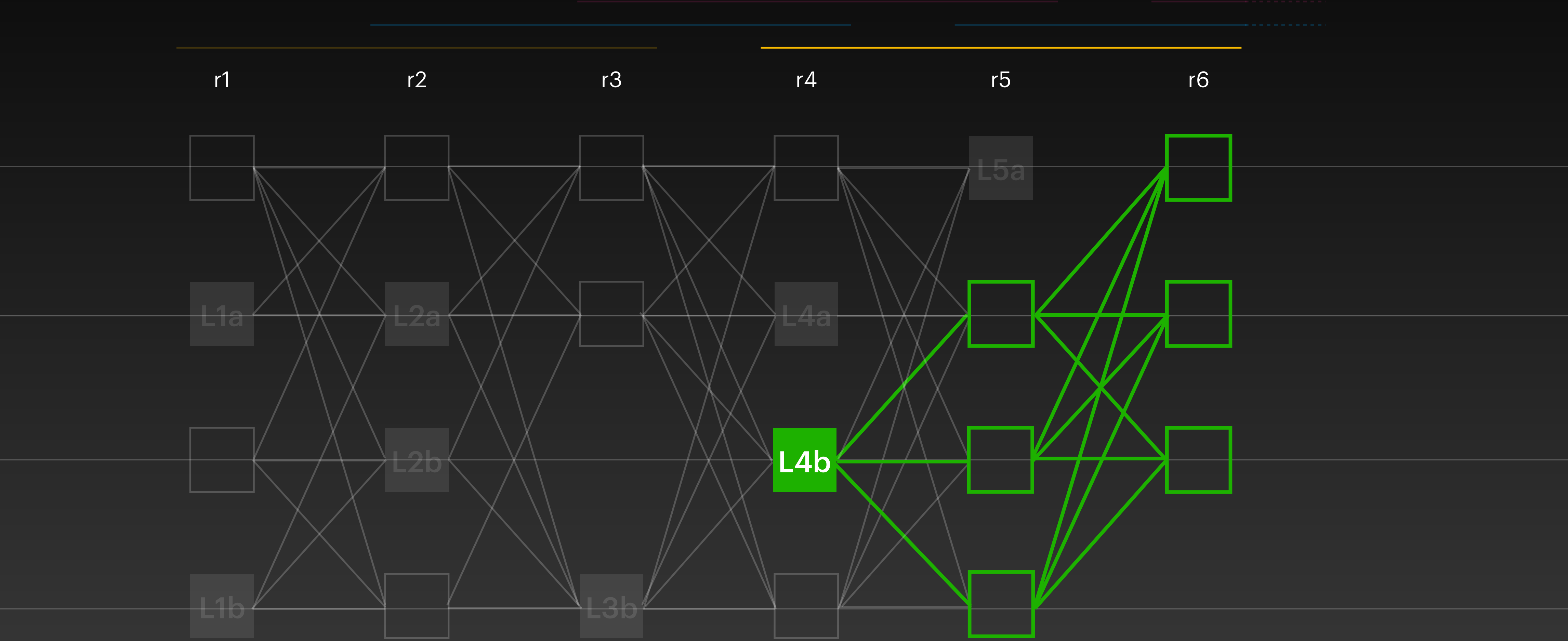
Ignore Incomplete Waves



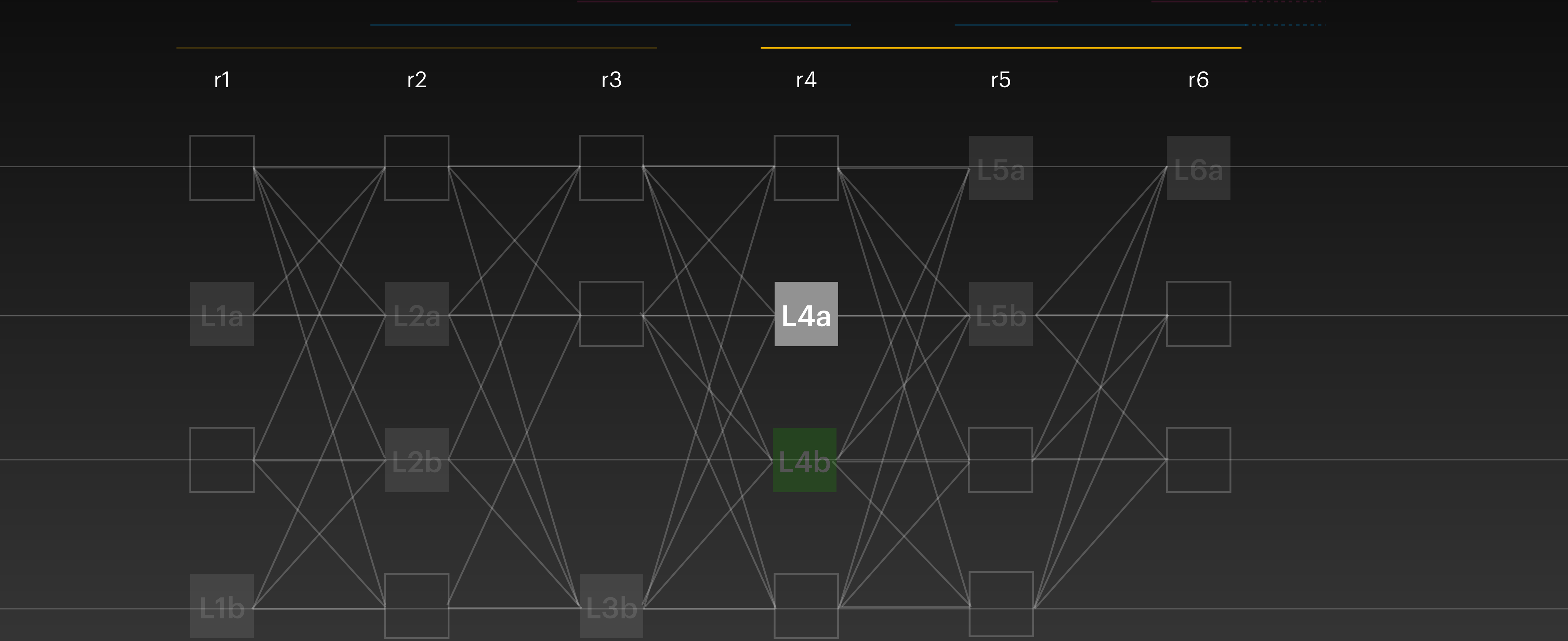
Apply Direct Rule



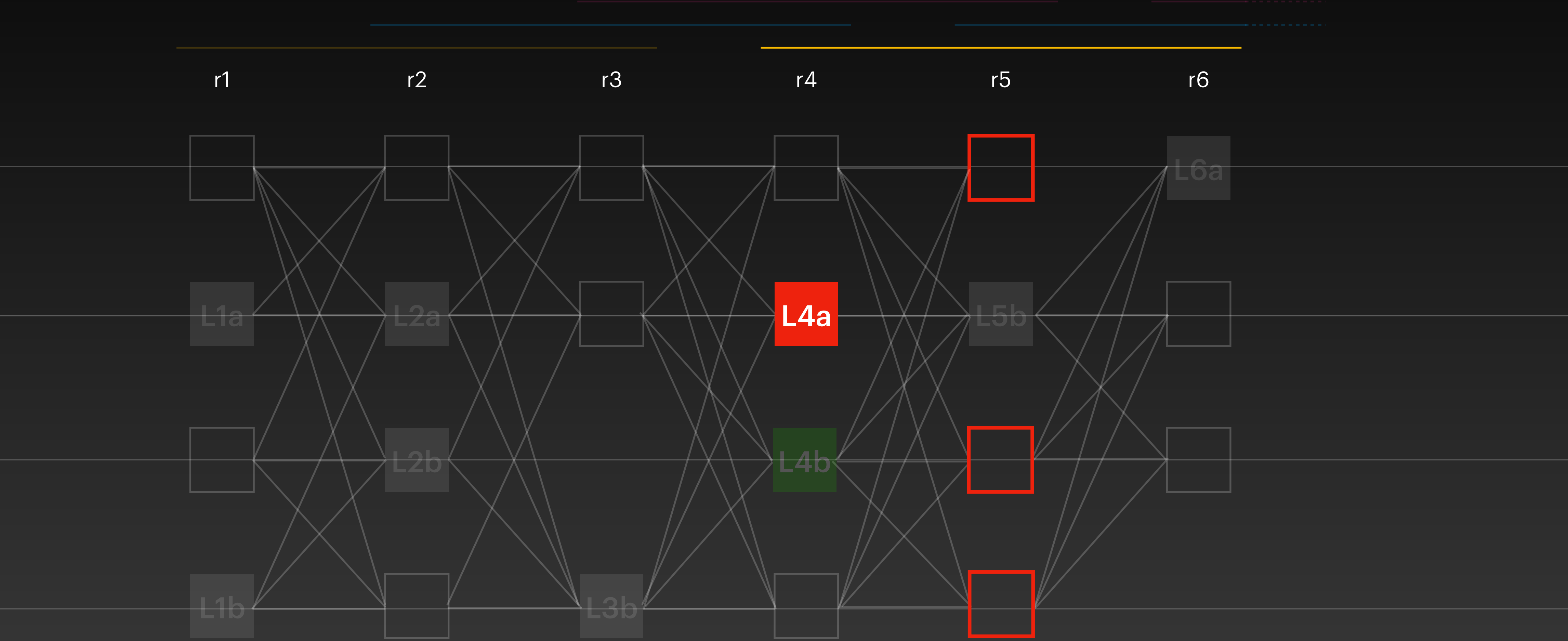
Apply Direct Rule



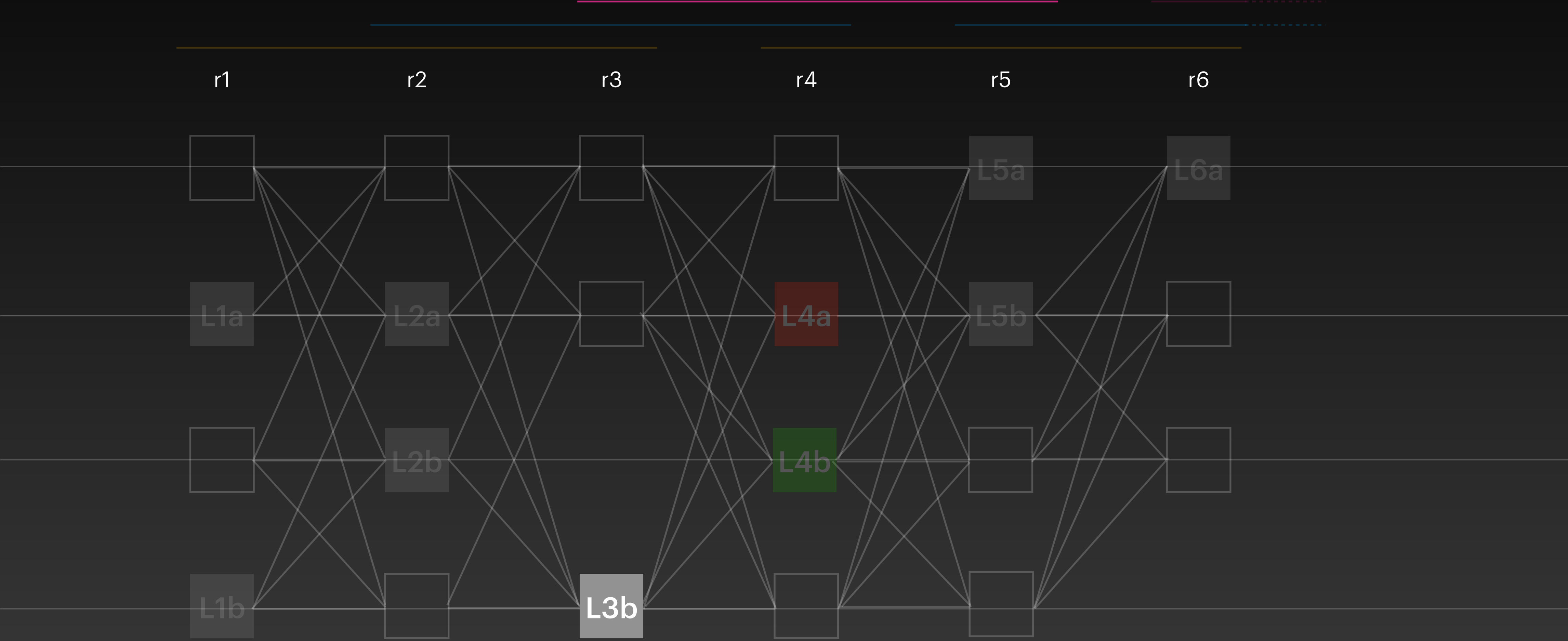
Apply Direct Rule



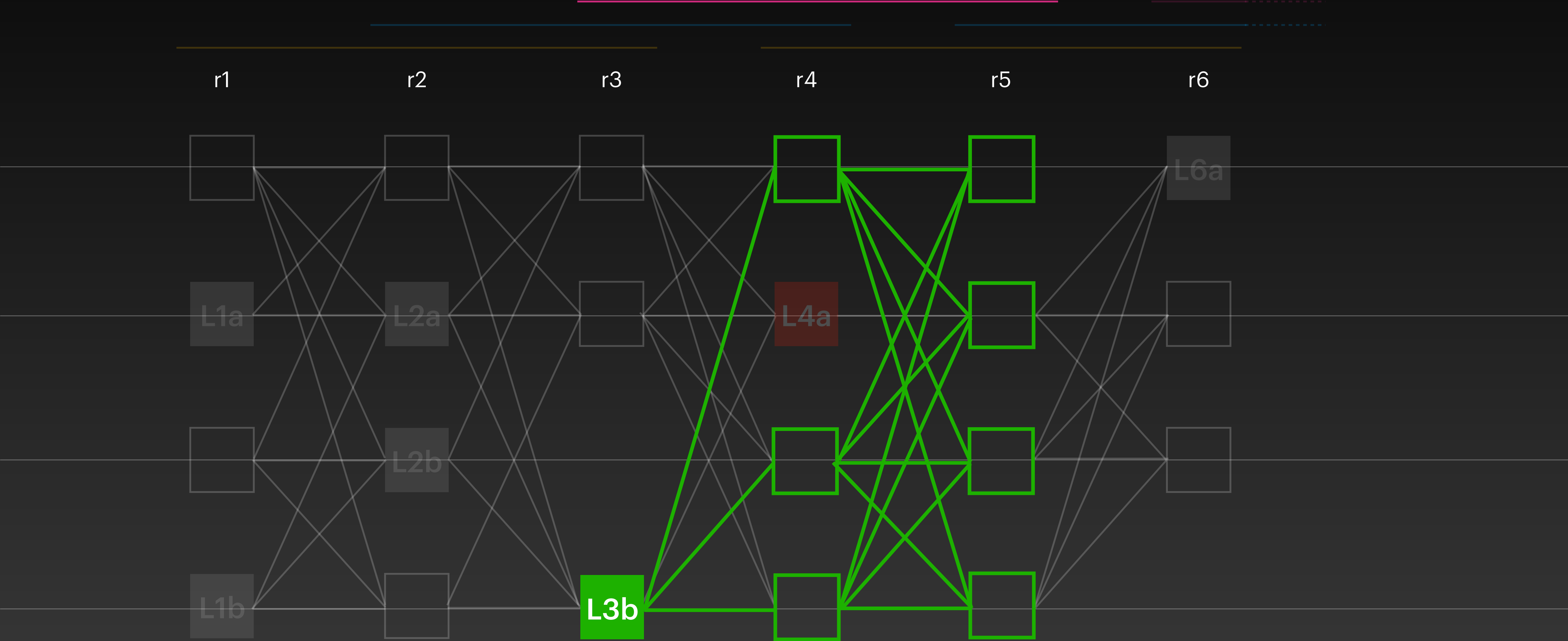
Apply Direct Rule



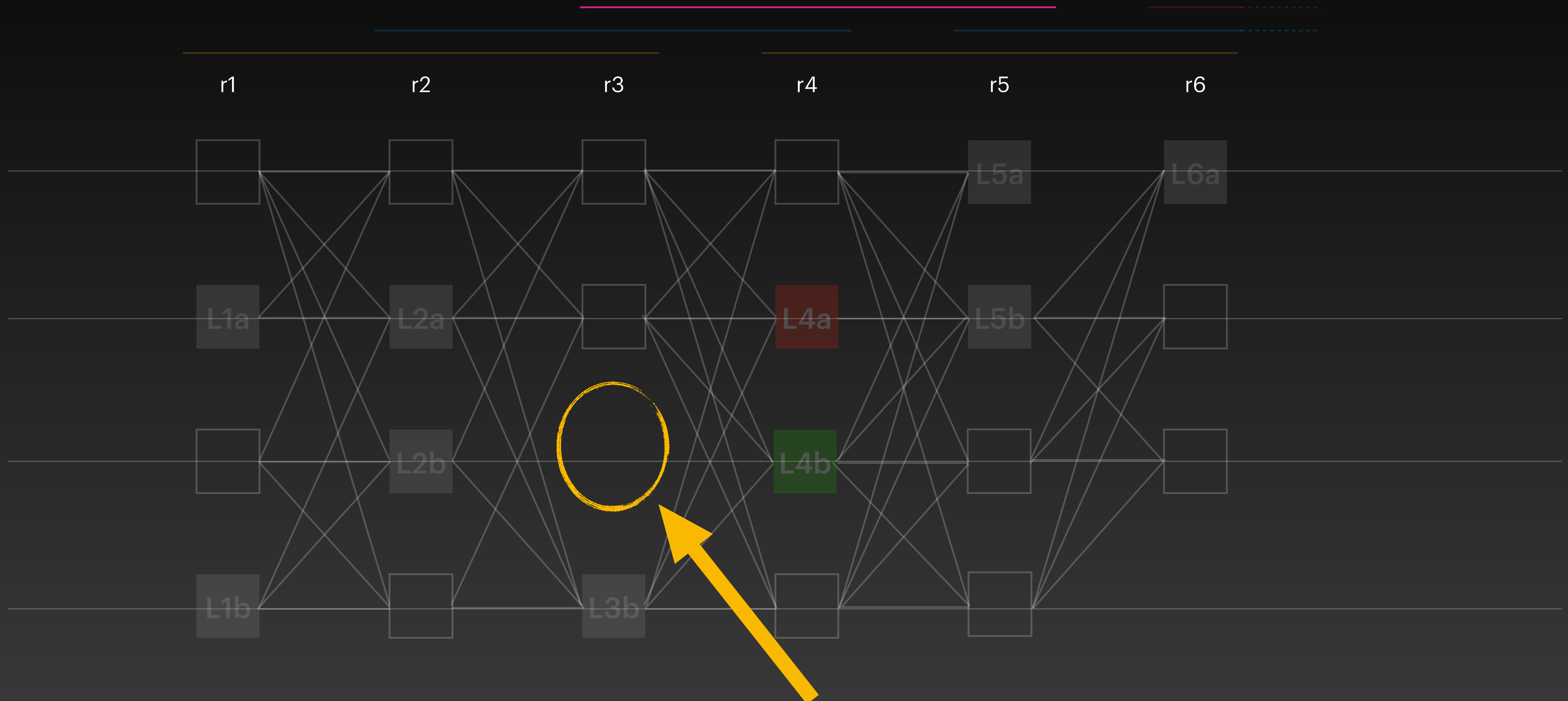
Apply Direct Rule



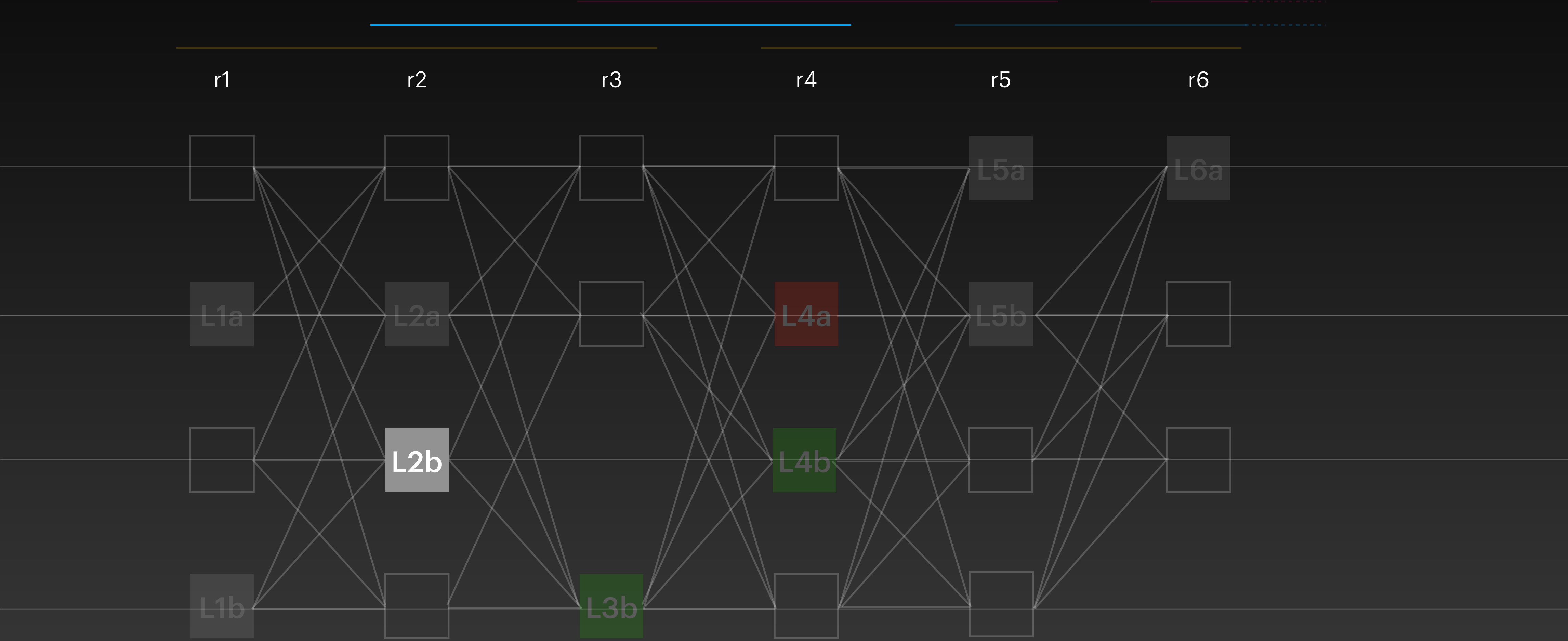
Apply Direct Rule



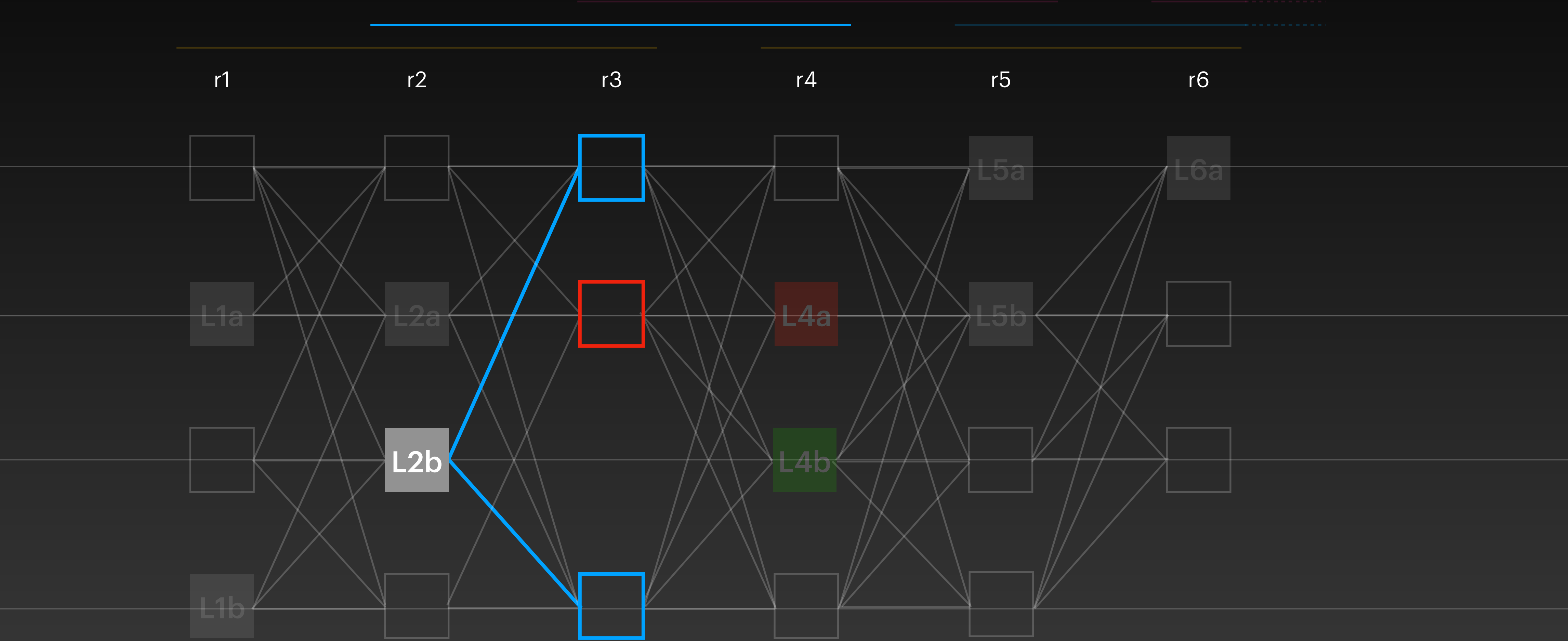
Ignore Missing Leader



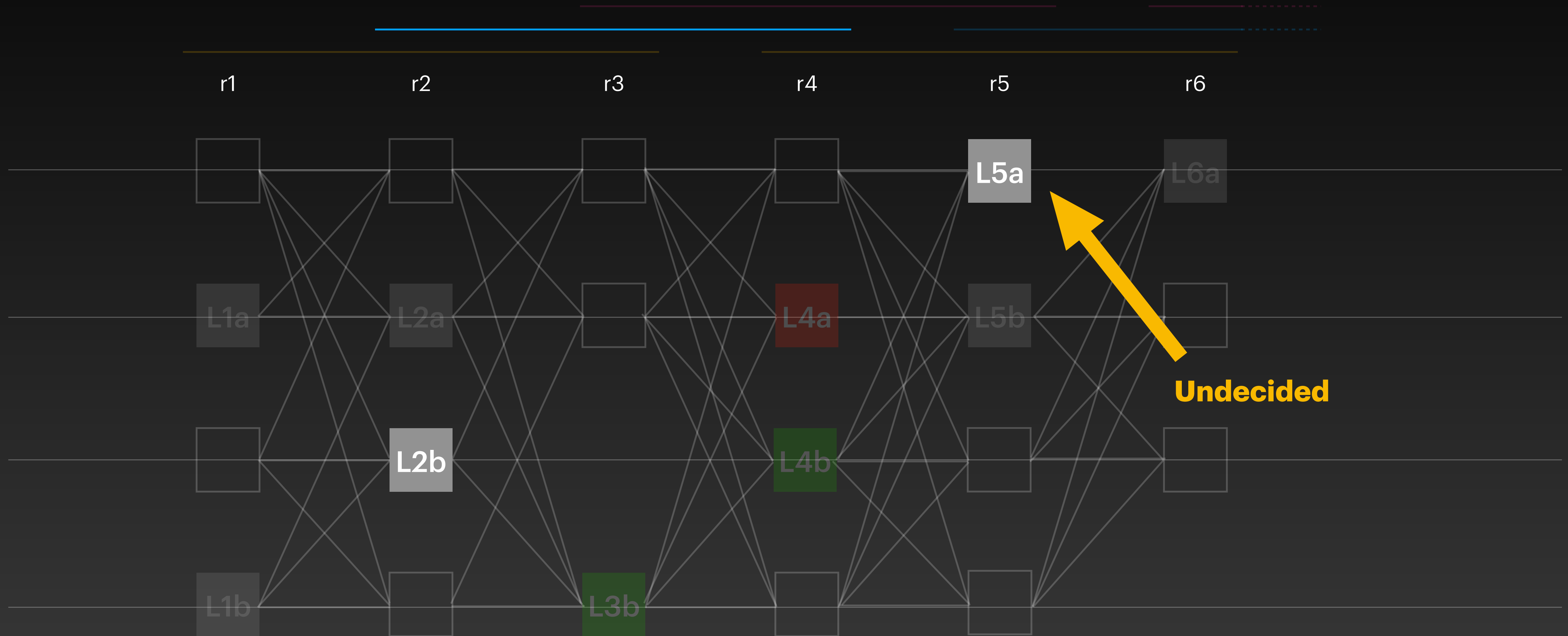
Apply Direct Rule



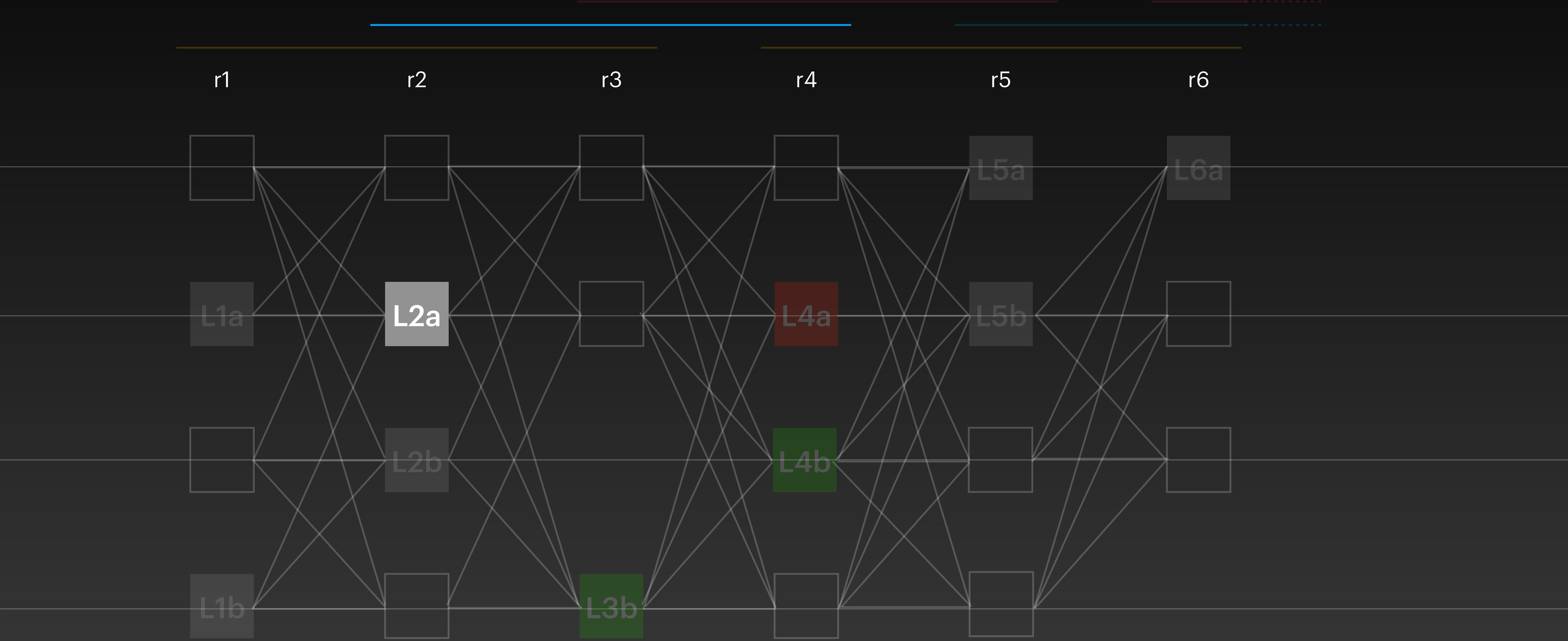
Apply Direct Rule



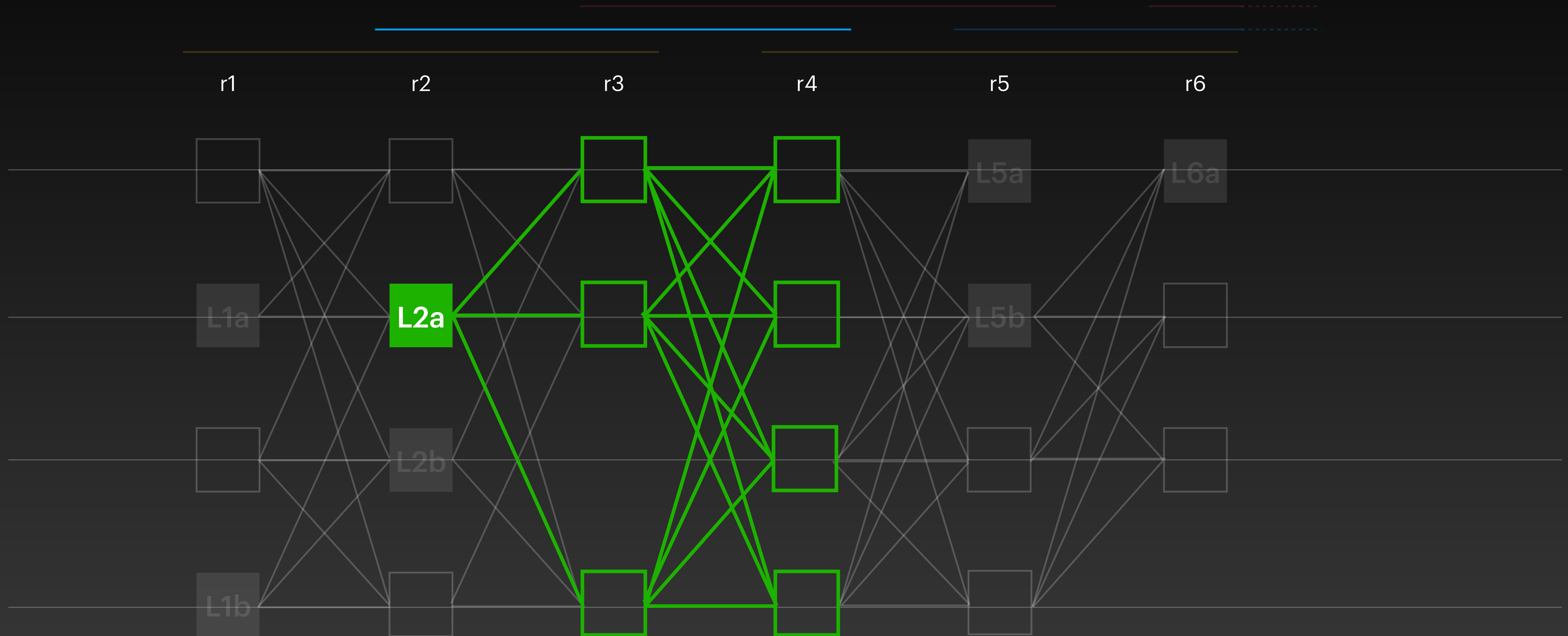
Apply Indirect Rule



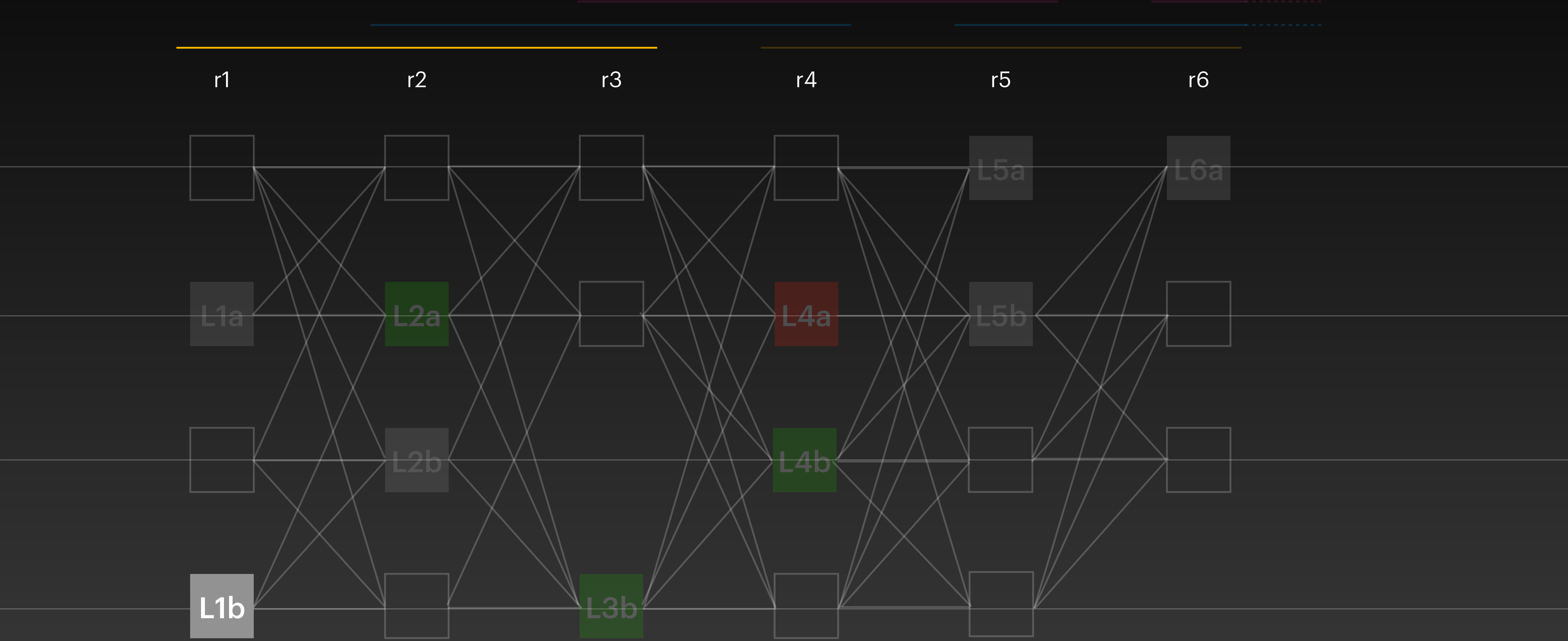
Apply Direct Rule



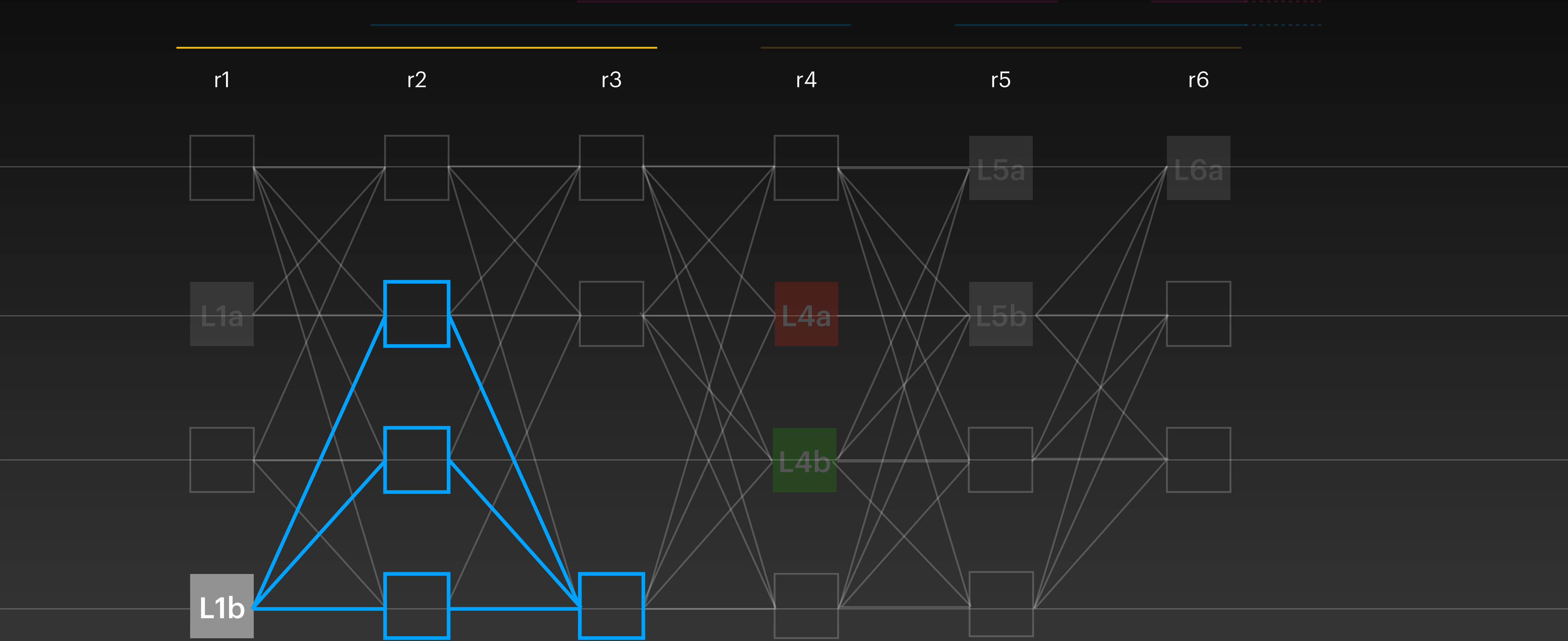
Apply Direct Rule



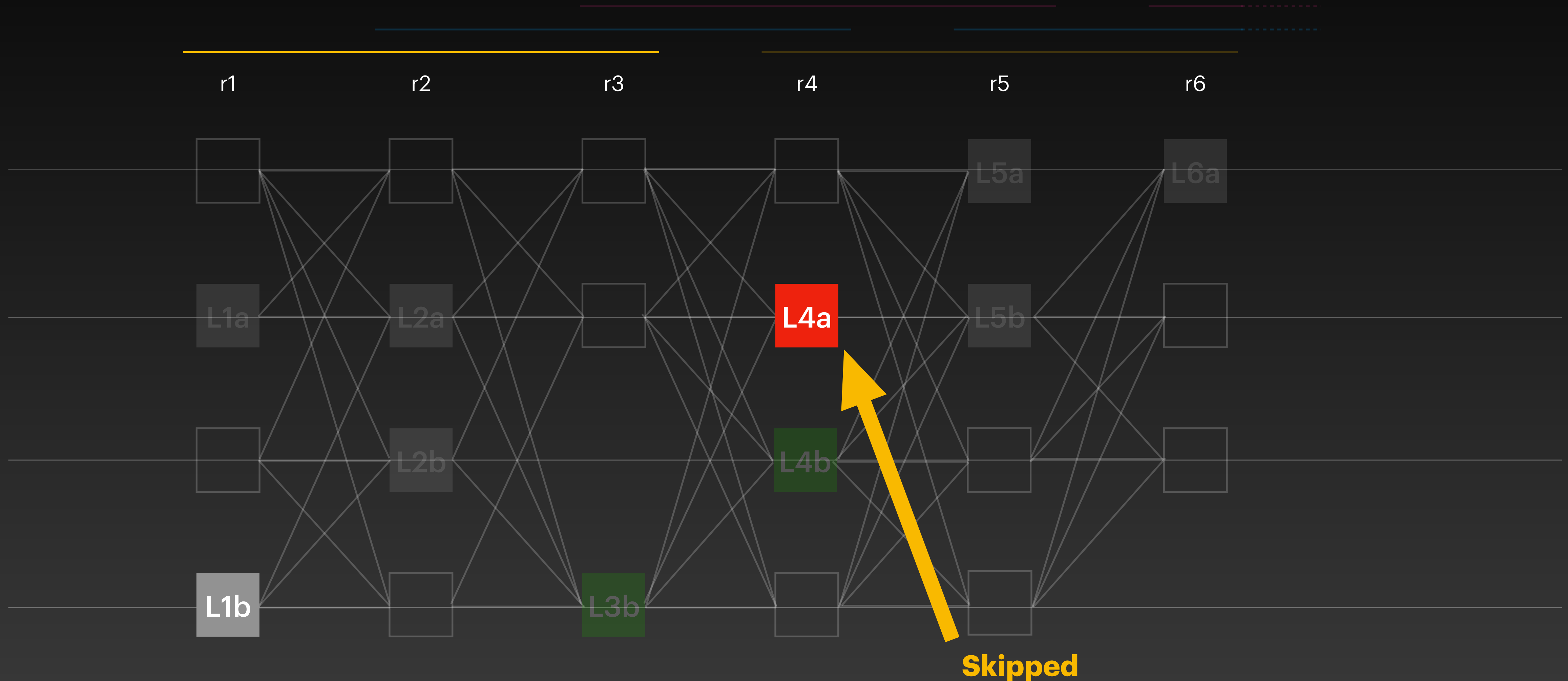
Apply Direct Rule



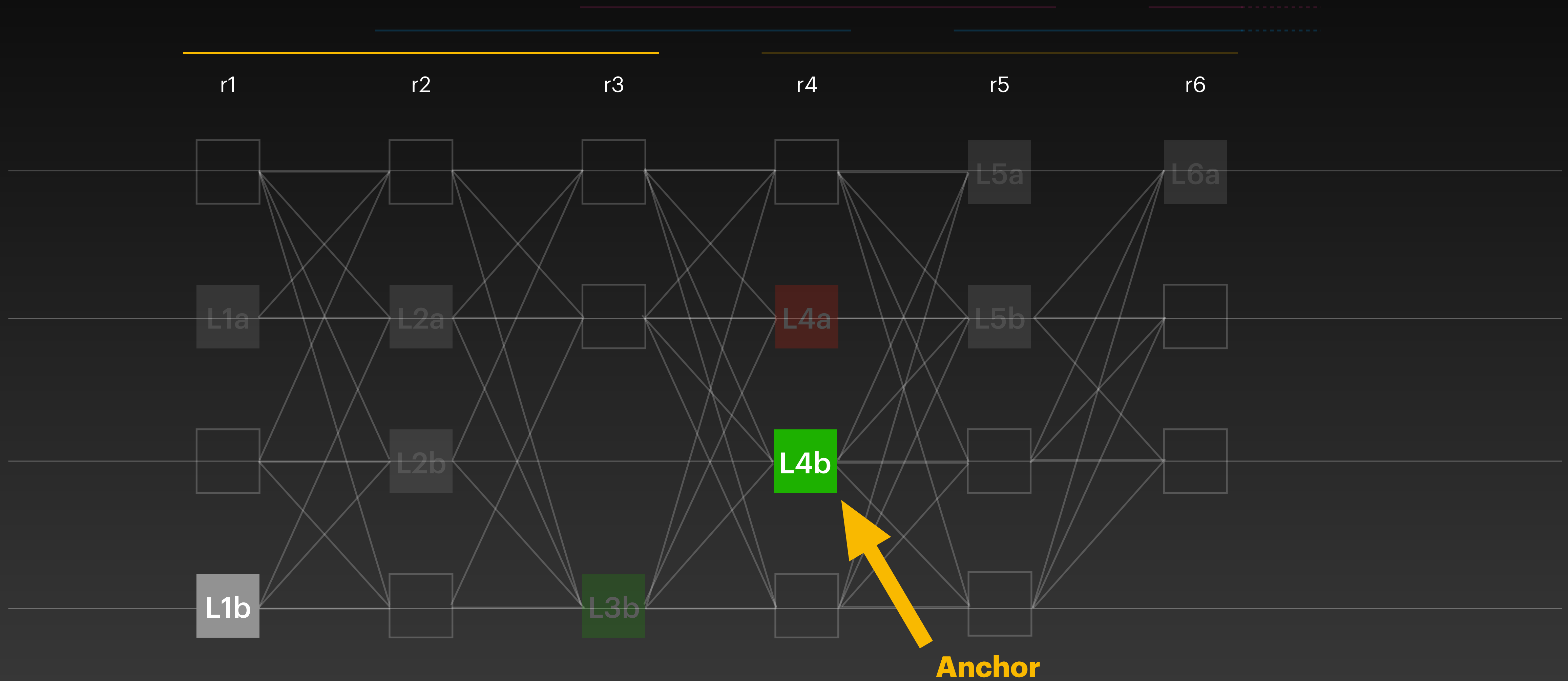
Apply Direct Rule



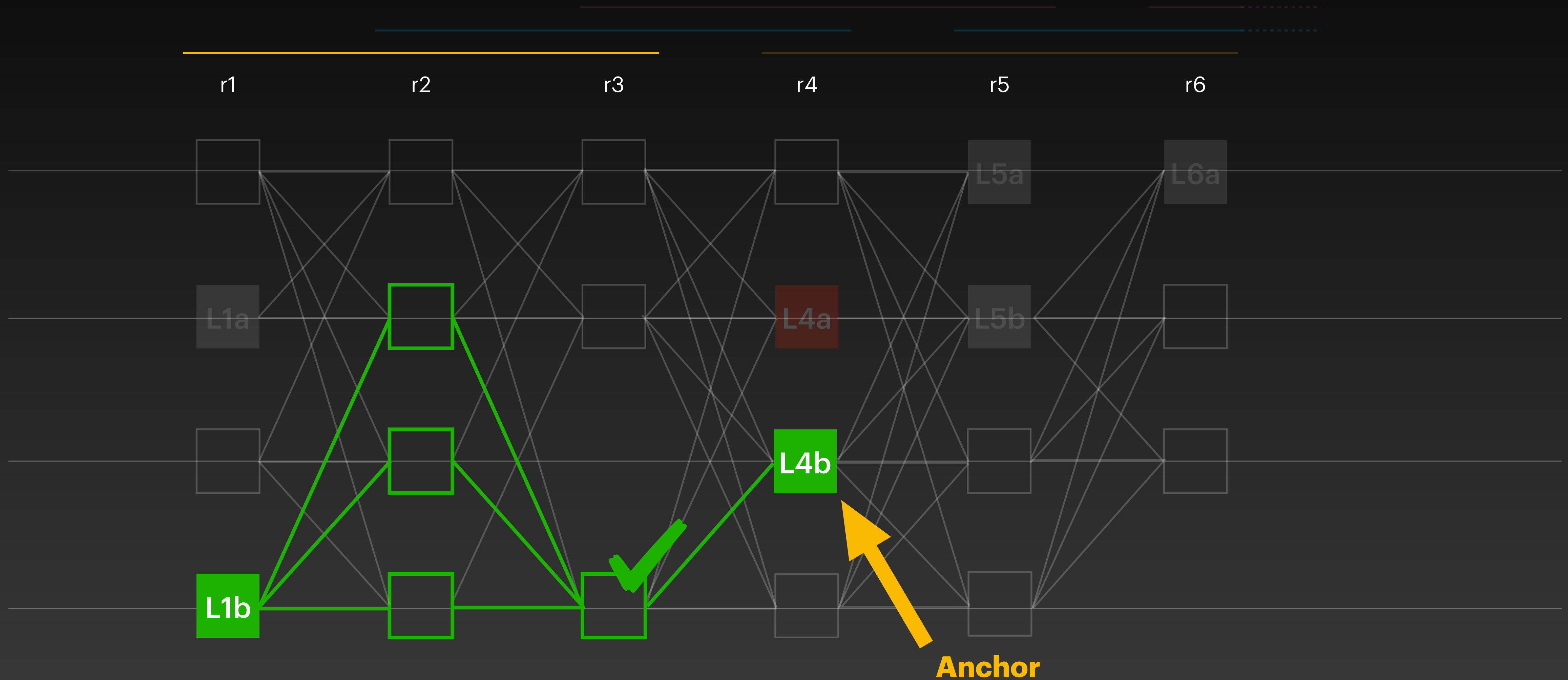
Apply Indirect Rule



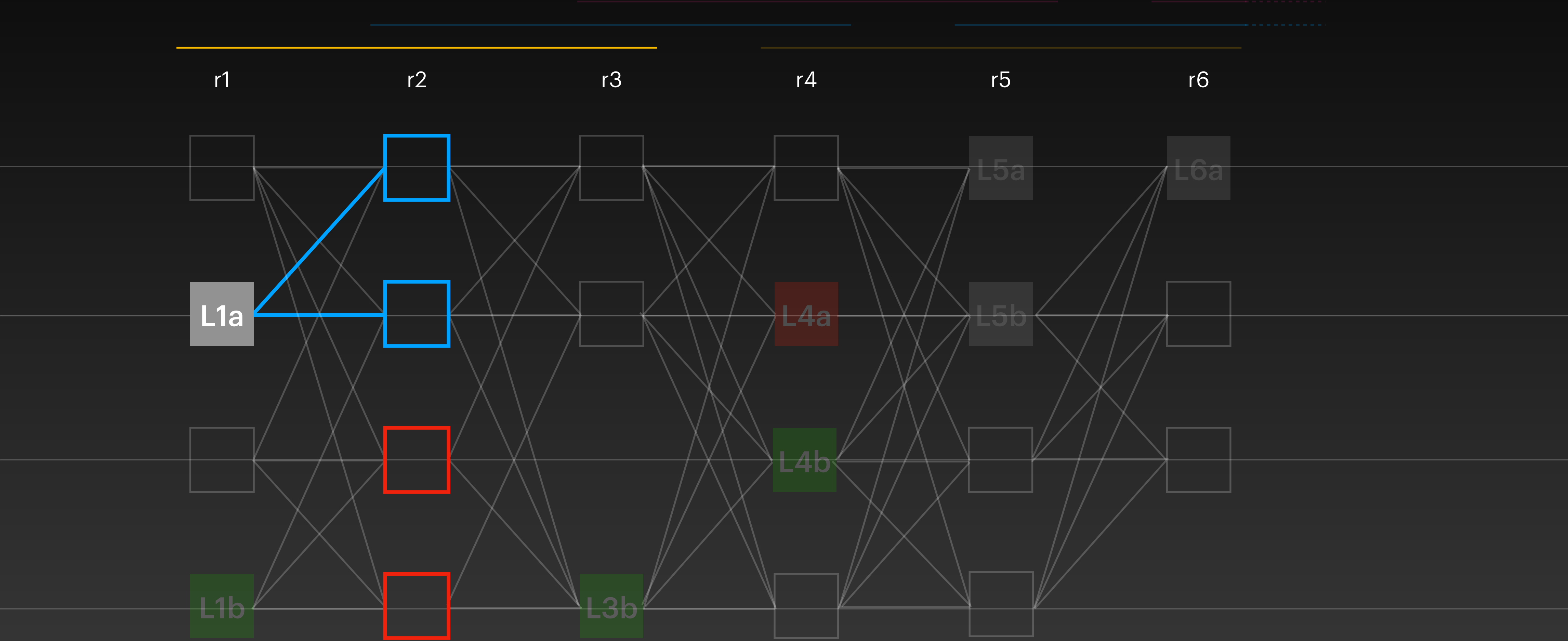
Apply Indirect Rule



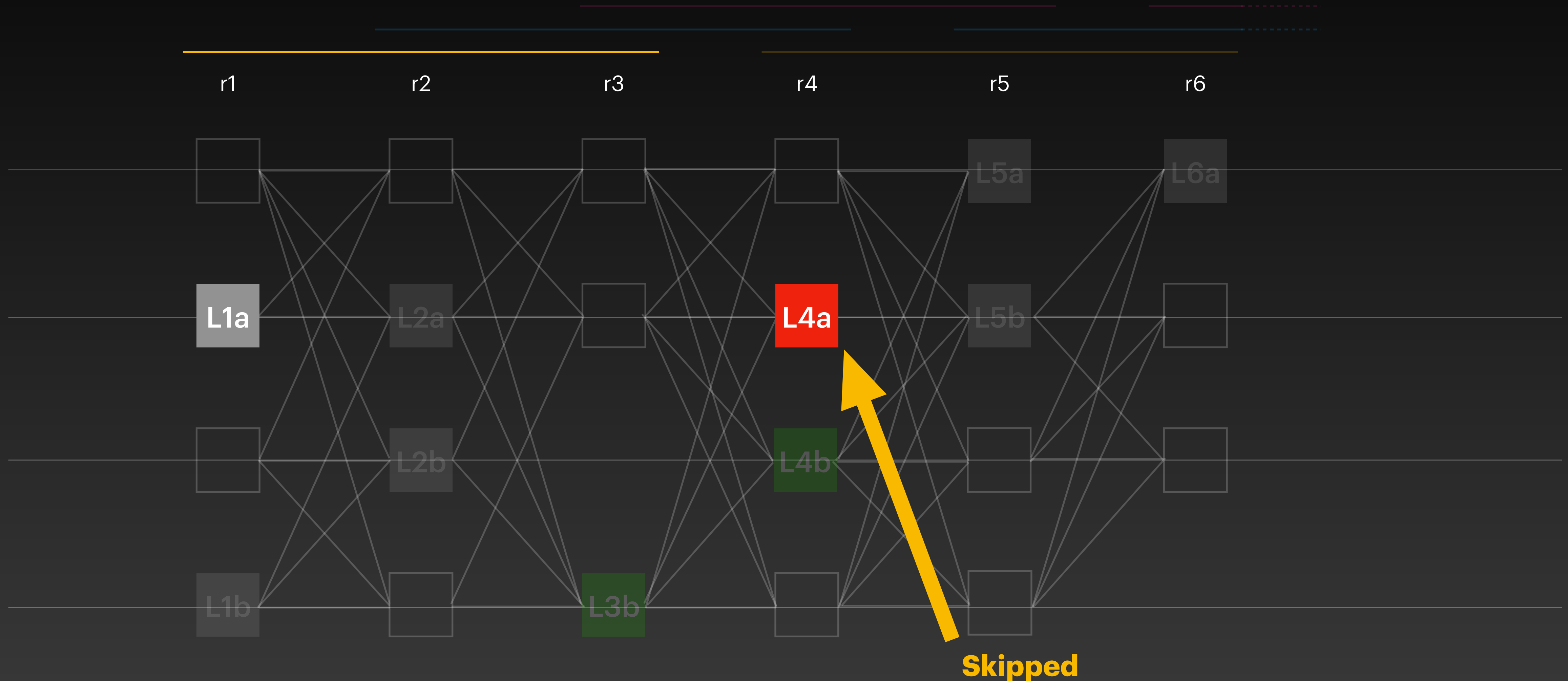
Apply Indirect Rule



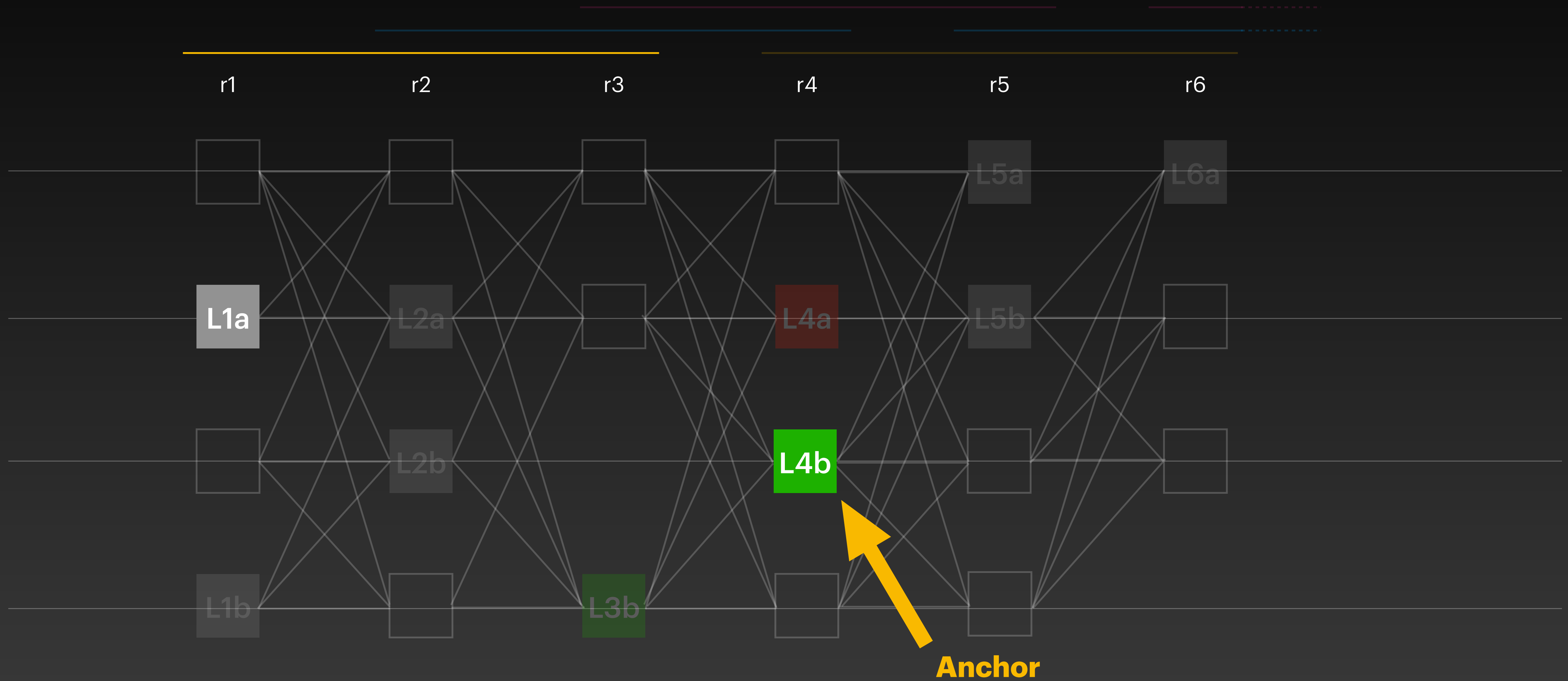
Apply Direct Rule



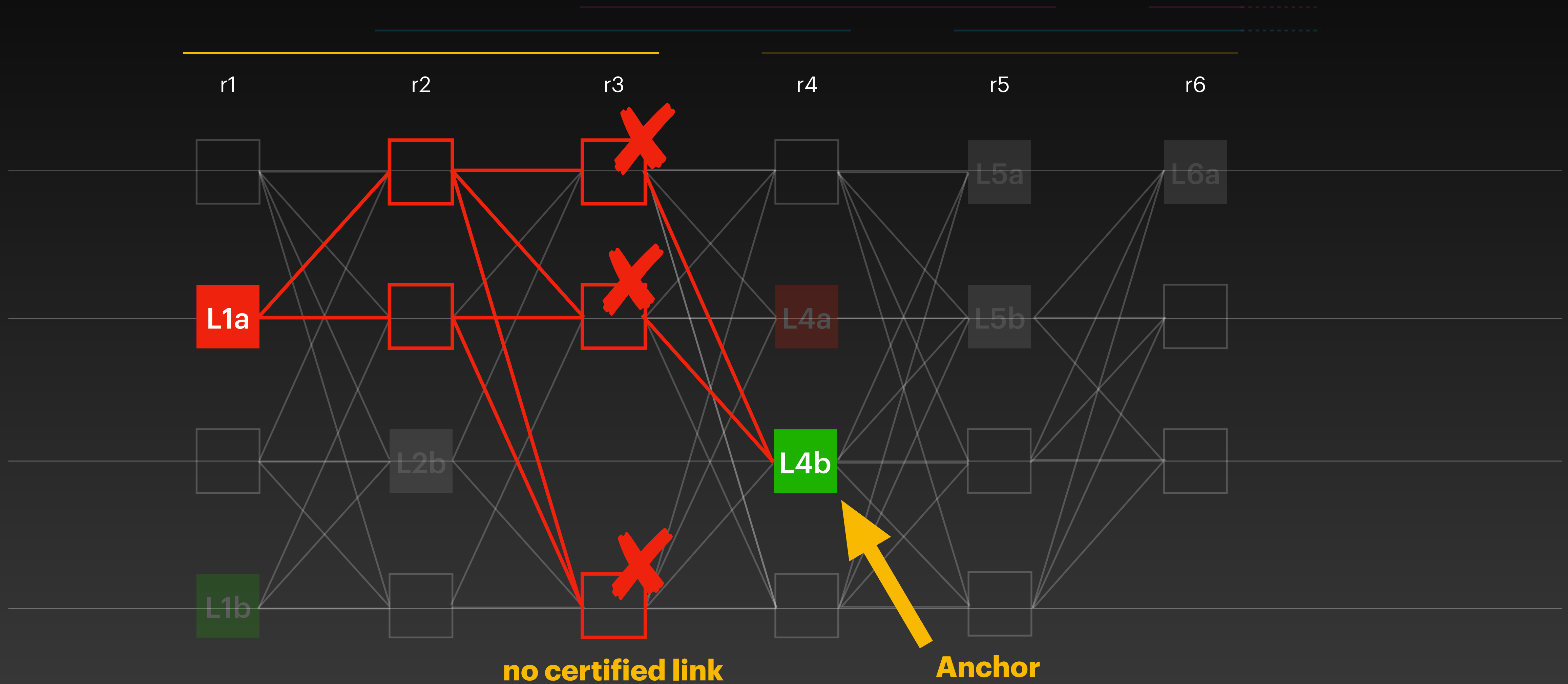
Apply Indirect Rule



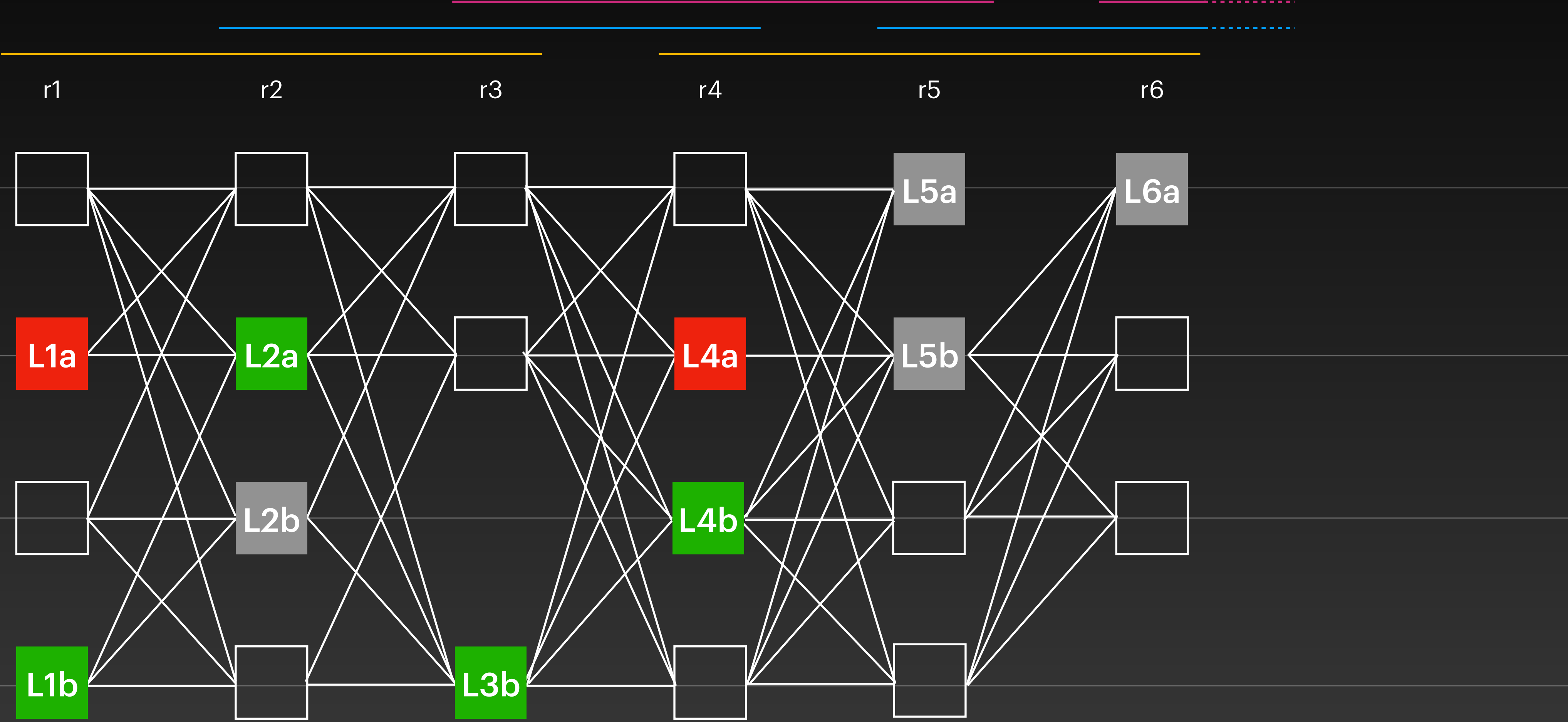
Apply Indirect Rule



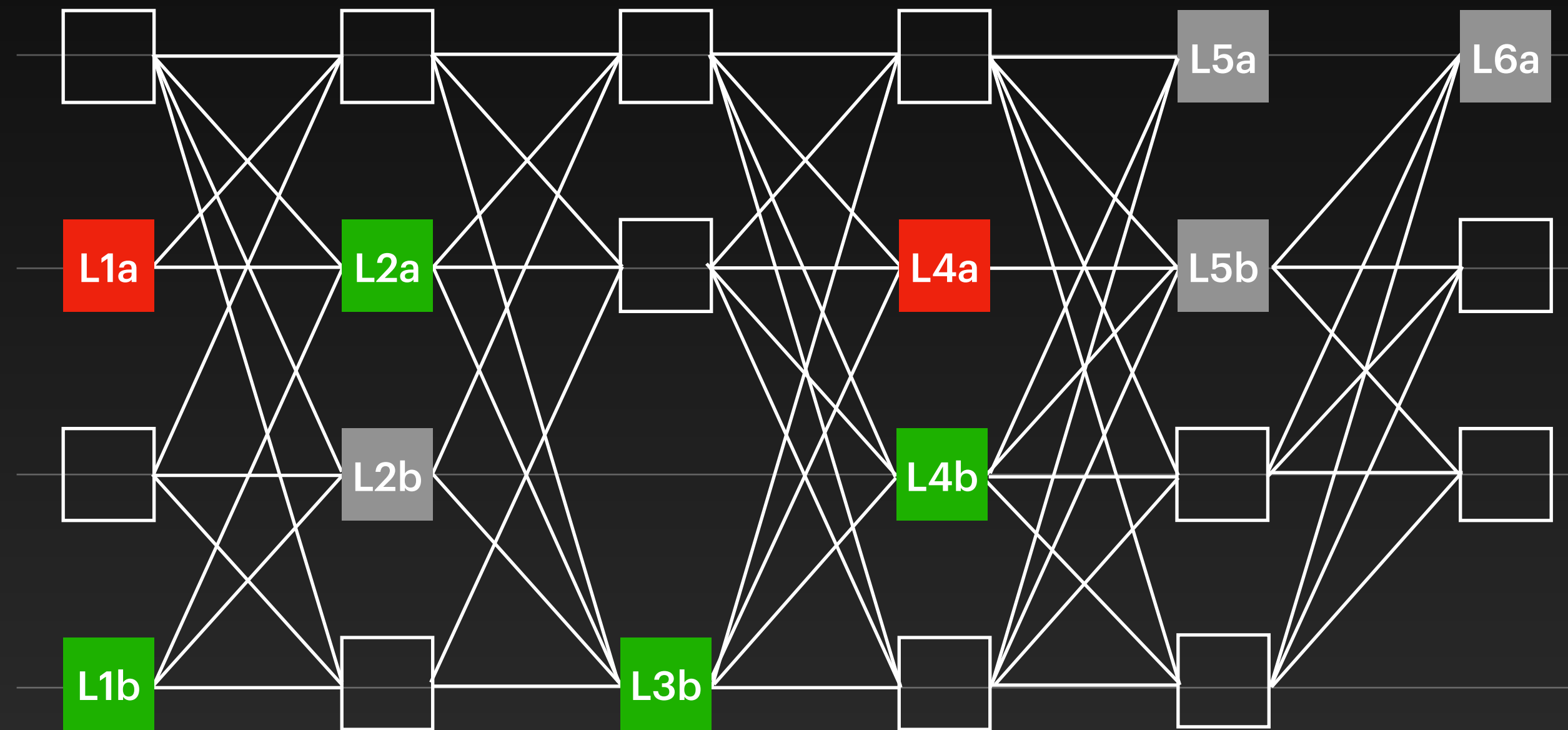
Apply Indirect Rule



Current Status



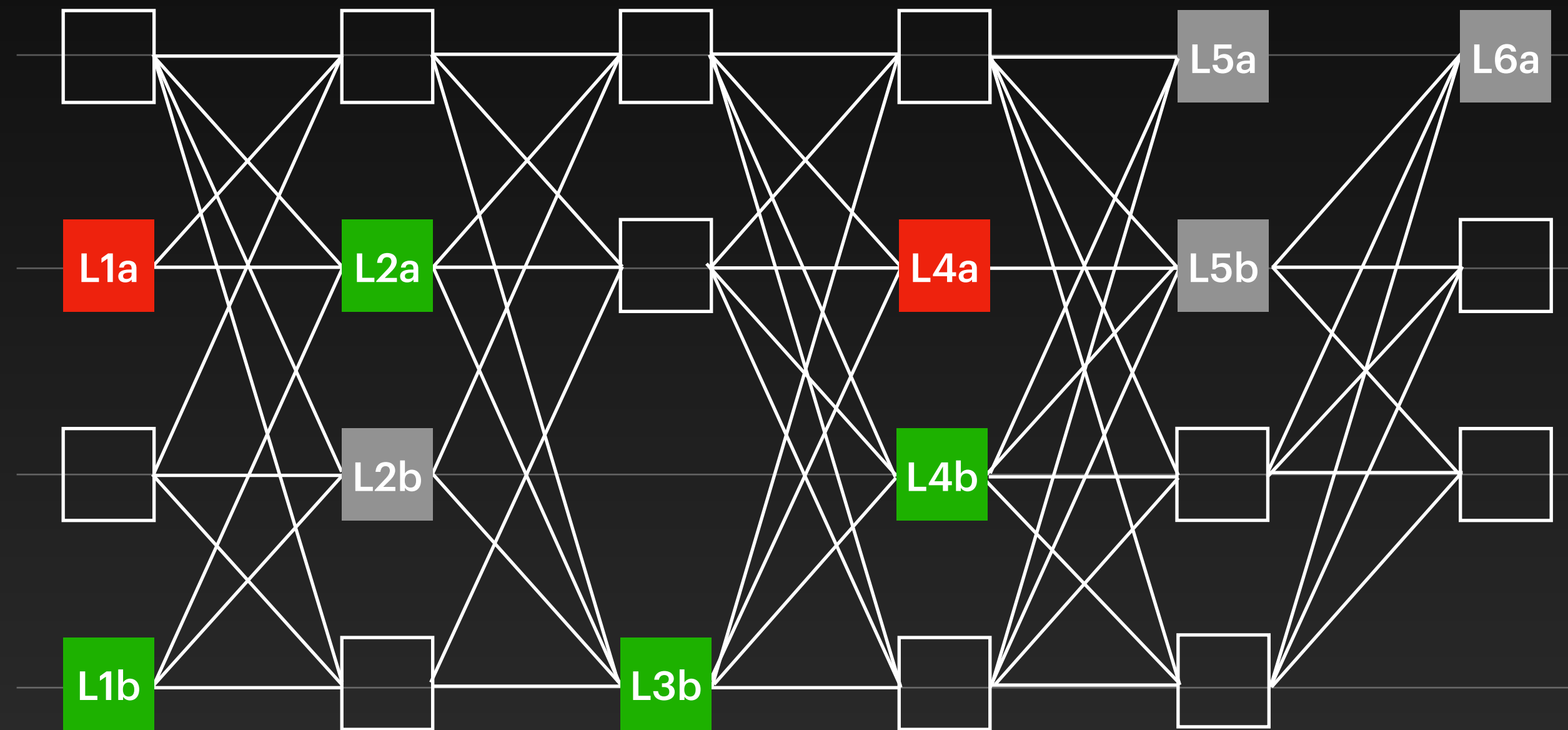
Take all leaders in order



leaders sequence:



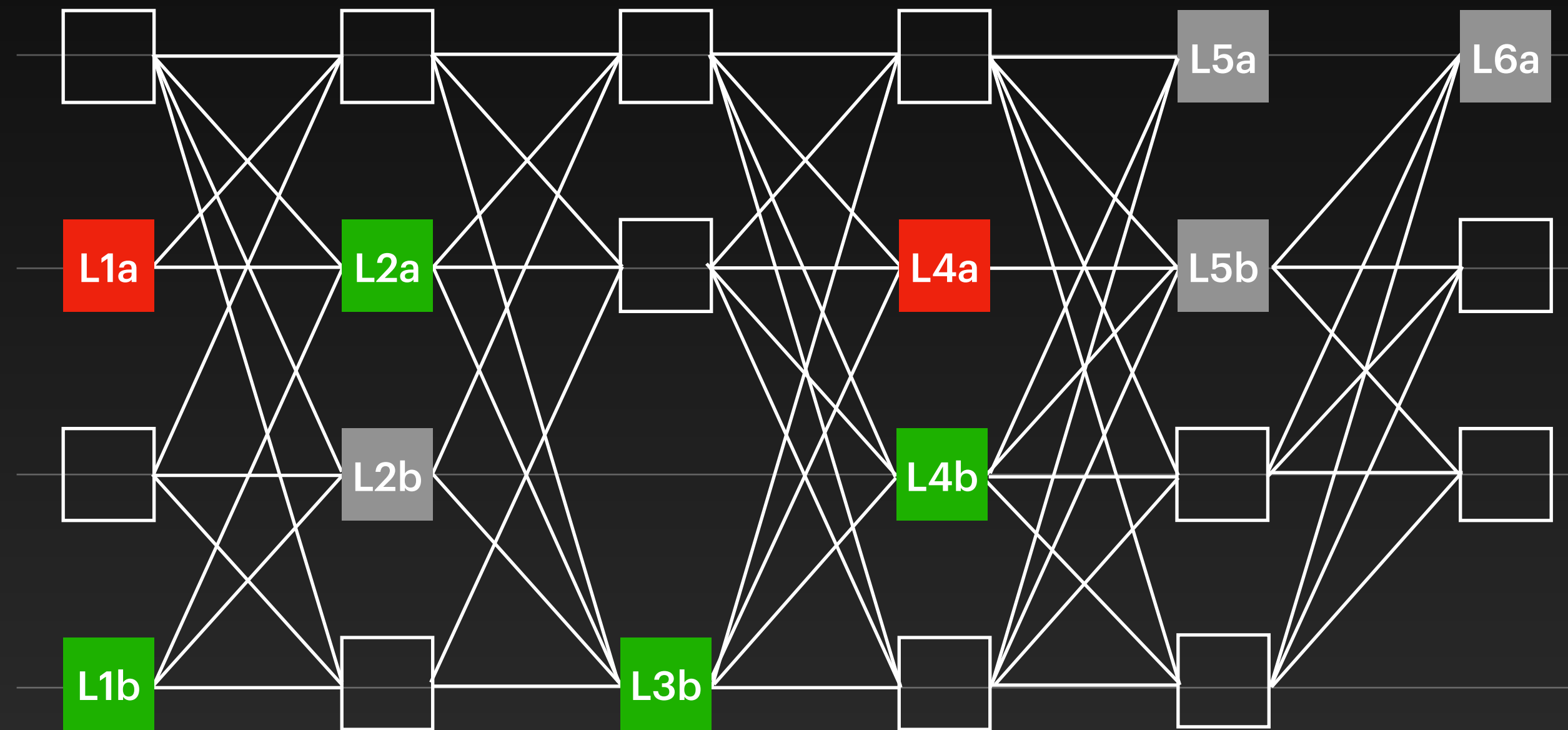
Stop at the first Undecided leader



leaders sequence:



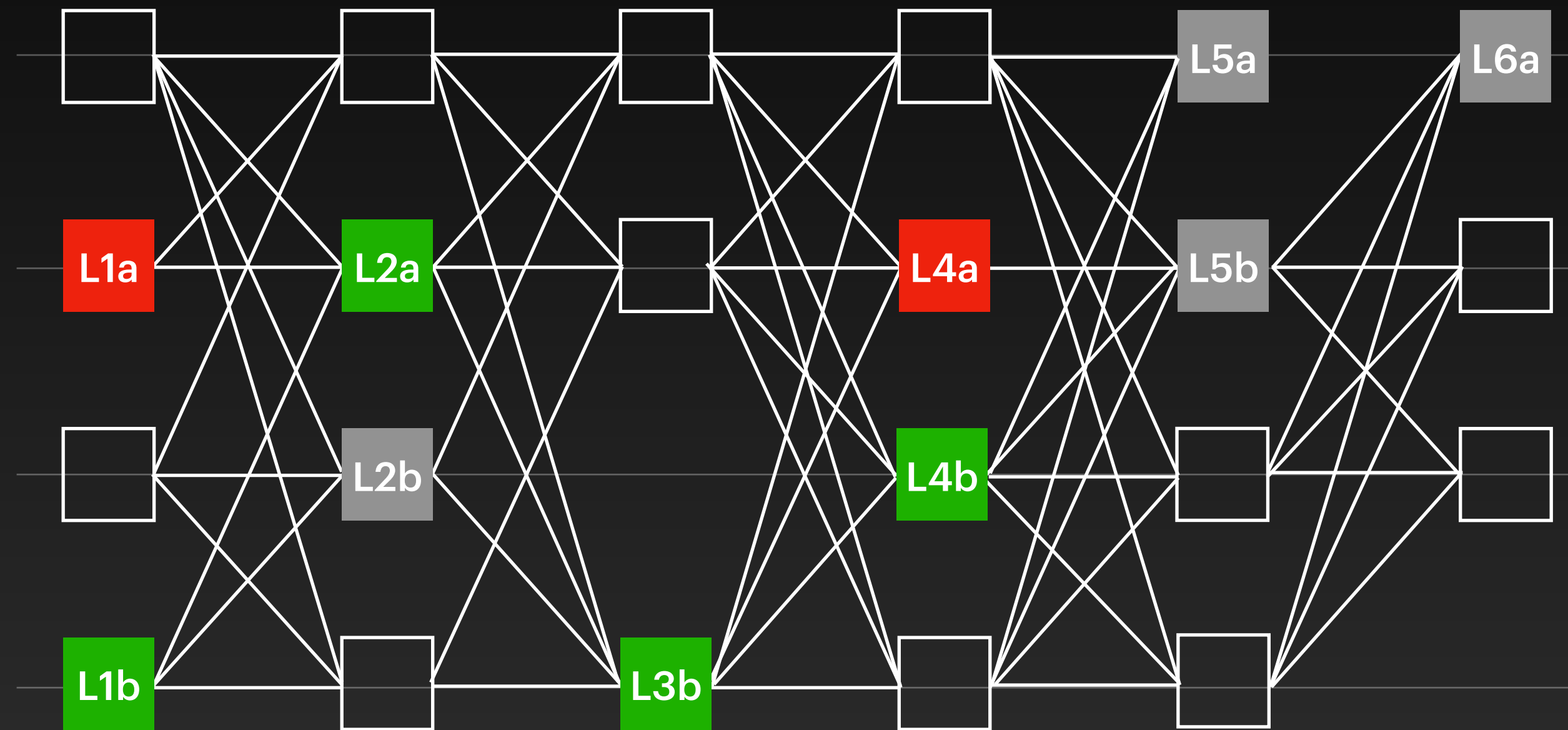
Remove skipped leaders



leaders sequence:



Final leader sequence

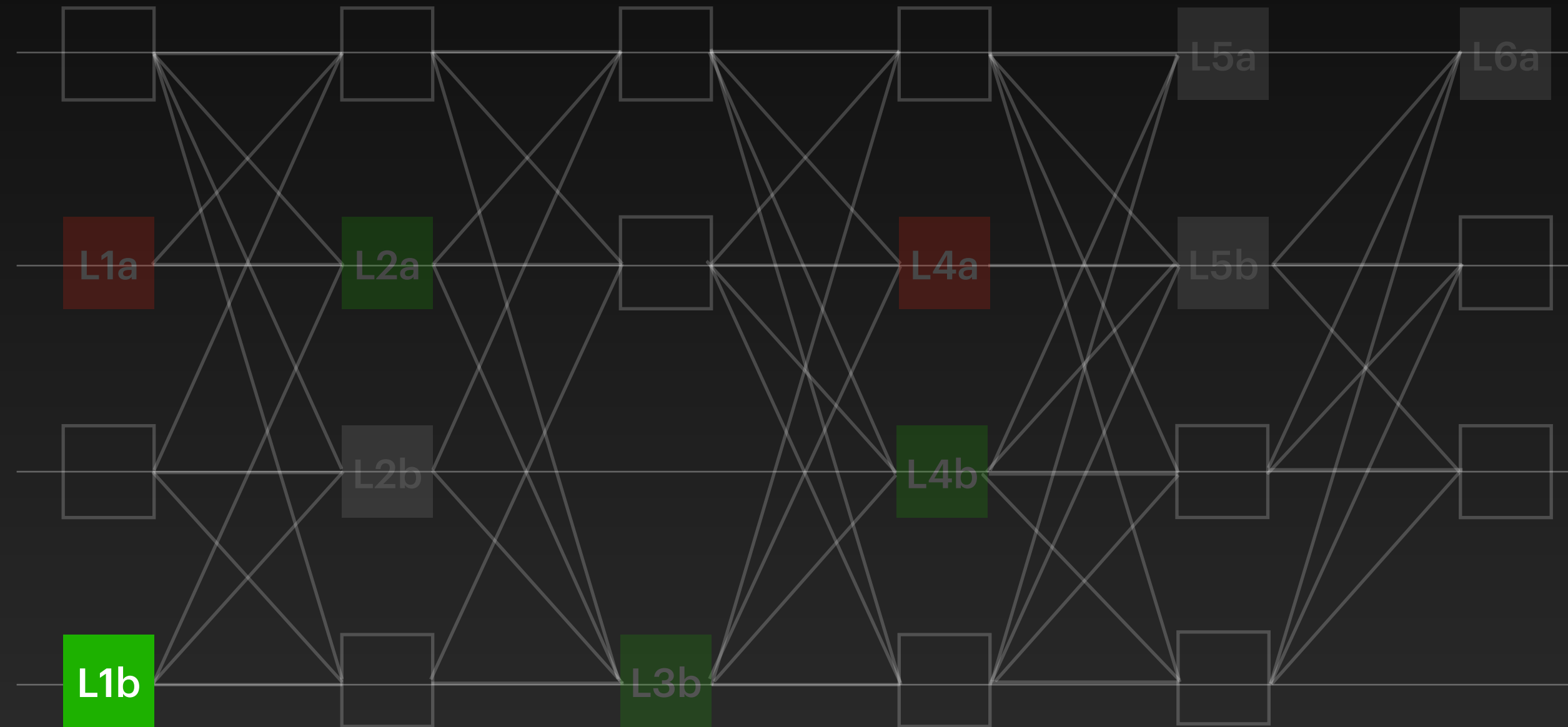


leaders sequence:

L1b

L2a

Commit sub-dag



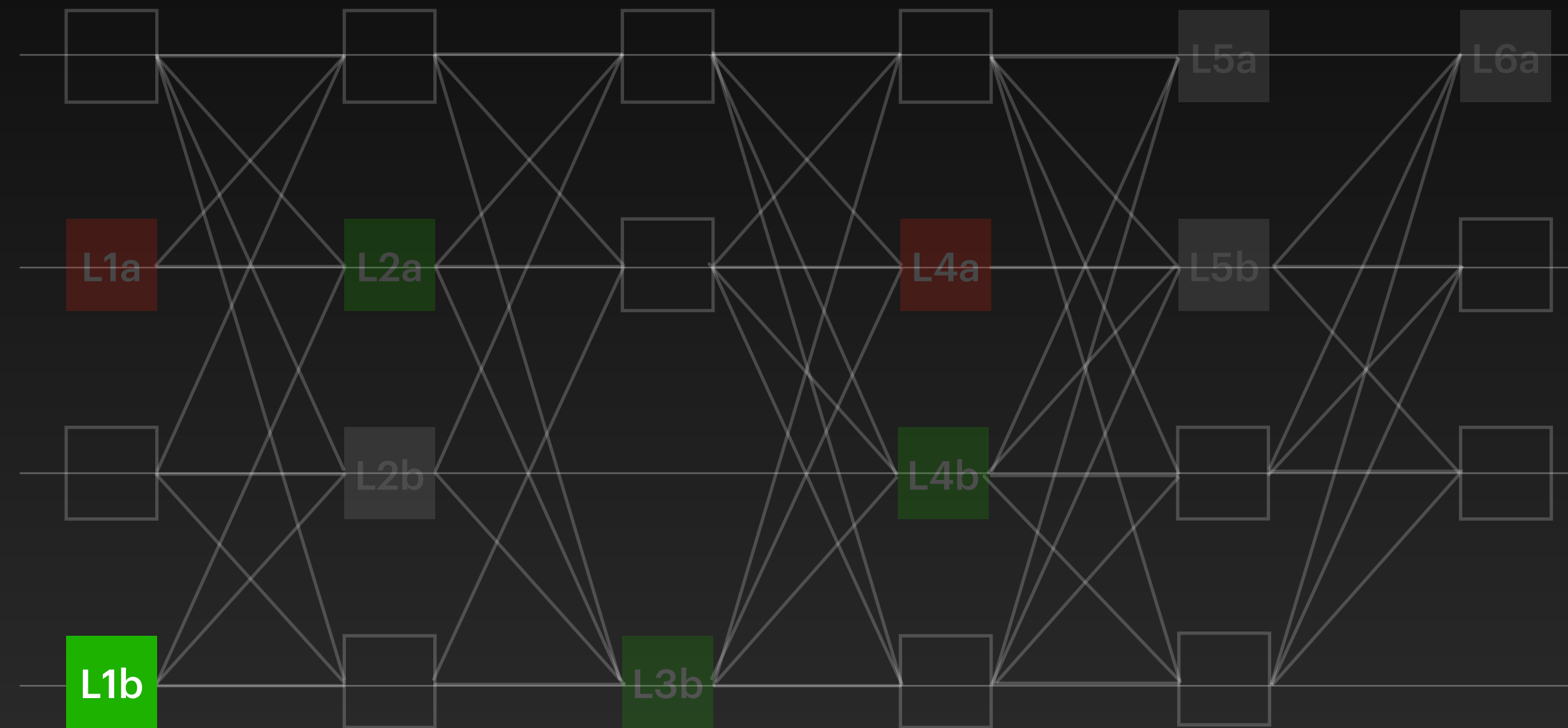
leaders sequence:

L1b

L2a

output sequence:

Commit sub-dag



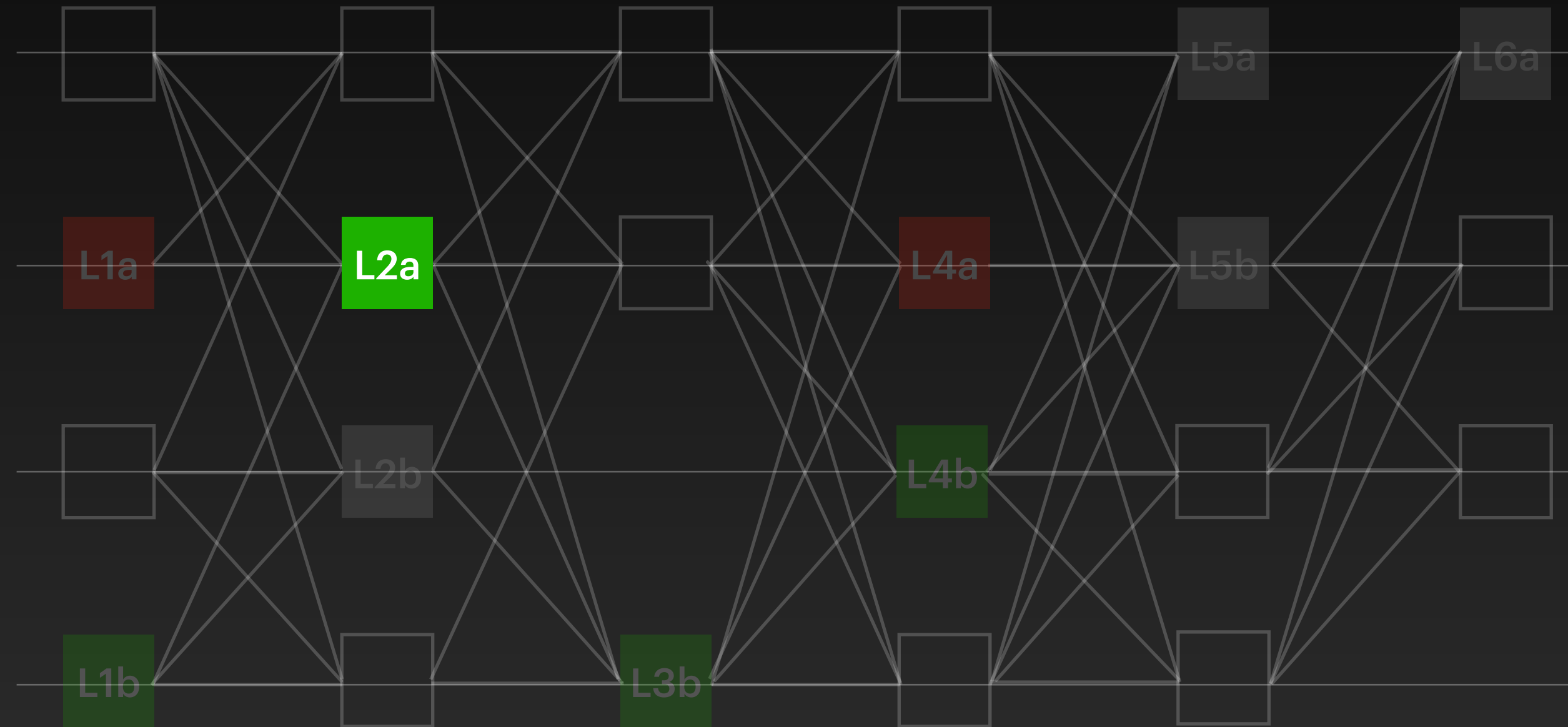
leaders sequence:

L2a

output sequence:

L1b

Commit sub-dag



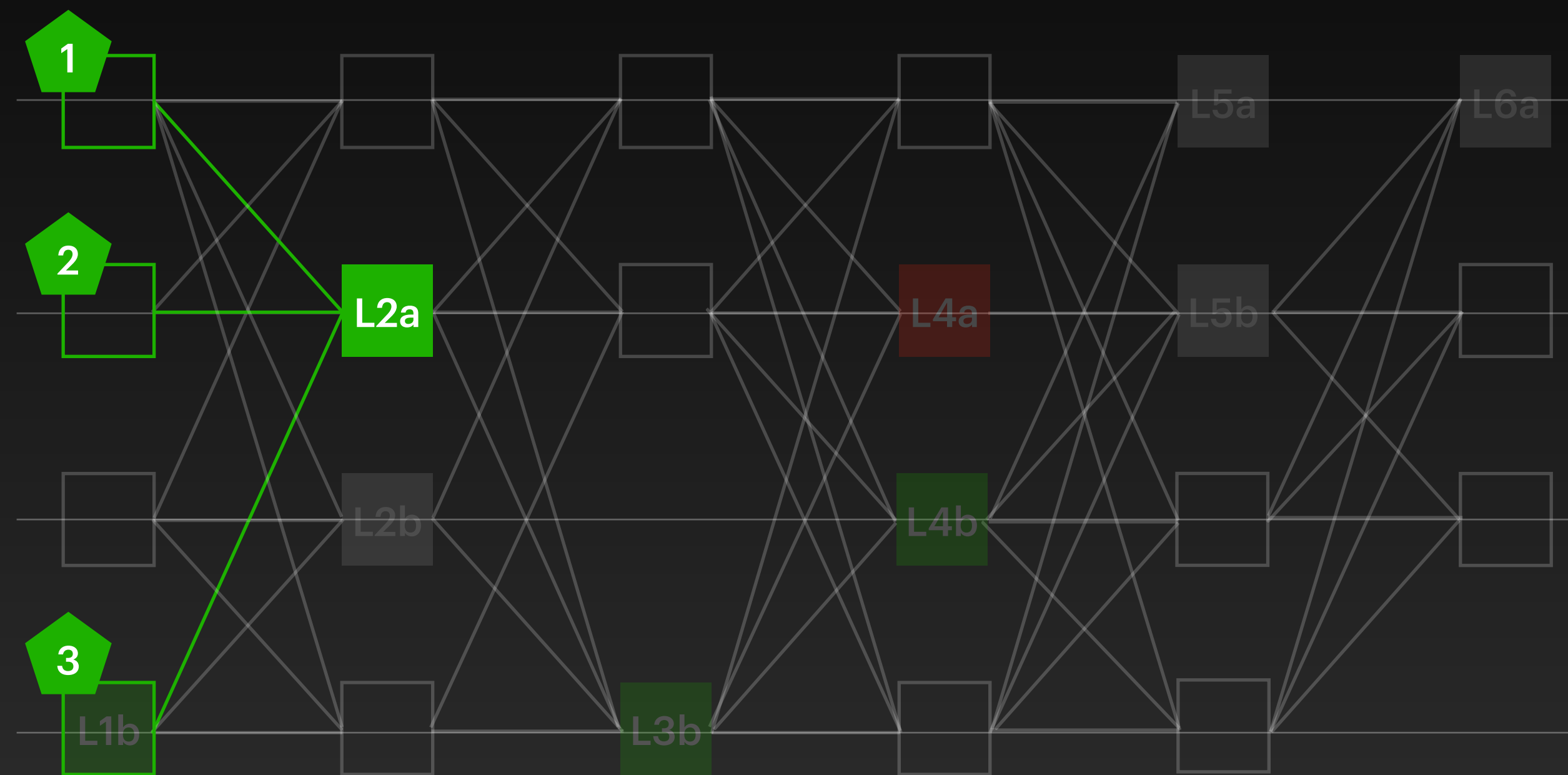
leaders sequence:

L2a

output sequence:

L1b

Commit sub-dag



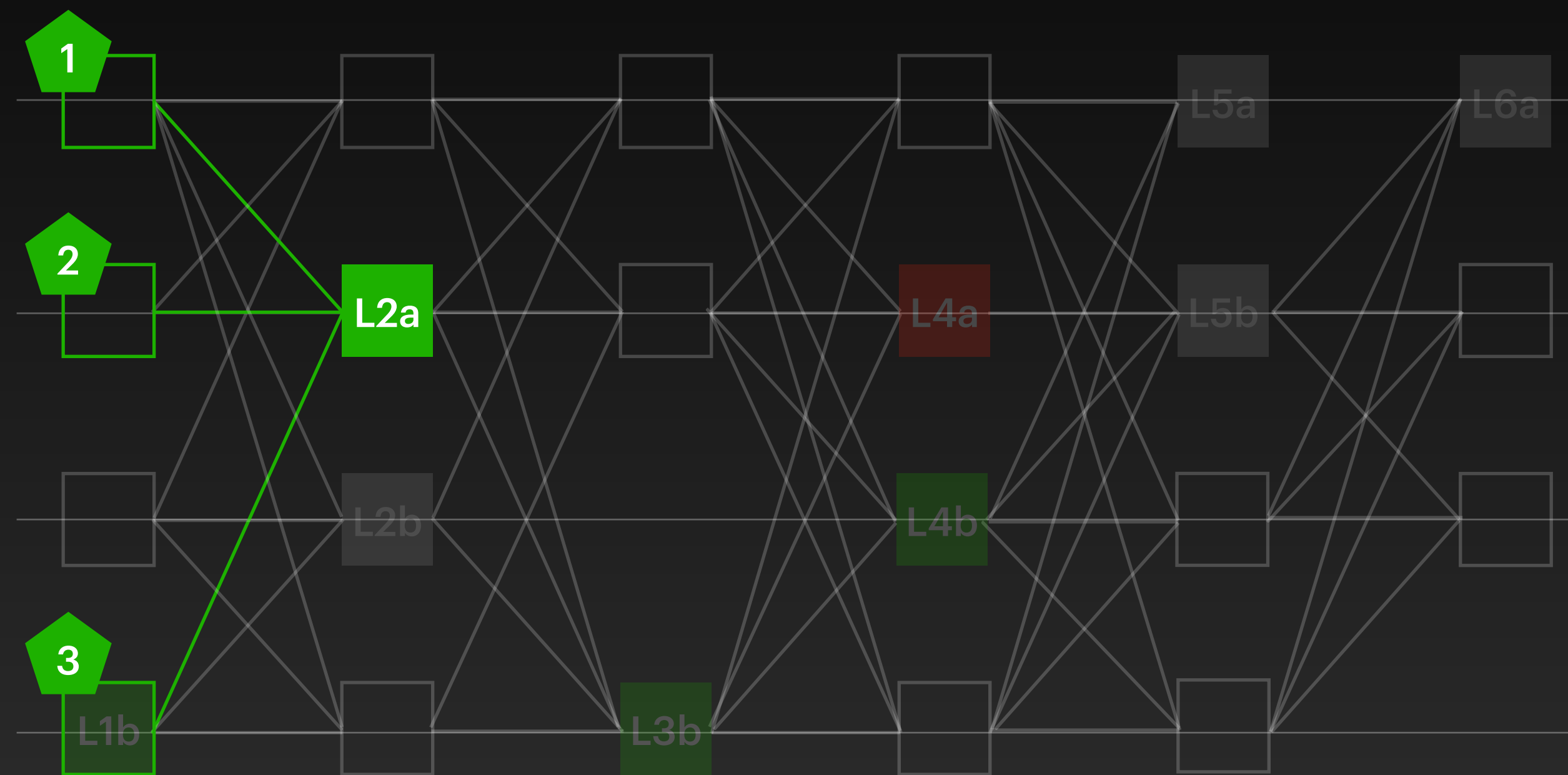
leaders sequence:

L2a

output sequence:

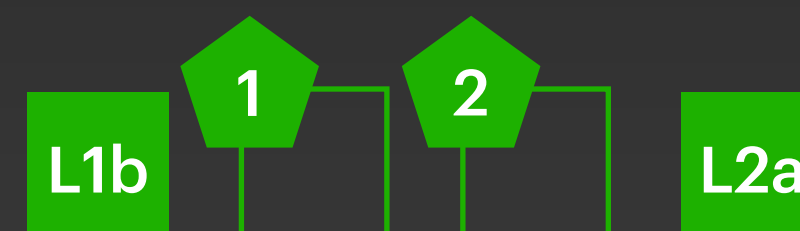
L1b

Commit sub-dag

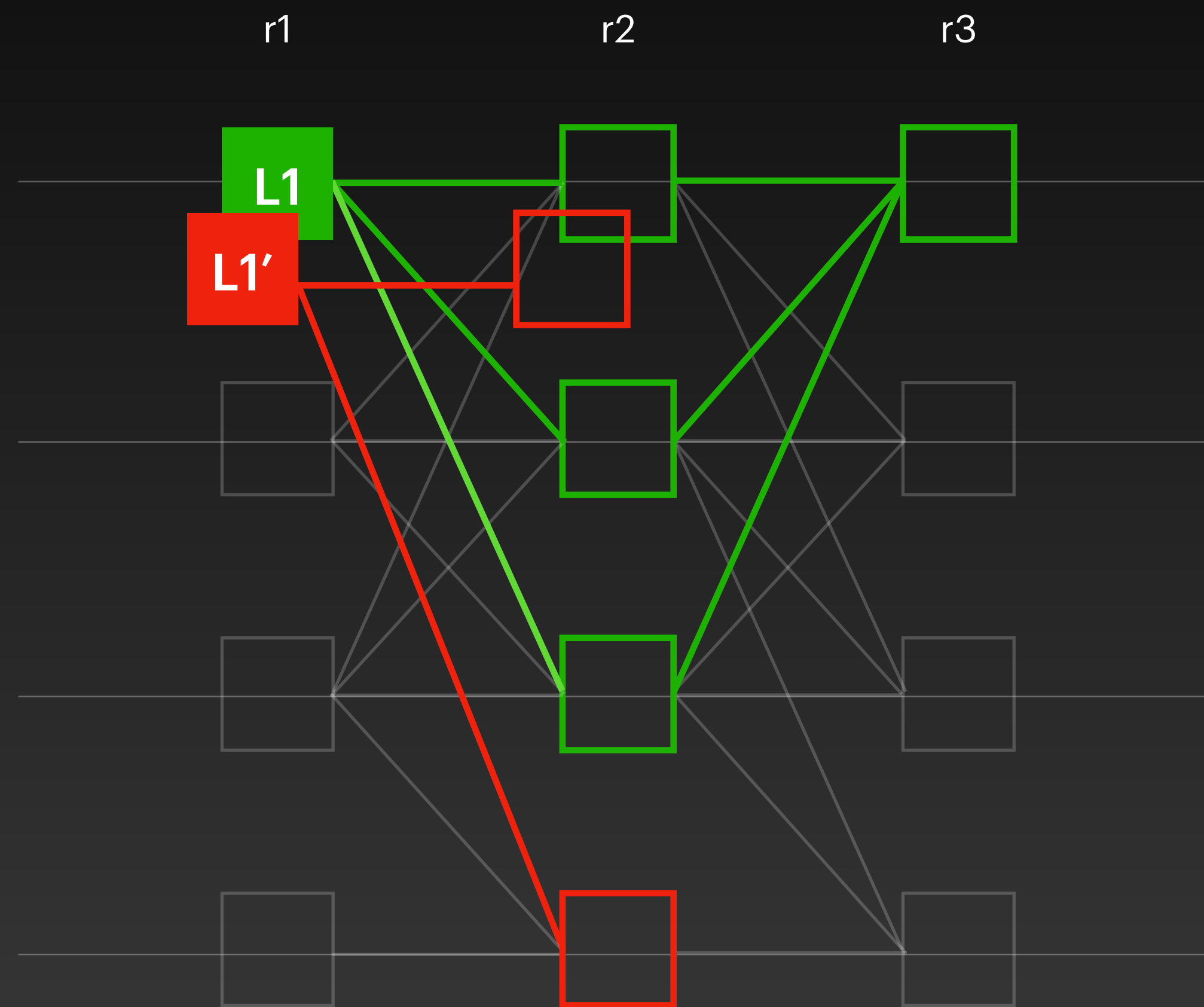


leaders sequence:

output sequence:

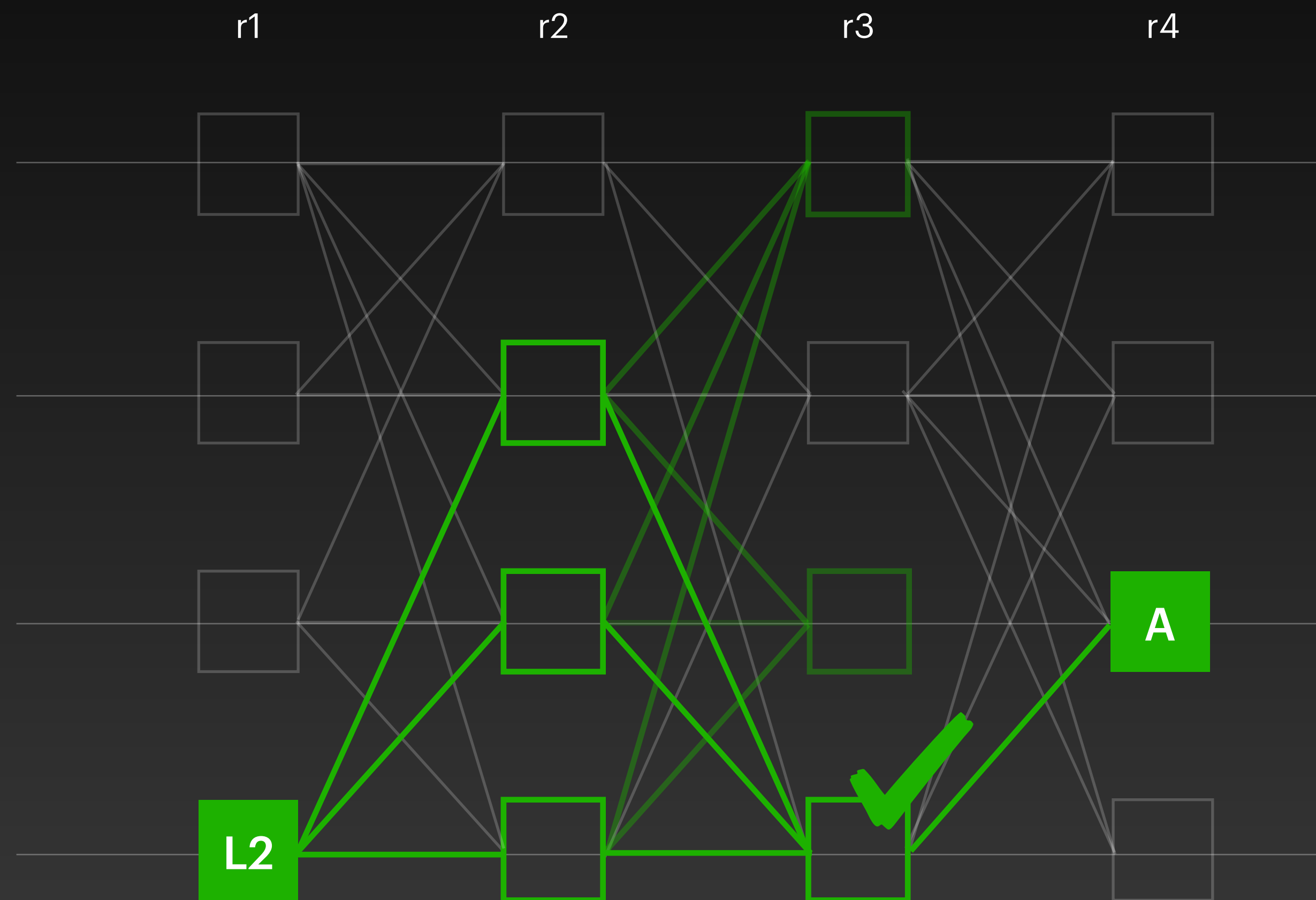


Safety



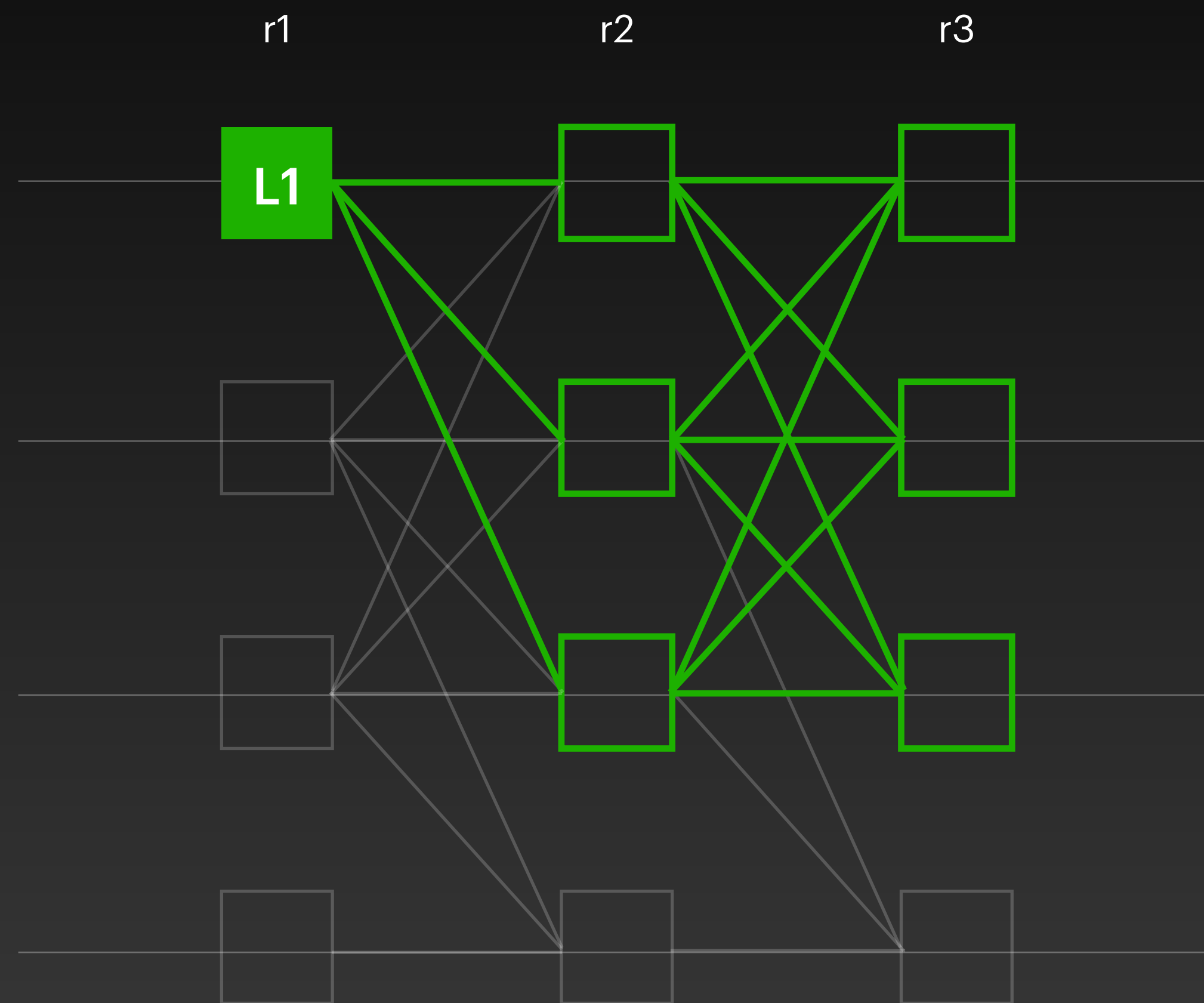
- At most **L1** or **L1'** can have a certificate pattern (quorum intersection)

Safety



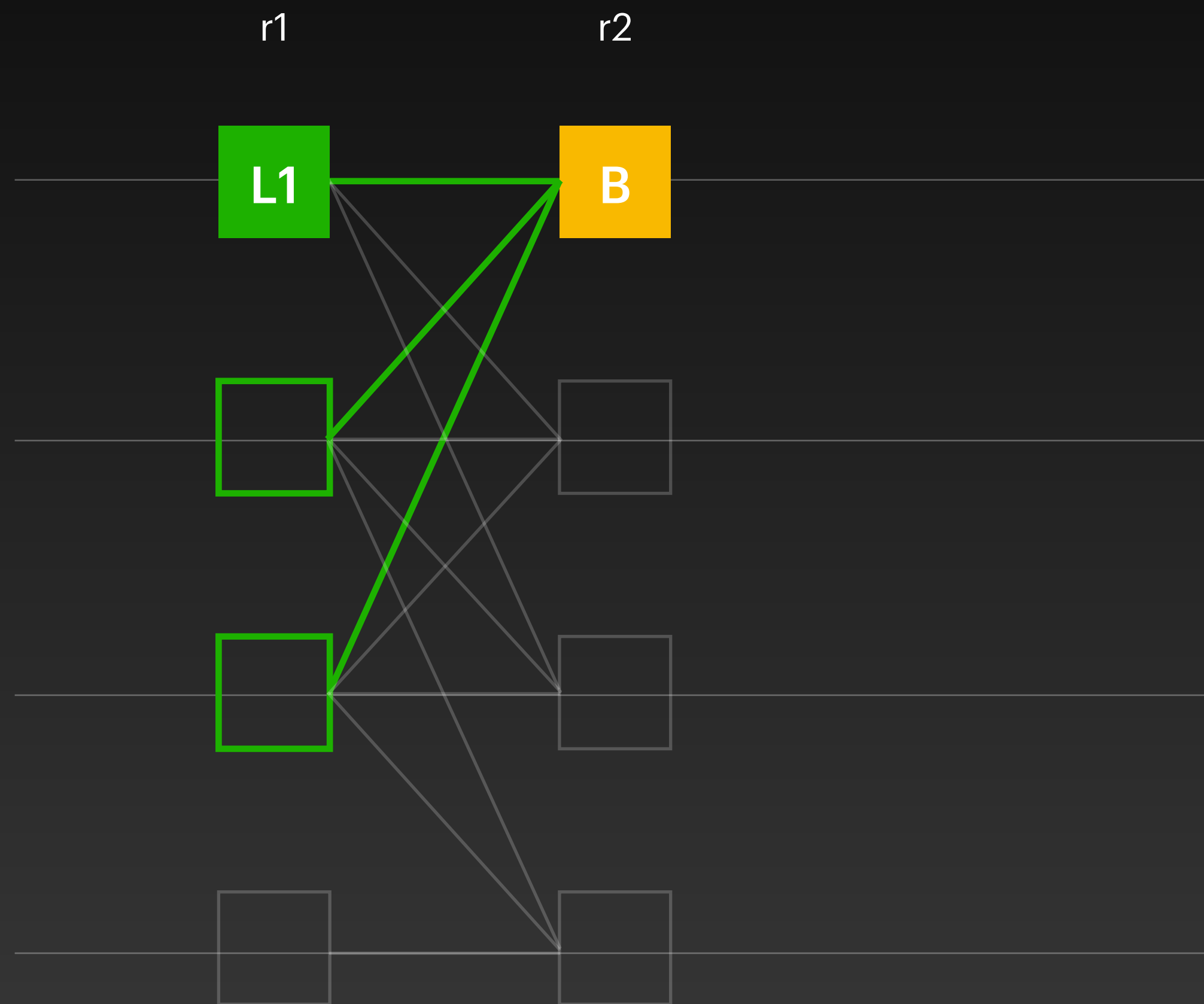
- At most **L1** or **L1'** can have a certificate pattern (quorum intersection)
- If **L2** has $2f+1$ certificate patterns, **A** always has a certified link to **L2**

Liveness



- After GST, the direct decision rule **commits** a block

Liveness



- After GST, the direct decision rule **commits** a block
- Leader timeout: **B** waits for $2f+1$ parents + 250 ms

**We are getting limited returns on
algorithmic improvements of BFT consensus**

controversial

High resource utilisation: throughput

Protocols commit often: latency

Engineering caught up with research

Change the assumptions (vs classic $3f+1$)

Specialised chains (vs one chain to rule them all)

Other components are bottleneck (network, execution, etc)

Block Synchroniser

Theory

Reliable Broadcast • Proactive dissemination

Prototype

Optimistic dissemination • Random pull on a timer (if needed)

Production

Optimistic dissemination • Proactive re-sharing on timer • Bulk-sync of past commits

Why DAGs

Narwhal (sections 1-4) • <https://sonnino.com/papers/narwhal-and-tusk.pdf>

Robustness & Throughput

Bullshark (partially synchronous) • <https://sonnino.com/papers/bullshark-simple.pdf>

Latency

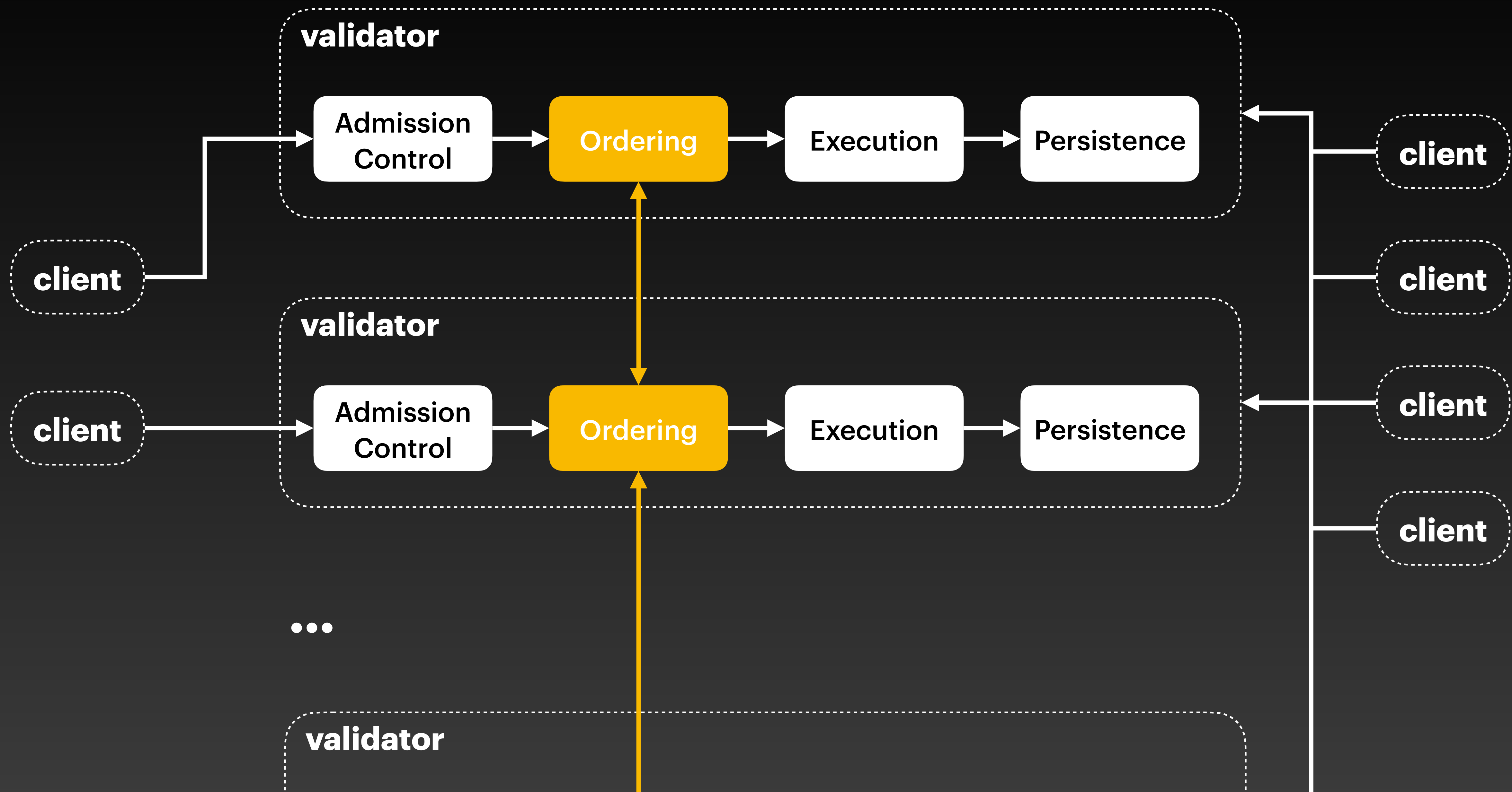
Shoal++ • <https://arxiv.org/pdf/2405.20488v1>

Uncertified DAG & Real World

Mysticeti (skip section 5) • <https://sonnino.com/papers/mysticeti.pdf>

Unstructured DAG

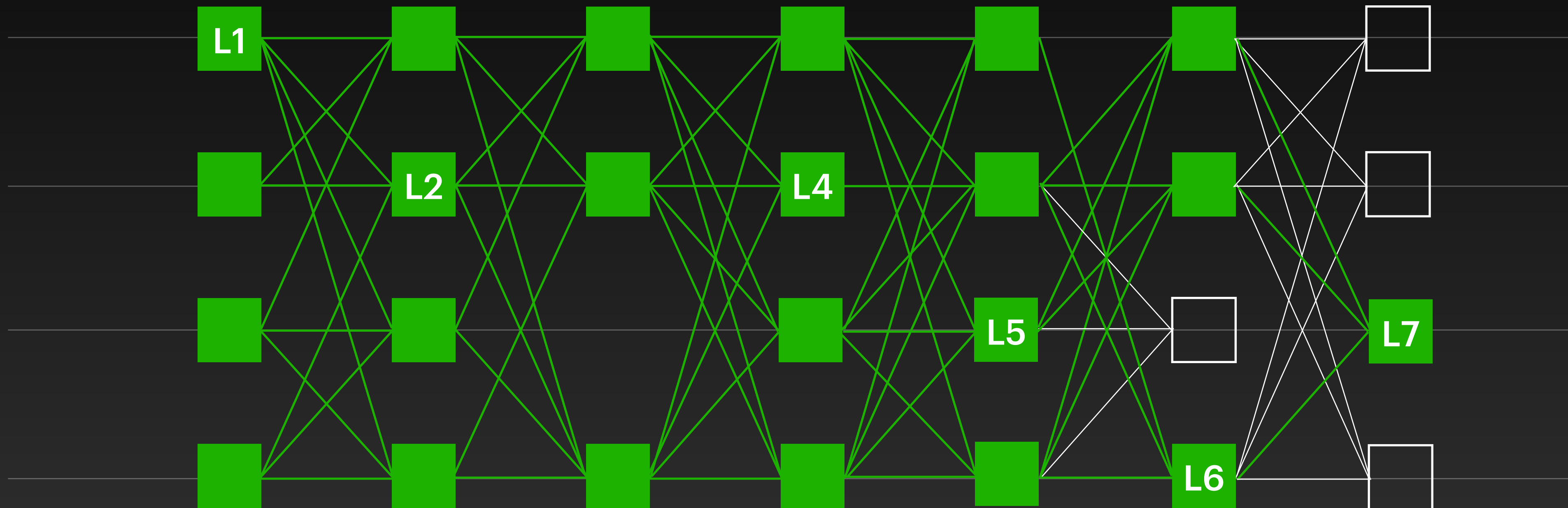
Autobahn • <https://arxiv.org/pdf/2401.10369>



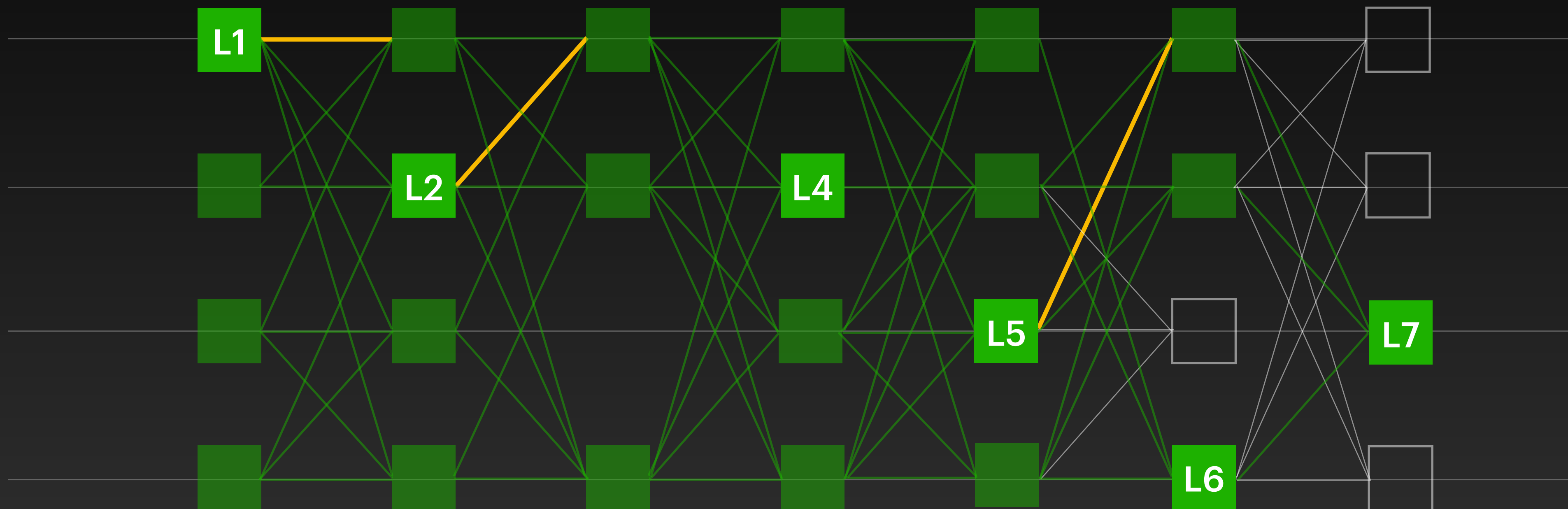
alberto@mystenlabs.com

EXTRA: **Leader Reputation**

Past Commits

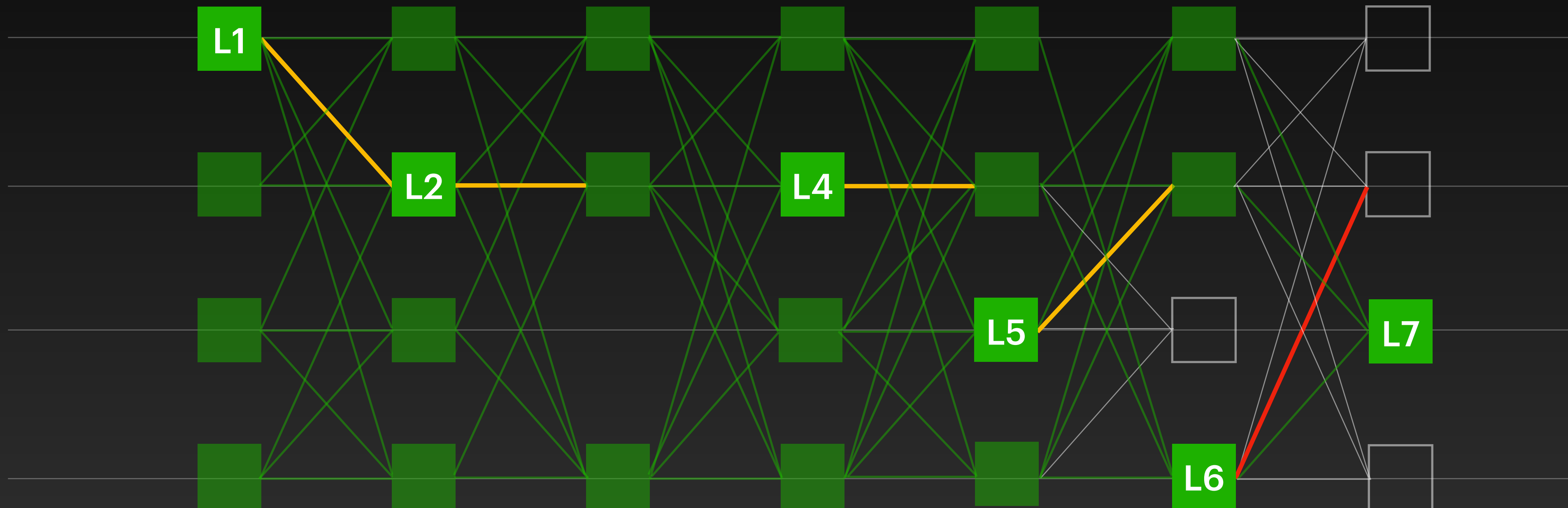


Compute Reputation Scores



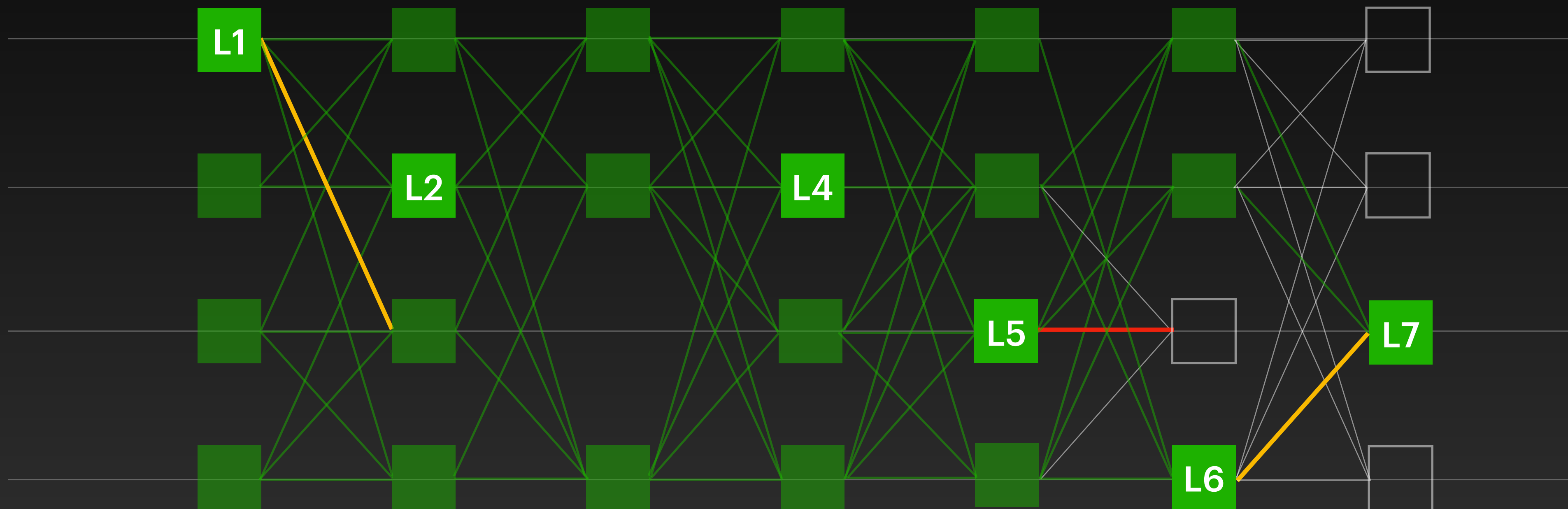
node 1: 3

Compute Reputation Scores



node 1: 3 node 2: 4

Compute Reputation Scores

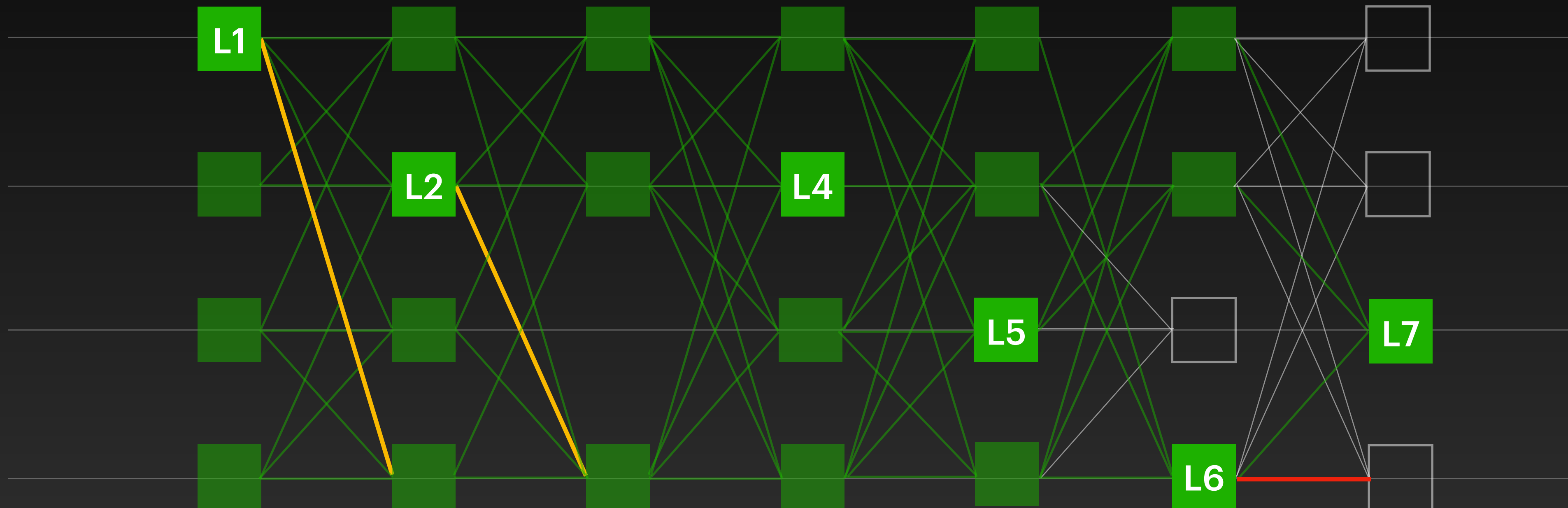


node 1: 3

node 2: 4

node 3: 2

Compute Reputation Scores



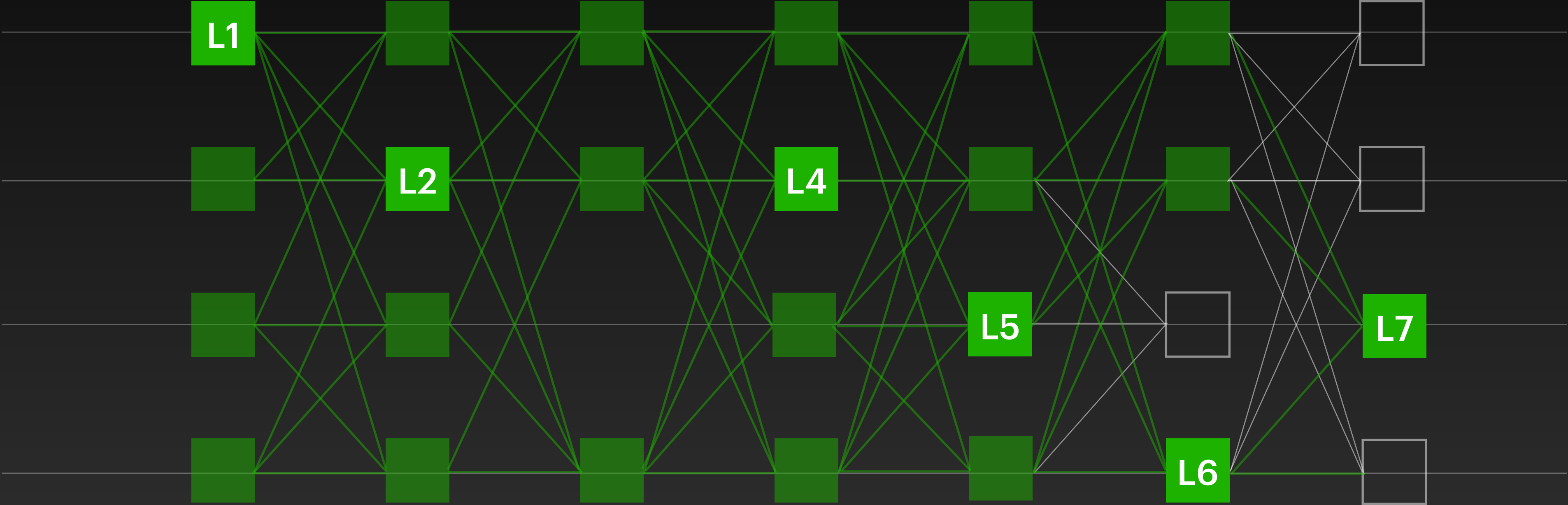
node 1: 3

node 2: 4

node 3: 2

node 4: 2

Future Leaders



node 1: 3

node 2: 4

node 3: 2

node 4: 2